

असली पुस्तक की
पहचान

SINCE
1986
40 YEARS

विश्वविद्यालय प्रकाशन (रजि.)

स्वातिथर
द्वारा प्रकाशित

20 प्रश्न

(परीक्षेपयोगी महत्वपूर्ण मॉडल प्रश्नोत्तर)

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Cost Accounting

[Major-2]

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Cost Accounting



MAJOR - II
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NEW SYLLABUS

Cost Accounting

Unit-I. Principle of best utilization of resources in ancient Indian tradition, Product costing in ancient business and economic models, Relationship between business ethics and cost control.

Meaning, Scope and Advantages of Cost Accounting. Difference between Cost Accounting and Financial Accounting, Cost Accounting Standard Costing Records and Accounting Auditing Rules.

Unit-II. Material Costing : Procurement of materials, Inventory Management and Control, Inventory Accounting. Cost price

method LIFO, FIFO, HIFO, Average cost method, Inflated price method. Physical Verification, Slow and Non-moving Stock and Treatment of Losses. Scrap, Spoilage, Defective and Normal-Abnormal Wastage.

Unit-III. Labour Costing : Accounting and control of labour cost, Time Keeping, Time Booking and Payroll, Overtime, Idle Time, Labour Turnovers and Fringe Benefits, Employee Cost Reporting, Methods of Wage Payments and Incentives Schemes - Halsey, Rowan, Taylor.

Unit-IV. Overhead Expenses and Unit Costing : Classification, allocation and absorption of Overhead Expenses. Under and Over Absorption, Capacity Level of Cost, treatment of certain items in costing like Interest on Capital, Packaging Expenses, Bad Debts, Research and Development Expenses. Calculation of unit costing and preparation of Cost Sheet. Job Costing.

Unit-V. Contract Costing : Contract Costing, Sub-Contract Costing, Process Costing (excluding Process Losses). Joint and By-products.

Unit-VI. Operating Cost Accounting : Transportation and hotel service costing, Integral and Non-Integral Cost Accounting, Reconciliation of cost accounting records with financial accounts.



Assessment and Evaluation

Suggested Continuous Evaluation Methods

Maximum Marks : 100

Continuous Comprehensive Evaluation (CCE): 30 Marks

University Exam (UE): 70 Marks

Internal Assessment: Continuous Comprehensive Evaluation (CCE): 30	Class Test Assignment/Presentation/Quiz/ Peer Teaching	30 Marks
External Assessment: University Exam: 70	Section (A) : Multiple Choice Questions Section (B) : Short Questions Section (C) : Long Questions	70 Marks
Total Marks 100		

COST ACCOUNTING

UNIT-I

Short Answer type Questions :

Q.1. What is the principle of best utilization of resources in ancient Indian tradition?

Ans. The principle of best utilization of resources in ancient Indian tradition focused on minimizing waste and ensuring that every resource was used efficiently for the welfare of society. Ancient texts like *Arthashastra* emphasized planned resource allocation, sustainable use, and ethical production. Craftsmen and traders ensured raw materials were fully utilized, and any by-products were repurposed. This approach not only reduced costs but also promoted environmental balance and economic self-reliance. It laid the foundation for modern cost control by encouraging efficiency and accountability. This principle reflects an early understanding of resource optimization, which remains relevant in cost accounting practices today.

Q.2. How was product costing applied in ancient business and economic models?

Ans. In ancient Indian business systems, product costing was applied through careful calculation of material, labor, and time used in production. Artisans and traders maintained records of inputs like raw materials, wages, and overheads to determine the cost of goods. Pricing was set after adding a fair profit margin, ensuring both ethical trade and sustainability. Guilds and local markets followed standard costing norms to maintain quality and control prices. Though informal compared to today's accounting systems, these practices represented a structured costing model. This ancient system is seen as a precursor to modern cost accounting and cost control mechanisms.

Q.3. What is the relationship between business ethics and cost control?

Ans. Business ethics and cost control are closely related. Ethical practices ensure that costs are reduced without compromising quality, employee welfare, or environmental standards. In ancient Indian tradition, ethical conduct in trade and production was essential to build trust and reputation. Modern cost control similarly focuses on efficiency while maintaining fair wages, honest reporting, and responsible use of resources. Unethical cost cutting may lead to short-term savings but harms business in the long run. Ethical cost control, however, leads to sustainable profits, customer trust, and long-term growth. Thus, ethics act as the guiding force for effective cost control.

Q.4. What is the meaning and scope of Cost Accounting?

Ans. Cost Accounting is a branch of accounting that deals with recording, classifying, analyzing, and controlling costs related to production and operations. Its main purpose is to determine the cost of goods or services, control expenses, and assist in decision-making. The scope of cost accounting includes cost ascertainment, cost control, budgeting, standard costing, cost reduction, and performance evaluation. It is applicable in manufacturing as well as service sectors. By providing detailed cost information, it supports managerial planning and helps improve efficiency. In short, cost accounting covers all activities related to tracking and managing business costs effectively.

Q.5. What are the advantages of Cost Accounting?

Ans. Cost Accounting offers several advantages to businesses. It helps in determining accurate product costs, which assists in setting proper prices. It improves cost control by identifying wastage and inefficiencies in the production process. It supports better budgeting, cost planning, and decision-making by management. Cost accounting also ensures optimal utilization of resources, increasing profitability. Additionally, it provides data for performance evaluation and cost comparison over time. By promoting transparency and accountability, it strengthens overall financial management. For these reasons, cost accounting is considered an essential tool for both operational and strategic business success.

Q.6. What is the difference between Cost Accounting and Financial Accounting?

Ans. Cost Accounting and Financial Accounting serve different purposes. Cost Accounting focuses on determining and controlling the cost of production and operations. It provides internal reports to management for planning and decision-making. Financial Accounting, on the other hand, records all business transactions and prepares financial statements like the Profit and Loss Account and Balance Sheet for external stakeholders. Cost Accounting is detailed, flexible, and future-oriented, whereas Financial Accounting is historical and follows standardized formats. Cost accounting helps in cost control and efficiency, while financial accounting ensures compliance and overall financial reporting. Both systems are complementary, together giving a complete view of business performance.

Q.7. What is meant by Cost Accounting Standards?

Ans. Cost Accounting Standards (CAS) are a set of guidelines issued to bring uniformity, consistency, and transparency in cost accounting practices across industries. They define principles and methods to be followed while determining various cost elements such as material, labor, overhead, and administrative expenses. By following CAS, organizations can ensure accuracy and comparability of cost data. These standards also support better pricing decisions, cost control, and compliance with

regulatory requirements. In India, CAS is issued by professional bodies like The Institute of Cost Accountants of India. Adopting cost accounting standards builds stakeholder trust and improves managerial decision-making.

Q.8. What are Costing Records?

Ans. Costing Records are systematic documents that contain detailed information about the cost of materials, labor, overheads, and production. These records help in ascertaining the total and per-unit cost of products or services. Typical costing records include material requisition slips, labor time sheets, overhead distribution sheets, cost sheets, and inventory records. Maintaining accurate costing records enables businesses to identify wastage, control expenses, and improve efficiency. It also provides reliable data for management decisions and compliance with statutory cost audit requirements. Costing records are an essential part of a good cost accounting system, ensuring transparency and accountability.

Q.9. What is meant by Cost Accounting Audit?

Ans. Cost Accounting Audit refers to the systematic examination of cost records and accounts to ensure their accuracy, compliance with standards, and efficiency of cost control measures. It verifies whether proper costing systems have been maintained and whether cost data truly reflect the operations. In India, certain companies are required to maintain cost records and undergo cost audits as per legal regulations. A cost audit helps in detecting errors, fraud, inefficiencies, and areas where costs can be reduced. It ultimately improves cost management, pricing strategies, and profitability. This makes cost audit a crucial part of modern business governance.

Q.10. Why are Cost Accounting Rules important in business?

Ans. Cost Accounting Rules provide a legal framework for maintaining cost records, conducting cost audits, and ensuring compliance with cost accounting standards. These rules ensure that companies follow uniform and transparent cost reporting practices, which support accurate pricing, fair competition, and effective cost control. They help regulators monitor cost structures in key industries and protect consumer interests. For businesses, following cost accounting rules builds credibility and enhances decision-making. In India, such rules are framed under the Companies Act and are applicable to specified industries. Overall, cost accounting rules strengthen both internal management and external accountability.

Long and short Answer type Questions :

Q.1. What were the principles followed in ancient Indian tradition for the best utilization of resources?

Or

Enumerate some principles followed in ancient India to utilize their given resources at its best.

Ans. Ancient Indian tradition emphasized the principle of Aparigraha (non-possessiveness) and Asteya (non-stealing), which promoted the best utilization of resources by discouraging over-consumption and waste. This philosophy extended to environmental stewardship, where nature was respected and resources were used judiciously, with an understanding of their finite nature and the interconnectedness of all living beings. Practices like water harvesting, organic farming, and recycling were integral to this approach. Here's a more detailed breakdown :

(1) Non-Possessiveness and Non-Stealing (Aparigraha and Asteya) :

- (i) These principles, fundamental to Jainism and also present in Hinduism, advocate for minimizing one's possessions and avoiding taking more than what is needed.
- (ii) This mindset naturally leads to reduced waste and a more sustainable approach to resource consumption.
- (iii) By practicing non-possessiveness, individuals are encouraged to value what they have and avoid hoarding, which frees up resources for others and minimizes the overall demand on the environment.

(2) Respect for Nature and Interconnectedness :

- (i) Ancient Indian philosophies, including those found in the Vedas, Upanishads, and Jain/Buddhist teachings, consistently emphasize the sanctity of nature and the interconnectedness of all living beings.
- (ii) This understanding fosters a sense of responsibility towards the environment, encouraging practices that minimize harm and promote ecological balance.
- (iii) The concept of Panchamahabhutas (five elements) in Ayurveda highlights the importance of maintaining harmony between the human body and the natural world, influencing sustainable practices in agriculture and urban planning.

(3) Sustainable Practices :

(i) Water Harvesting : Ancient Indians developed sophisticated systems for collecting and storing rainwater, such as step wells, tanks, and reservoirs.

(ii) Organic Farming : Practices like using cow dung and urine as natural fertilizers, promoting biodiversity, and conserving indigenous seeds were common.

(iii) Recycling and Upcycling : Broken pottery was repurposed as planters, and old clothes were used to make new items.

(iv) Efficient Resource Use : Traditional architecture focused on natural light and ventilation, reducing the need for artificial energy.

(v) Community-Based Resource Management : Water resources were often managed collectively by local communities, ensuring equitable access and sustainable use.

(4) Examples in Daily Life :

(i) Traditional Utensils : Clay pots, metal utensils, and other items made from natural, biodegradable materials were used, minimizing reliance on plastics and promoting a circular economy.

(ii) Food Preparation : Banana leaves were used for serving food, reducing waste and promoting a connection with nature.

(iii) Respect for Animals : Cows were revered, and their byproducts were used in agriculture. Other animals were also treated with respect, reflecting a deep understanding of the importance of biodiversity.

In essence, ancient Indian tradition viewed the best utilization of resources not just as an economic or practical matter, but as a spiritual and ethical obligation, deeply intertwined with the principles of non-violence, respect for nature, and the interconnectedness of all living beings. This holistic approach offers valuable lessons for modern society as it seeks to address the challenges of sustainability and environmental conservation.

Q.2. Throw some light upon product costing in ancient business and economic models.

Ans. While formal, systematic product costing as understood today is modern, ancient societies still engaged in fundamental cost considerations for goods, including direct costs for materials and labour, and the allocation of indirect costs like the "overhead" of skilled artisans or workshop maintenance, often with pricing based on these perceived costs, scarcity, or perceived value rather than a formal system. For example, in ancient Rome, the "cost" of a luxury item would include the expense of rare materials, the specialized labour of artisans, and the risks of transport, with the final price reflecting these factors.

Ancient Cost Concepts -

(1) Direct Material Costs : Even without formal accounting, the cost of raw materials was a primary concern. A potter would consider the cost of clay and other additives, a weaver the cost of wool or flax, and a builder the cost of timber and stone.

(2) Direct Labour Costs : The skill and time of the artisan or labourer was a significant factor. A craftsman would expect compensation for their work, whether in the form of payment or goods, reflecting the value of their specialized abilities.

(3) Indirect Costs (Overhead) : While less formalized, ancient societies recognized factors beyond direct material and labour. This included:

- (i) **Workshop and Tool Costs:** The upkeep of tools, space, and any communal resources needed for production was implicitly accounted for.
- (ii) **Transport Costs:** The expense of bringing raw materials and distributing finished goods, particularly over long distances or through dangerous routes, was a key factor.
- (iii) **Skill and Scarcity:** Highly skilled labour, like that of a master silversmith or a renowned sculptor, was more expensive due to scarcity and its associated value. This was an implicit form of cost allocation.

Economic Models and Pricing :

- (1) **Scarcity and Value :** The cost of goods was heavily influenced by their scarcity. Rare items, whether due to difficult production or limited availability of materials, commanded higher prices.
- (2) **Perceived Value :** The worth of a product was often tied to its perceived value, which could include its utility, prestige, or symbolism.
- (3) **Barter and Exchange :** In early economies, the "cost" of an item was often determined through direct barter, with the value established by what a person was willing to give up in return.

Example from Ancient Rome :

Consider the production of a fine bronze statue in ancient Rome:

- (1) **Materials:** The cost of the bronze itself, including its acquisition.
- (2) **Labour:** The wages or sustenance of the skilled bronzesmiths and artisans.
- (3) **Overhead:** The cost of the foundry, tools, and any required fuel or resources.
- (4) **Transport:** The cost of moving the metal to the workshop and the finished statue to market.
- (5) **Price:** The final price would incorporate all these factors, along with the statue's rarity and the prestige associated with a work of art by a master craftsman.

Q.3. Discuss the relationship between business ethics and cost control.

Or

Is there any relationship between business ethics and cost control? If yes, discuss.

Ans. Business ethics and cost control are intertwined. While cost control focuses on minimizing expenses, it's crucial to ensure ethical practices are maintained throughout the process. Unethical cost-cutting can lead to various problems, including damaged reputation, decreased employee morale, legal issues, and ultimately, higher long-term costs.

Here's a more detailed look at the relationship:

Ethical Considerations in Cost Control :

- (1) **Fair Labour Practices :** Cost control shouldn't come at the expense of fair wages, safe working conditions, and employee benefits.
- (2) **Sustainable Resource Management :** Ethical cost control includes reducing waste, conserving energy, and using sustainable materials.
- (3) **Transparency and Integrity :** Companies should be transparent about their cost-saving measures and how they impact stakeholders.
- (4) **Avoiding Unethical Cost-Cutting :** This includes not compromising on product quality, safety standards, or ethical sourcing.

Consequences of Unethical Cost Control :

- (1) **Reputational Damage :** Unethical practices can severely damage a company's reputation, leading to loss of customer trust and sales.
- (2) **Legal Issues :** Unethical cost-cutting can lead to legal challenges and penalties.
- (3) **Decreased Employee Morale :** When employees feel exploited or undervalued, it can lead to decreased productivity and increased turnover.
- (4) **Long-Term Costs :** While unethical cost-cutting might save money in the short term, it can lead to higher costs in the long run due to legal issues, damaged reputation, and decreased employee morale.
- (5) **Loss of Market Share :** Companies known for unethical practices may lose customers to competitors who prioritize ethical conduct.

Benefits of Ethical Cost Control :

- (1) **Improved Reputation :** Ethical businesses build trust and loyalty with customers, investors, and employees.
- (2) **Increased Productivity :** A positive work environment with fair labour practices can lead to increased productivity.
- (3) **Reduced Risk :** Ethical practices minimize the risk of legal issues and reputational damage.
- (4) **Long-Term Sustainability :** Ethical businesses are more likely to thrive in the long run due to their strong reputation and engaged workforce.

Examples :

- (i) **Primark and child labour :** Primark faced a boycott after reports of child labour in their supply chain, demonstrating the potential cost of unethical cost-cutting.

- (ii) **Volkswagen emission scandal** : The Volkswagen scandal highlighted the dangers of unethical practices and the potential for legal and financial repercussions.

In conclusion, ethical considerations are essential for effective and sustainable cost control. Companies must strive to find cost-saving solutions that do not compromise their ethical obligations to employees, customers, and the environment.

Q.4. What is Cost Accounting? Explain the functions of Cost Accounting?

Or

What is Cost Accounting? Explain briefly its objects and advantages.

Or

What do you mean by Cost Accounting? Explain the functions, objectives and advantages of Cost Accounting?

Or

What do you mean by Cost Accounting? What are the objects of instituting a system of Cost Accounting?

Or

What is Cost Accounting? Explain its nature and scope?

Or

Cost Accountancy 'Costing' and 'Cost Accounting' are used to mean the same in general. Give their precise distinction and define them.

Or

What is the concept of Cost Accounting? Explain its importance, main objects and functions?

Or

Describe the meaning of Cost Accounting and its functions.

Or

Define Costing. State briefly the objectives and advantages of Cost Accounting?

Or

Define 'Cost Accounting'? Explain briefly its nature and subject matter or scope.

Or

Describe the concept of Cost Accounting. Discuss its main objects and mention the importance of ideal Cost Accounting System.

Or

'The object of Cost Accounting is not only to control, but there are certain other objects also.' Examine this statement.

Ans. Meaning of Cost Accounting : Cost accounting is concerned with analysis and accounting of cost for the purpose of managerial

decision making. The information provided by the cost accounting is useful for cost control and cost reduction.

The term 'Cost Accounting' is defined by different authors as follows :

Definitions

Stallman and Russell, 'Cost accounting is that portion of the accounting discipline concerned with the development of cost information related to the activities of the economic entity.'

According to **I.C.W.A. London**, 'Cost Accounting is the technique and process of ascertainment of Cost.'

According to the **Institute of Costs and Works Accountants of the U.K.**, 'Cost Accounting is the application of costing and cost accounting principles, methods and techniques to the science, art and practice of cost control and ascertainment of profitability as well as presentation of information for the purpose of managerial decision making.'

According to **Walter W. Bigg**, 'Cost Accounting is the provision of such analysis and Classification of expenditure as will enable the total cost of any particular unit of production to ascertain with reasonable degree of accuracy, and at the same time to disclose exactly how such total cost is constituted.'

From the above definitions of Cost Accounting, we conclude that 'Cost Accounting is the process of accounting for costs. It is the formal mechanism by means of which costs of products or services are ascertained and controlled.'

Scope or Objects or Functions of Cost Accounting :

- (i) **Cost Ascertainment** : Cost is collected for each product manufactured, service rendered or process. The cost is analysed into material, labour and expense. Further, direct and indirect elements are also identified. The cost is allocated to cost centres or cost units. In case of indirect costs, apportionment of costs to cost centres or costs units is carried out on appropriate basis.
- (ii) **Cost Control** : For the fulfilment of this objective cost accounting uses budgetary control, standard costing, etc. Firstly each item of cost (viz. materials, labour and expenses) is budgeted and then actual expenses incurred is compared with the budget and variations are obtained. Lastly, the reasons for variations are inquired and steps are taken to present the unfavourable variations.
- (iii) **Cost Statement** : Cost Sheet or Cost statement discloses prime cost, works cost, cost of production, cost of sales etc., for a particular product. The analysis of cost is possible for cost statement.

(iv) **Profit** : Cost statement provides profit as per costing records. The profit or (loss) is the basis for future expansion and diversification plants.

(v) **Determining Selling Price** : This is also an important objective of cost accounting. It provides information regarding the cost to make and sell such products. Many other factors such as the conditions of market, the area of distribution, the quantity which can be supplied etc. are also be given due consideration by management before deciding the price but the costs plays an important role.

(vi) **Financial Statements** : In order to operate the business at a high level of efficiency, it is essential for the management to have a frequent review of production, sales and operating results. It can be achieved by cost accounting. A developed cost accounting system provides immediate information regarding the stock of raw materials, work-in progress and finished products. [This helps in the preparation of financial statements.]

(vii) **Decision Making** : Cost accounting provides the information for decision making in following areas :

- (a) Fixation of selling price (based upon cost phase percentage of profit)
- (b) Wastages, abnormal losses indicate the areas in which efficiency can be improved. Further it provides the base for responsibility accounting.
- (c) Cost and profit on each product determining future expansion, diversification programmes.
- (d) Cost Control, cost reduction or cost savings can be implemented if cost control systems (of budgetary control or standard costing) are introduced by the concern.
- (e) Amount invested in stocks (raw materials, work-in-progress, finished goods) is scrutinised to determine that excessive amount is not locked up in stocks.
- (f) It provides system of prompt reporting of relevant information. Analysis of prompt information increases efficiency of management by taking corrective actions.

Nature of Cost Accounting : Cost Accounting, is a technique of planning and controlling various business activities in such a manner that the costs are kept to their minimum and are accounted and recorded in a proper manner. Its aim is to furnish the management of an organisation with the necessary information to plan, control and evaluate operations.

The nature of Cost accounting can be presented in two phases viz, (i) Planning phase and (ii) Control Phase. In the planning phase, Cost accounting deals with the future. It helps management to budget the costs of material, labour and overheads. Once if the costs are budgeted properly

they help the management in tackling the problems relating to product pricing. Capital expenditure decisions, expansion plans for increased sales or production, manufacturing or buying decisions etc.

In the Control phase, Cost accounting deals with the present and it aims at comparing current results with predetermined standard and budgets for better cost controls. Past costs for the purpose of profit determination and allocation of historical costs to periods of time are dealt with under this phase of cost accounting.

We can understand the nature of cost accounting by studying the tasks that are performed by Cost Accounting. They

- (1) Provide information to the management for creation and execution of plans and budgets.
- (2) Provide information to resolve problems that involve choice from among two or more alternative courses viz. decision making.
- (3) Help the management in the establishment of methods and procedures for reduction or improvement of costs.
- (4) Help management in controlling and maintaining inventory for costing and pricing purposes.
- (5) Provide the information to the management relating to costs and profits for an accounting period.

Q.5. Explain the essentials of a good Costing System.

Or

'A good system of Costing is an invaluable aid to the management.' Explain.

Or

Give characteristics of an Ideal System of Cost Accounting.

Ans. A good system of Costing should have certain characteristics to fulfil its objectives. The following are the main Characteristics of a good system of Costing.

- (1) **Simplicity** : The system should be simple and easy to understand. The Costs should be classified, analysed and presented in a simplified form so that the costs calculated at different stages of manufacture may be easily understood.
- (2) **Suitability to the Business** : The system should be suitable to the business, i.e. it must meet the requirements, nature, size and manufacturing conditions of the business. A system which helps the business to run efficiently and improve profitability is an ideal system.
- (3) **Economical** : The system should not be more expensive. The running expenses of the system should be economical, and optimistic to the benefits derived from the system.
- (4) **Flexibility** : The system should be flexible in nature. It should be flexible to the needs of the business. If the business is small, a small system of cost accounting should be capable to meet its

requirements but as and when business grows or becomes larger, that system may be modified and enlarged to suit the changed condition.

- (5) **Comparability** : The system should have the quality of comparability. It should allow the present costs to be compared with the costs of previous corresponding periods to enable the management to evaluate the performances and efficiency.
- (6) **Accountability** : The system should make each and everyone accountable to job and responsibility assigned to him..
- (7) **Cost ascertainment and Cost Control** : The system must necessarily be such as may enable the management to ascertain cost, control cost, ascertain profitability and make decisions.
- (8) **Prompt Cost Information** : The system should be such as it may provide prompt, reliable and accurate information as and when desired by the management.
- (9) **Usage of forms and proformas** : Various forms and proformas used in the system should be properly designed and instructions for use should be clearly mentioned over leaf to avoid ambiguity.
- (10) **Reconciliation of Cost and Financial Accounts** : Where two sets of books of accounts one for cost accounting and the other for financial accounting are kept, there should be reconciliation of the profits shown by them. The system should be such that on reconciliation accuracy of both the books can be ensured to a great extent.

Q.6. What are the main benefits that may be expected by the installation of a costing system in a manufacturing business?

Or

It has been said that 'Good costing system is an invaluable aid to the management.' Enumerate the chief points in support of this statement.

Or

What do you mean by Cost Accounting? Explain its benefits for manufacturers.

Or

'The Amount spent on establishment of cost accounting is not an expense but an investment'. Explain?

Or

Give the meaning and advantages of Cost Accounting?

Or

State the advantages of Cost Accounting to various segments of the society.

Or

Discuss the advantages of cost accounting in detail.

Or

Mention the different benefits obtained from cost accounting. Also describe the various demerits of cost accounting.

Or

Give the merits of Cost Accounting to various parties.

Ans. [Note : Refer Q.4. for the meaning of Cost Accounting]

Advantages of Cost Accounting : Cost accounting as a tool of management, is an integral part of the management process and, helps the management in efficient utilisation of resources and achieving the goal of the business. Advantages of Cost Accounting may be classified into the following categories.

- (a) Advantages to Management. (b) Advantages to Creditors.
- (c) Advantages to Employees. (d) Advantages to National Economy.

(a) Advantages to Management : The main advantages of cost accounting to management are as under :

1. **Useful in Recession** : Costing came into existence on account of recession that prevailed at the end of first world war. Thus costing indicates how a business can best survive in condition of financial set backs.
2. **Elimination of Wastages** : In a good system of costing, wastes of time, expense, equipment, tools etc. are pin-pointed and their causes are explained. Remedial measures can be taken to eliminate them.
3. **Data Available** : Cost records provide sufficient data to prepare profit and loss accounts and determine the strategies of business.
4. **Control of Labour Costs** : With the help of Cost Accounts, labour cost can be controlled, minimising the average cost of production.
5. **Managerial Control** : Managerial control is facilitated by cost control. It infuses a spirit of cost consciousness among the members of the management team.
6. **Inventory Control** : Cost accounting introduces a system in the matter of inventory control which improves the efficiency of the plant and removes bottlenecks.
7. **Measuring Efficiency** : Cost Accounting allows for proper measurement of cost and efficiency. Ways and means can be devised to remove inefficiency.
8. **Aids in preparing Estimates** : Well maintained costing records help in preparing accurate estimates and tenders. These estimates can be prepared on a competitive basis.

(b) Advantages to Creditors : There are many types of creditors who invest large sums in a business enterprise. They are benefited by the installation of a costing system. These lenders of funds have started relying on reports and data submitted by the cost accountants. It has become a policy matter of the banks, that no loans to industries are given unless such

units have complete cost accounting systems. In the case of new enterprises, the banks require cost investigations, budgets and estimated profit and loss statements etc.

(c) **Advantages to Employees** : Cost accounting is very much beneficial to employees also. It benefits the employees in more than one way. The employees of all types get more 'take home' pay. It promotes continuous employment of employees. The workers get more pay and remain employed all through.

(d) **Benefits to National Economy** : Cost Accounts has benefited the economy as a whole. The main benefits to the economy are as under :

- (i) It brings stability in the functioning of resources of the firms.
- (ii) Efficiency has led to reduce prices for the consumers.
- (iii) Stable output of firms employing sound costing systems has brought more taxes to the government.
- (iv) The investment climate improves and stability of business firms has led to lesser business failure.

Demerits or Disadvantages or Drawbacks of Cost Accounting:

Some unprogressive conservatives state the following demerits or disadvantages or drawbacks of Cost Accounting :

1. **It is unnecessary** : Some of the critics argue that when we account for all the events and transactions under the financial accounting then what is the need for recording them in the cost accounts. As such these critics hold that the cost accounts are unnecessary.
2. **Cost system is highly expensive** : The expenditure incurred on the installation of a Cost system is quite heavy and the operational part is also expensive. Therefore, it goes only to increase the cost of production.
3. **Results from Cost Accounts are incorrect and not trustworthy** : Some of the people also conceive that the results and the conclusions drawn from the cost accounts are based upon the estimated figures, hence they are not dependable and trustworthy.
4. **Unsuitable for some industries** : Some of the critics are also of the view that the cost accounting system is unsuitable for some industries. Really speaking this criticism is true to only such industrial units which do not actually know the correct or appropriate use of the cost accounting system.
5. **On adoption of cost accounting system accounting work-load increases** : Some of the scholars say that when some concerns take recourse to the cost accounting system alongwith the financial accounting, it unnecessarily increases work load of accounting work. Its main cause is that many other registers,

forms, statements, and documents etc. are required to be maintained.

Q.7. How is Cost Accounting related to Financial Accounting?

Or

Compare and contrast Cost Accounting and Financial Accounting.

Or

What do understand by Cost Accounting and Financial Accounting. Differentiate between the two?

Or

Explain the similarity and dissimilarity in Cost Accounting and Financial Accounting.

Or

What is Cost Accounting? In what essential respects does Cost Accounting differ from Financial Accounting?

Or

Distinguish between Cost Accounting and Financial Accounting.

Or

Explain the differences between financial accounting and cost accounting.

Ans. Before we relate 'Cost accounting to financial accounting and make distinction between the two, we must know the meaning of cost accounting and financial accounting.'

Cost Accounting : Cost Accounting is the technique and process of ascertainment of costs. It is the processing and evaluation of monetary and non-monetary data to provide information for external reporting internal planning and control of business, operations and special analysis and decisions. In its widest usage, it embraces the preparation of statistical data, the application of cost control methods and the ascertainment of the profitability of activities carried out or planned.

Financial Accounting : Financial accounting is concerned with recording, classifying and summarising financial transactions and preparing statements relating to the business according to generally accepted accounting concepts and conventions. Financial accounting information is intended primarily to serve external users.

Similarities : The similarities between cost accounting and financial accounting are as under :

- (i) Both the systems are based on Double Entry System of Book keeping.
- (ii) Both the accounts show the profit and loss position of the business.

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- (iii) The base of both the accounts is similar. All entries in cost accounting and financial accounting are recorded with the help of invoice, vouchers and other letters.
- (iv) In both the accounts, direct and indirect expenses are included and both the accounts help in the fixation of price of the product.
- (v) Both the accounts are subsidiary to each other and complementary to each.
- (vi) Errors can be located easily in both the accountings.

Dissimilarity : The distinction between cost accounting and financial accounting may be mentioned as under :

Basis of Difference	Cost Accounting	Financial Accounting
1. Scope	Cost Accounts are maintained by Productive and Service institutions.	These Accounts are maintained by all the trading institutions.
2. Efficiency of Different Departments	The efficiency of various departments can be judged by cost accounting.	The efficiency can not be ascertained by it.
3. Position of Accounts	These accounts are helpful to financial accounting.	These are the main accounts of the organisation.
4. Internal and External Transactions	It keeps the internal matter accounts.	It maintains the accounts of external transactions.
5. Estimates of Expenses.	The expenses are assumary and included in cost.	Only real expenses are included in cost.
6. Accounts of expenses	Only expenses relating to cost are recorded.	All types of expenses are recorded in it.
7. Division of Expenses	The expenses are divided in such a manner, so that the cost can be calculated.	It can give the full information about the total cost.
8. Accounting Period.	Cost accounts or statements are prepared according to products, services rendered. The period covered by cost accounts may extend beyond the financial accounting year.	Financial accounts are prepared for a specific period.

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9. Profit

Cost Accounts disclose profit or loss for each product job or process. Financial accounts disclose profit or loss for the entire business.

10. Control on Goods and Cash

More control is placed on material rather than on cash. More control is placed on cash rather than on material.

11. Auditing

Auditing is not compulsory in it. Auditing is compulsory.

12. Tender Price

Tender price can easily be ascertained. Tender price cannot be ascertained easily.

13. Cost Control

Cost accounts helps in reducing the cost of material and labour etc. Financial Accounts cannot control costs.

Q.8. What types of records are kept in Cost Accounting?

Or

What records are kept in standard costing in Cost Accounting?

Ans. Standard costing records in cost accounting involve documenting predetermined costs for direct materials, direct labour, and overhead, which are then compared to actual costs incurred during production. These comparisons highlight variances, which trigger management analysis to identify inefficiencies and implement corrective measures, thereby controlling costs and improving performance. Key records include the standard cost sheet, which details expected costs, and variance accounts where discrepancies between standard and actual costs are recorded.

Types of Records :

- (1) **Standard Cost Sheets :** These documents lay out the detailed predetermined costs for a product or service, including standard quantities and costs per unit for direct materials, labour, and overhead.
- (2) **Material Requisitions and Time Sheets :** Records for actual material usage and labour hours are used to calculate actual costs.
- (3) **Variance Accounts :** These accounts are used in the general ledger to summarize the differences (variances) between standard and actual costs.

Process and Recording :

- (1) **Establish Standards :** Set predetermined standard costs based on historical data, engineering studies, and future trends for materials, labour, and overhead.
- (2) **Record Actual Costs :** Track actual material purchases and consumption, labour hours and rates, and overhead expenditures.

(3) **Calculate Variances** : Compute the differences between standard and actual costs.

(4) **Record Variances in the Ledger** :

(i) **Materials Variance** : The materials price variance might be recorded when materials are purchased, while materials usage variance is recorded when materials are used in production.

(ii) **Labour Variance** : The labour rate and efficiency variances are calculated and recorded when labour is utilized.

(iii) **Overhead Variance** : Variances related to manufacturing overhead are also calculated and recorded.

(5) **Analyze and Act** : Compare the variances to actuals to pinpoint reasons for deviations, such as price changes or inefficient labour, and take necessary corrective actions.

Purpose of Standard Costing Records :

(1) **Cost Control** : Identify where costs are deviating from the plan to take timely corrective action.

(2) **Performance Evaluation** : Measure the efficiency of procurement, production, and other departments by comparing performance against established standards.

(3) **Pricing Decisions** : Provide a solid foundation for pricing products, ensuring profitability.

(4) **Budgeting and Planning** : Assist in future budgeting and financial planning by providing a clear understanding of expected costs.

Q.9. What are the rules for auditing of Cost Accounting?

Ans. Cost accounting auditing rules require companies to maintain accurate, detailed cost records according to Cost Accounting Standards (CAS) and obtain a cost audit from a qualified cost auditor if they meet certain turnover thresholds, as defined by the Companies Act, 2013, and the Companies (Cost Records and Audit) Rules, 2014. The audit focuses on the accuracy of inventory valuation, physical control, documentation, and recording of costs, with the auditor submitting a report in a prescribed form to the Central Government.

Key Aspects of Cost Accounting Auditing Rules :

(1) **Maintenance of Cost Records** : Companies are mandated to maintain detailed cost records, including raw materials, labour, overhead, and other direct and indirect costs, following a standardized format.

(2) **Cost Accounting Standards** : The maintenance of these records must adhere to the principles outlined in Cost Accounting Standards (CAS) issued by the Institute of Cost Accountants of India (ICAI).

(3) **Cost Audit Applicability** : A cost audit is applicable for companies that meet specific turnover criteria, detailed in the Companies

(Cost Records and Audit) Rules, 2014, after the immediately preceding financial year.

(4) **Cost Auditor Appointment** : Qualified individuals must be appointed as cost auditors, with the appointment process often involving a recommendation from the audit committee and a letter of consent from the auditor.

(5) **Audit Scope** : The audit typically covers several aspects of cost accounting, including:

(i) **Inventory Valuation**: Verifying the accuracy of how inventory is valued.

(ii) **Physical Control**: Assessing the security measures over raw materials, work-in-process, and finished goods to prevent loss or misuse.

(iii) **Documentation**: Examining the adequacy and authorization of documents for inventory transfers.

(iv) **Record Accuracy**: Verifying that all inventory movements are accurately recorded in the appropriate master files.

(6) **Auditor's Report** : The cost auditor submits a report containing observations and suggestions, in a form prescribed by the rules, to the Central Government and provides a copy to the company.

(7) **Compliance** : The entire process emphasizes compliance with various laws and regulations related to cost accounting.

UNIT-2

Short Answer type Questions :

Q.1. What is meant by procurement of materials?

Ans. Procurement of materials refers to the process of acquiring the raw materials and supplies required for production. It involves identifying the need, selecting suppliers, negotiating prices, placing purchase orders, receiving materials, and ensuring quality and quantity standards. An efficient procurement system ensures that materials are available at the right time, right price, and right quality, avoiding production delays and cost overruns. Proper procurement also involves maintaining supplier relationships and ensuring compliance with organizational policies. This function plays a vital role in cost control because material cost often forms a significant portion of total production cost.

Q.2. What is Inventory Management and Control?

Ans. Inventory Management and Control is the systematic approach to ordering, storing, and using an organization's inventory—raw materials, work-in-progress, and finished goods. Its main goal is to maintain adequate stock to meet production and sales requirements without overstocking or understocking. Effective inventory control uses techniques like Economic Order Quantity (EOQ), reorder levels, and ABC analysis. It

helps minimize carrying costs, prevent stockouts, and reduce wastage. Proper inventory management ensures continuous production flow and better cost control. It also provides reliable data for decision-making and supports efficient working capital management in businesses.

Q.3. What is Inventory Accounting?

Ans. Inventory Accounting is the process of valuing and recording the cost of materials and goods held in stock. It includes raw materials, work-in-progress, and finished goods. The method of inventory accounting affects cost of goods sold and profit calculation. Common methods used include FIFO, LIFO, HIFO, and Average Cost Method. Accurate inventory accounting ensures proper financial reporting, helps track material usage, and supports cost control. It also enables businesses to detect wastage, pilferage, and slow-moving stock. Proper inventory valuation provides a true picture of the company's assets and profitability.

Q.4. What is the FIFO method of pricing materials?

Ans. FIFO (First In, First Out) is a cost price method where materials received first are issued first. Under this method, the oldest stock is used first in production, and closing stock consists of the most recent purchases. This method is simple and logical because it matches the natural flow of materials. It results in closing inventory valued at recent prices, giving a more accurate reflection of current costs. FIFO is useful during periods of rising prices as it shows lower cost of production and higher profit. However, it may not match actual replacement cost for pricing decisions.

Q.5. What is the LIFO method of pricing materials?

Ans. LIFO (Last In, First Out) is a cost price method where the latest materials purchased are issued first to production. Under this method, closing stock is valued at older prices, while cost of production reflects the most recent costs. LIFO is useful in times of rising prices as it results in higher cost of goods sold and lower taxable profit. However, it does not show the actual current value of inventory in the balance sheet. Though it helps match current costs with current revenues, LIFO is less commonly used in some countries due to accounting regulations.

Q.6. What is the Average Cost and HIFO method of pricing materials?

Ans. The **Average Cost Method** determines the issue price of materials by calculating the average of total cost and total quantity in stock. It smooths out price fluctuations and is simple to apply. It is useful when material prices change frequently.

The **HIFO Method (Highest In, First Out)** issues materials at the highest price in stock first, leaving lower-priced items in closing stock. This method helps charge higher costs to production, reducing taxable income during inflationary periods. However, it is less commonly used due to its complexity and lack of logical material flow. Both methods influence profit calculation, inventory valuation, and cost control strategies differently.

Q.7. What is Inflated Price Method and when is it used?

Ans. The **Inflated Price Method** involves issuing materials at a price slightly higher than the actual cost. This inflated price is used to cover additional indirect costs such as freight, transportation, storage, handling, or losses due to wastage. It ensures that the total material cost charged to production includes both direct and related overhead costs. This method is generally used in industries where incidental costs form a significant portion of total material cost. By including such costs in the issue price, the company ensures more accurate product costing and better cost control.

Q.8. What is Physical Verification of Inventory?

Ans. **Physical Verification** is the process of physically counting and checking the quantity of materials in stock against the inventory records. It ensures that actual stock matches the book records, helping detect errors, theft, or pilferage. Physical verification is usually done periodically—either annually, quarterly, or through continuous stock-taking. It enhances internal control, improves accuracy in inventory accounting, and ensures proper valuation of stock. Any discrepancies found are adjusted in the records, and corrective actions are taken. It is an essential control tool in material management.

Q.9. What is Slow and Non-moving Stock and how is it treated?

Ans. **Slow-moving stock** refers to inventory items that move or are used at a very slow rate, whereas **non-moving stock** refers to items that have not moved or been used for a long time. These stocks block capital and increase carrying costs. Proper treatment includes identifying such stock through inventory analysis, discounting or disposing of obsolete items, and avoiding unnecessary reordering. Businesses may also reallocate or repurpose such materials. Regular review of inventory helps in minimizing slow and non-moving stock, improving inventory turnover, and enhancing working capital efficiency.

Q.10. What is the difference between Scrap, Spoilage, Defective and Normal/Abnormal Wastage?

Ans.

1. **Scrap:** Small pieces of leftover material with little or no value. It can sometimes be sold or reused.
2. **Spoilage:** Materials damaged during production, not usable for their intended purpose. May be normal (unavoidable) or abnormal (avoidable).
3. **Defective:** Items that can be corrected or reworked at some cost.
4. **Normal Wastage:** Expected loss during production (e.g., evaporation, trimming). Its cost is included in production cost.

5. **Abnormal Wastage:** Unexpected or avoidable loss. Its cost is charged to the profit and loss account separately. These distinctions are crucial for accurate cost calculation and control.

Long Answer type Questions :

Q.1. What is procurement of raw materials in Cost Accounting? What are the steps involved in procurement process?

Ans. In cost accounting, material procurement is a strategic process for acquiring necessary raw materials, which includes identifying needs, sourcing suppliers, negotiating terms, placing purchase orders, receiving goods, inspecting quality, and processing payments. The objective is to ensure a reliable supply of quality materials at the lowest possible cost, with all associated acquisition and handling expenses properly accounted for to determine the true material cost.

Key Steps in Material Procurement-

The procurement process for materials typically involves these steps:

- (1) **Identify Material Needs :** Determine the specific materials and quantities required for a project or production run.
- (2) **Source Suppliers :** Research and identify potential vendors based on their ability to supply the needed materials, their reputation, financial stability, and quality control processes.
- (3) **Request for Quotation (RFQ) :** Formally request price quotes and terms from potential suppliers, providing them with detailed specifications, quantities, delivery requirements, and quality standards.
- (4) **Evaluate Proposals and Negotiate :** Review and compare supplier proposals, considering factors such as price, quality, delivery capabilities, payment terms, and warranties.
- (5) **Create Purchase Order :** After selecting a supplier, issue a formal, legally binding purchase order (PO) detailing the agreed-upon terms and conditions.
- (6) **Receive and Inspect Materials :** Upon delivery, goods are received, unpacked, and inspected for quantity accuracy, quality, and damage against the information on the purchase order.
- (7) **Process Payment :** Verify the received goods against the invoice and process the payment to the supplier.

Q.2. What are the considerations to be taken in material procurement in Cost Accounting? Give the importance of procurement in Cost Accounting?

Ans. For cost accounting, the goal is to include all costs associated with bringing a unit of material into factory production :

- (1) **Direct Acquisition Costs :** This includes the vendor's invoice price and transportation charges.

- (2) **Indirect Acquisition Costs :** Other expenses such as purchasing, receiving, inspection, insurance, and storage costs are also part of the material's total acquisition cost.
- (3) **Cost Allocation :** These indirect costs may be added to the material's invoice price through an applied rate to ensure that all associated costs are captured in the cost of materials.

Importance of Procurement in Cost Accounting : Effective material procurement is crucial for profitability and operational efficiency. That results in :

- (1) **Cost Minimization :** It helps secure materials at the best possible prices, directly impacting profit margins.
- (2) **Supply Chain Stability :** Ensures a steady flow of raw materials, preventing production disruptions.
- (3) **Risk Reduction :** By vetting suppliers and carefully negotiating terms, procurement mitigates supply chain risks.
- (4) **Accurate Costing :** Proper accounting for all acquisition costs provides an accurate understanding of the total material cost, enabling better pricing and profit analysis.

Q.3. What is 'Material Control'? Describe its objectives and basic principles.

Or

Explain 'Material Control'. State briefly its objectives.

Or

Discuss the concept of material control. Also mention the various objectives of material control.

Ans. Meaning of Material Control : Material control may be defined as the systematic control over the procurement, storage and usage of materials so as to maintain an even flow of material and avoiding at the same time excessive investment in inventories.

Materials control involves efficient functioning of the following operations :

- | | |
|--|------------------------------|
| (i) Purchasing of Materials. | (ii) Receiving of Materials. |
| (iii) Inspection of Materials | (iv) Storage of Materials |
| (v) Issuing of Materials | |
| (vi) Maintenance of Inventory Records. | |
| (vii) Stock Audit. | |

Objectives of Materials Control : The following may be listed as broad objectives of material control :

- (1) The various materials and stores necessary for carrying on production smoothly should always be available.
- (2) While purchasing materials and stores care should be taken to see that requisite quality and quantity of materials are purchased and that the price paid reasonably low.

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| (vii) Stock Audit. | |

Objectives of Materials Control : The following may be listed as broad objectives of material control :

- (1) The various materials and stores necessary for carrying on production smoothly should always be available.
- (2) While purchasing materials and stores care should be taken to see that requisite quality and quantity of materials are purchased and that the price paid reasonably low.

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- (3) While various materials are stored in the godowns, the loss due to pilferage or damage should be minimum, wastage, theft or leakage should be detected easily.
- (4) Stocks of materials should be kept according to the need of production. There should be no overstocking or under-stocking. Minimum quantity of stores should always be maintained.
- (5) Information about the availability of stocks must be continuously available.

A careful record of the receipt and issue of stores should be kept in order to achieve the above objectives.

Basic or Salient Principles : The salient principles of materials control are the following :

- (1) Accurate sales and production forecasting determine the size of inventory.
- (2) The investment that the management can afford to make for the inventory at one time determines the size of the order to be placed for the purchases, but the size of the order has to be economic.
- (3) The purchase should be made at the right time, of the right quantity, at the right prices and from the right suppliers.
- (4) Scheduling of production controls inventory (Materials).
- (5) How much to purchase and when to purchase are two basic matters to be decided judiciously by the management to control materials for the efficient operations.
- (6) Control is a relative term, and the experience of the persons engaged in the process, and their judgements too are varied according to time, place, and circumstances. So rules and procedures be laid down to maintain uniformity in action and for making evaluation and taking decisions.
- (7) The investment in materials should be maintained at the reduced minimum levels consistent with operation requirements.

Q.4. Explain in brief Bin Card System.

Or

Write a short note on Bin Card System.

Ans. Bin Card System : 'Bin' is meant an almirah, rack or drawer in which the materials are kept. Bin Card is a sort of card in which complete record of the materials kept in the bin is mentioned. In other words, bin card indicates the receipts of the materials kept in the bin, the issue and the balance of the remaining materials. The bin Card for every materials is kept separately. In one and the same almirah, two commodities or items are kept separately in different drawers, the bin cards too are separately maintained for the separate materials.

In case of the issue or receipt of the materials, necessary entries are made in the bin card and the balance in hand is mentioned. Merely by looking at the bin cards, every thing could be known about the materials. It

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is also mentioned in the bin cards as to how much of the maximum or minimum materials must be there in the concerned bins. By it, on one hand, the store-keeper finds it convenient to check the materials; on the other hand, it is easy to submit the purchase indent.

Specimen of Bin Card is as follows:

BIN CARD

Material.....

Code No.....

Stores Ledger Folio.....

Bin No.....

Maximum Limit.....

Minimum Limit.....

Ordering Level.....

	Received		Issued		Balance	
Date	D/N or Quantity Challan No. or L/R No.	Date	Requisition Slip No.	Quantity	Quantity	Checked by & Remark

Q.5. Write the main objectives of Material Control.

Ans. Objectives of Materials Control : The following may be listed as broad objectives of material control :

- (1) The various materials and stores necessary for carrying on production smoothly should always be available.
- (2) While purchasing materials and stores care should be taken to see that requisite quality and quantity of materials are purchased and that the price paid reasonably low.
- (3) While various materials are stored in the godowns, the loss due to pilferage or damage should be minimum, wastage, theft or leakage should be detected easily.
- (4) Stocks of materials should be kept according to the need of production. There should be no overstocking or under-stocking. Minimum quantity of stores should always be maintained.

Q.6. What is Reordering Level? Explain.

Or

Write short notes on – (i) Ordering Level (ii) Minimum stock level, and (iii) Maximum stock level.

Ans. To avoid over-stocking or under-stocking each item of inventory has the order point, minimum level and maximum level.

(i) Ordering Level or Re-order Stock Level : Re-order Level represents that level of stock, at which orders should be placed for further replenishment of stock. The reordering level is fixed somewhere between minimum level and maximum level. The re-order level depends on (i) Minimum level (ii) Rate of usage of materials, and (iii) Lead time (The anticipated time-lag between the dates of issuing orders and the receipt of materials).

Reorder Level = Maximum usage × Maximum re-order period.

(ii) **Minimum Stock level** : It represents the minimum quantity of stock that should be held at all times. Such level is fixed for each item of materials and is revised in accordance with the changing circumstances. If the level of stock falls below the minimum, the management must arrange fresh supplies at once on top priority.

Formula :
Minimum level = Re-order Level - (Normal Consumption × Normal reorder period)

(iii) **Maximum Level** : It represents maximum quantity above which stock should not be held at any time. It is higher than the order level.

Formula :
Maximum Level = Re-order Level + Re-order Quantity - (Minimum Consumption × Minimum Re-order Period).

Q. What is meant by 'Economic Order Quantity' and what factors shall be considered in determining the size of an economic order? Illustrate with an example.

Or
What do you mean by Economic order Quantity? Give some examples.

Or
What do you understand by 'Economic order Quantity'? How is it determined?

Or
Explain Economic Order Quantity and give suitable examples.

Ans. It is the most economical purchase order quantity or lot size which keeps, balance between inventory carrying costs and ordering costs. Purchase department in manufacturing concerns is usually faced with the problem of deciding the 'quantity of various items' which they should purchase. If purchases of material are made in bulk then inventory carrying cost will be high. On the other hand, if order size is small each time, then the ordering cost will be high. In order to minimise ordering and carrying costs it is necessary to determine the order quantity which minimises these two costs. The size of the order for which both ordering and carrying cost are minimum is known as economic order quantity.

The calculation of economic order, quantity of material to be purchased is subject to the following assumptions :

- Ordering cost per order and carrying cost per unit per annum are known and they are fixed.
- Anticipated usage of material in units is known.
- Cost per unit of material is constant and is known as well.
- The quantity of material ordered is received immediately i.e., the lead time is zero.

Determination of Economic Order Quantity : The famous mathematician Wilson derived a formula which is used for determining the

size of order for each of purchases at minimum ordering and carrying costs.

$$EOQ = \sqrt{\frac{2AS}{C}}$$

Where,

A = Annual usage units/Annual consumption

S = Ordering Cost per unit

C = Annual Carrying cost of one unit in a year

Example -1. Calculate EOQ from the following information

Consumption of material per annum	10,000 Kg.
Order Placing cost per order	Rs. 50
Cost per kg. of raw materials	Rs. 2
Storage Cost	8 %

on average inventory

Solution :

$$\begin{aligned} EOQ &= \sqrt{\frac{2 \times A \times S}{C}} \\ &= \sqrt{\frac{2 \times 10,000 \times 50}{\left(2 \times \frac{8}{100}\right)}} \\ &= \sqrt{\frac{10,00,000}{0.16}} \\ &= \sqrt{62,50,000} = 2,500 \text{ kg} \end{aligned}$$

No. of orders to be placed in a year = $\frac{10,000}{2,500} = 4$ orders

Example 2. A manufacturer uses 3,000 units of a subassembly each year. The subassembly costs Rs 5 each. Other cost data are given below :

Cost of placing and receiving one order : Rs. 30 Inventory carrying cost : 10% inventory value. Find economic order quantity for sub assemblies.

$$\begin{aligned} EOQ &= \sqrt{\frac{2AS}{C}} \\ A &= 3,000 ; S = 30 ; C = \left(5 \times \frac{10}{100}\right) = 0.5 \\ EOQ &= \sqrt{\frac{2 \times 3,000 \times 30}{0.5}} \\ &= \sqrt{3,60,000} \\ &= 600 \text{ units} \end{aligned}$$

Hence EOQ is 600 units

Orders per year at the given level of activity

$$= \frac{3,000}{600} = 5 \text{ orders}$$

Q.8. Explain the various methods of pricing the issues of materials. State the method to be adopted for pricing the issue of materials for an industry where their cost is fluctuating constantly.

Or
Explain with suitable examples the following methods of pricing issue of materials: (a) FIFO (b) LIFO

Or
Which of these methods would you recommend under conditions of rising prices and why?

Or
Explain briefly, the various methods of issue of materials.

Or
Critically Examine the method of pricing of material issues.

Or
Discuss the different methods of pricing material issued from stores for production and state which of these methods is in your opinion, most accurate.

Ans. Methods of Issue of Material : The various methods of issue of materials are as under :

1. First In First Out (FIFO) : In this system, the materials which are received first are issued first. The pricing of the issue of the first lot is done at the cost at which that lot was purchased. The price of the second purchase is charged on that issue which are made after the first lot is exhausted. The closing stock is valued at latest purchase price and it will represent current condition. This method is suitable for those materials whose prices do not fluctuate very frequently.

Advantages

1. This method is simple to understand and easy to operate.
2. This method is suitable where : (i) the material is slow moving and (ii) the prices are fluctuating very frequently.
3. The closing stock will be reciprocated at current prices.
4. Actual cost of material is charged from the various jobs of production in the order in which it is incurred.

Disadvantages

1. In times of rising prices, the charge to production is unduly low as the cost of replacing the stock will be higher.
2. Comparison between some time of jobs would be difficult if the two similar jobs are charged with materials at different prices.
3. If there are too many material issues at different price, it will make the stores ledger more complicated.

Example : Suppose material received and issued are as under :

June 1	Received 500 kg. rate Rs. 10
June 10	Received 300 kg. rate Rs. 15
June 16	Issued 400 kg.
June 20	Issued 300 kg.

In this case the issue of 400 kg. material on June 16, will be charged at the rate of Rs. 10. Similarly the issue of material 300 kg. of June 10 will be charged as first 100 kg. @ Rs. 10 and the balance 200 kg. @ Rs. 15.

2. Last In First Out (LIFO) : It operates on the assumption that the material received last is issued first for production. The price of the last purchase is used for pricing materials issued until all units from this purchase have been exhausted, after which the price of the previous purchase becomes the issue price.

Advantages

1. This system is simple to understand and easy to operate.
2. Current costs are correlated with current sales price and enables the management in arriving at the correct decisions.
3. In case of rising prices, the higher prices of latest purchases are charged to production. In this way the profit can be earned in a secret way by increasing the price of the finished product.
4. It evens out the effect of fluctuations in profit over a period of changing price levels.
5. Closing stock is valued at the earlier price and will not show unrealised profits.

Disadvantages

1. The closing stock is valued at a price of the oldest consignment on hand.
2. Comparison between similar jobs would be difficult.
3. It involves considerable amount of clerical work.

Examples :

June 1	Received 500 kg. @ Rs. 10
June 10	Received 300 kg. @ Rs. 15
June 15	Issued 250 kg.

The issue of 250 kg. of material on June 15 will be charged at Rs. 15 till the material of June 10 is not exhausted.

3. Average Cost Method : It is an average of prices without having regard to the qualities involved. It can be expressed as :

$$\text{Issue Price} = \frac{\text{Total of Different Price}}{\text{Total Purchases}}$$

Weighted Average Price : Under this method, the quantity of each receipt is also taken into account for finding out the unit price of material to be issued. The issue prices are calculated each time whenever a new purchase is made.

Advantages

1. It smoothes out the effect of widely varying prices of different stock.
2. This method is more scientific, since it takes into consideration the quantity of materials purchased each time.
3. The need for calculation of new issue prices of material arises when the new lot of material is purchased.

Disadvantages

1. The material cost are not charged out at actual cost and a profit or loss will arise out of such pricing of issue.
2. A considerable amount of operation is involved for frequent calculations of new issue price of material.

4. Market Price Method : Under this method, material issued for any jobs should be charged at the rate at which another lot of material that could be available from the market. Thus materials issued are valued at price prevailing in the market at that time.

Advantages

1. The result of good or bad purchasing is disclosed in this method.
2. The sale prices are fixed on competition basis on the basis of prevailing market price of materials.
3. It ensures that the firm prices are always competitive.

Disadvantages

1. Where no exact market exists difficulty will be experienced in ascertaining correct market prices.
2. The replacement price is to be determined at each time when the material is to be issued.
3. It is a difficult and inconvenient method of calculating the issue price.

5. Inflated Price Method : When materials loss is unavoidable on account of their inherent nature, as losses arising out of evaporation and breaking up bulk etc., the issue price is inflated to cover this type of losses.

Advantages

1. The issue price is calculated at reasonable rates and the correct cost of production can be ascertained.
2. It is an appropriate method of calculation and losses are adjusted in this method.

Disadvantages

1. The actual position of each department cannot be known if this method of issue of material is followed.
2. This method is against the principles of costing, because issue price is charged more than the market price.

6. Standard Price : Under it the standard price of each material is fixed and all the issues are made at that standard price. The standard price is a predetermined price fixed on the basis of a number of factors.

Advantages

1. This method is relatively easy and convenient to operate.
2. Since all the issues are priced at the same price, any variation in material cost will be due to efficiency or inefficiency in the use of such materials.
3. It determines the efficiency of the purchase department.

Disadvantages

1. Materials are not charged out at actual cost and true cost of production is not known.
2. It becomes difficult to fix the standard price of materials when market conditions are changed heavily.

7. Cost Price Method : Under this system issue price is calculated on the basis of actual cost. Actual cost includes the purchase price of material plus the expenses on carriage, octroi, railway fare etc. It is never a stable price.

Advantages

1. The actual cost can be ascertained easily and it gives the true information about cost of production.
2. Replacement cost of production can easily be determined in this system.

Disadvantages

1. This method is seldom used and it is not a practical method of issue of material.
2. It is supposed to be against the principles of costing.

Q.9. Explain the Highest-in First-out method (HIFO) of costing and inventory valuation technique?

Ans. The Highest-In, First-Out (HIFO) method is an inventory valuation technique in cost accounting where the most expensive inventory items are assumed to be sold first, thereby increasing the cost of goods sold and lowering reported taxable income in a given period. Unlike FIFO and LIFO, HIFO is not recognized by major accounting standards like GAAP and IFRS, making it less common for official financial reporting but sometimes used for managing tax liabilities or specific inventory needs.

(1) How the HIFO Method Works?

- (i) Identify High-Cost Inventory :** When materials are issued or items are sold, the system identifies the inventory with the highest unit cost.
- (ii) Issue from Highest-Cost Lots :** The materials with the highest price are issued first.
- (iii) Allocate Remaining Quantity :** If the required quantity exceeds the available units at the highest price, the remaining units are issued from the next highest-priced lots.

(2) Purpose and Benefits :

- (i) Tax Minimization :** The primary benefit of HIFO is its ability to reduce taxable income by increasing the cost of goods sold.
- (ii) Specific Industries :** It is particularly useful in industries like cryptocurrency, where managing taxable capital gains is crucial.

(3) Drawbacks :

- (i) Lack of Standards :** HIFO is not a GAAP or IFRS recognized method, making it unsuitable for official financial reporting.
- (ii) Inventory Valuation :** It results in a lower ending inventory value.

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(iii) **Operational Complexity** : The method requires maintaining detailed records of all purchase prices to identify the highest-cost items, which can be complex.

Q.10. Show the following information of material in stores ledger, if FIFO method is used:

Dates of Receipt	Units	Rate(Rs.)	Dates of Issue	Units
2026, April 1 (Balance)	450	2.20	2026, April 10	400
2026, April 5 (Purchase)	500	2.10	2026, April 15	200
2026, April 12 (Purchase)	350	2.40	2026, April 18	300
2026, April 20 (Purchase)	300	2.25	2026, April 25	350

Solution : Stores Ledger Account

(FIFO Method)

Material Received				Material Issued			
Date	Units	Rate	Amounts	Date	Units	Rate	Amounts
2026		Rs.	Rs.	2026		Rs.	Rs.
April 1 (Balance)	450	2.20	990	April 10	400	2.20	880
April 5 (Purchase)	500	2.10	1,050	April 15	200	2.40	480
				April 18	150	2.40	360
April 12 „	350	2.40	840		50	2.20	110
April 20 „	300	2.25	675		100	2.10	210
				April 25	300	2.25	675
					50	2.10	105
				April 30	Balance		
					350	2.10	735
	1,600		3,555		1,600		3,555

Q.11. With the help of the following particulars, Prepare stores Account showing issue of materials on the basis of Last in first out:

Purchases

August 3, 2025 750 Kg @ Rs. 2.00
August 18, 2025 350 Kg @ Rs. 2.10
August 25, 2025 600 Kg @ Rs. 2.20
August 28, 2025 500 Kg @ Rs. 2.30

Issues

August 19, 2025 850 Kg.
August 26, 2025 450 Kg
August 29, 2025 510kg
August 30, 2025 150 Kg

Solution :

Stores Ledger

Purchases				Issue			Balance		
Date	Qty	Rate	Amt	Qty	Rate	Amt	Qty	Rate	Amt
2025									
Aug.3	750	2.00	15.00	-	-	-	750	2.00	1500
Aug. 18	350	2.10	735	-	-	-	750	2.00	500
							350	2.10	735
Aug. 19	-	-	-	350	2.10	735			
				500	2.00	1,000			
Aug 25	600	2.20	1320	-	-	-	250	2.00	500
							250	2.00	500
							600	2.20	1320

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Aug. 26	-	-	-	450	2.20	990	250	2.00	500
Aug. 28	500	2.30	1150	-	-	-	150	2.20	330
							250	2.00	500
							150	2.20	330
Aug 29	-	-	-	500	2.30	11.50	500	2.30	1150
				10	2.20	22	250	2.00	500
Aug 30	-	-	-	140	2.20	308	140	2.20	308
				10	2.00	20	240	2.00	480
	2,200		4,705	1,960		4225	4880		10,148

Q.12. The following transactions took place in respect of a Material item :

Date	Receipts (Quantity)	Rate Rs.	Issue (Quantity)
2. 3. 2026	200	2.00	-
10. 3. 2026	300	2.40	-
15. 3. 2026	-	-	250
18. 3. 2026	250	2.60	-
20. 3. 2026	-	-	200

Prepare a priced ledger Sheet, Pricing the issue at

(a) Simple Average Rate; and (b) Weighted Average Rate.

Solution :

Stores Ledger Sheet

(a) (Simple Average Rate)

Date	Receipts			Issues			Balance		
	Quantity	Rate Rs.	Amount Rs.	Quantity	Rate Rs.	Amount Rs.	Quantity	Rate Rs.	Amount Rs.
March 2	200	2.00	400	-	-	-	200	-	400
March 10	300	2.40	720	-	-	-	500	-	1,120
March 15	-	-	-	250	2.20	550	250	-	570
March 18	250	2.60	650	-	-	-	500	-	1,220
March 20	-	-	-	200	2.50	500	300	-	720

Note : Average Price for 15.3.2026: $\frac{2.00 + 2.40}{2} = \text{Rs. } 2.20$

Average Price for 20.3.2026: $\frac{2.40 + 2.60}{2} = \text{Rs. } 2.50$

While calculating average price for 20.3.2026 the price of 2.3.2026 purchase has not been considered because it is assumed that issues of 15.3.2026 completely exhausted the materials purchased on 2.3.2026.

Stores Ledger Sheet

(b) (Weight Average Rate)

Date	Receipts			Issues			Balance		
	Quantity	Rate Rs.	Amount Rs.	Quantity	Rate Rs.	Amount Rs.	Quantity	Rate Rs.	Amount Rs.
March 2	200	2.00	400	-	-	-	200	2.00	400
March 10	300	2.40	720	-	-	-	500	2.24	1,120
March 15	-	-	-	250	2.24	560	250	2.24	560
March 18	250	2.60	650	-	-	-	500	2.42	1,210
March 20	-	-	-	200	2.42	484	300	2.42	726

Note : Issue Rate :

$$\text{On 15.3.2026} = \frac{\text{Cost of Stock}}{\text{Quantity}} = \frac{1,120}{500} = \text{Rs. 2.24}$$

$$\text{On 20.3.2026} = \frac{1,210}{500} = \text{Rs. 2.42}$$

Q.13. What is physical verification of Inventory? What is its importance?

Or

How physical verification of inventory is conducted?

Ans. Physical verification of inventory is the process of physically counting, weighing, and inspecting a company's stock to ensure its existence, accuracy, and condition match the recorded financial records. This procedure is crucial for maintaining accurate records, detecting theft or damage, complying with regulations, optimizing inventory levels, and supporting informed business decisions. Key steps include planning the count, preparing the inventory area, conducting the count (using methods like barcode scanning or manual counting), investigating discrepancies, and reconciling the physical count with the accounting system to update records.

Importance of Physical verification of inventory :

- (1) **Accuracy** : Ensures the physical count of items aligns with the company's accounting records.
- (2) **Loss Detection** : Helps uncover losses due to theft, damage, or misplacement.
- (3) **Regulatory Compliance** : Meets legal and regulatory requirements for inventory accuracy.
- (4) **Operational Efficiency** : Leads to better planning, timely order fulfillment, and optimized stock levels.
- (5) **Financial Statement Reliability** : Provides assurance that financial statements reflect the true and fair value of inventory.

How to Conduct Physical Verification?

- (1) **Planning** : Develop a detailed plan, including the scope, timing, and resources needed for the count.
- (2) **Preparation** : Organize the inventory storage area, clear aisles, and prepare any necessary counting documents or devices.
- (3) **Counting** : Physically count, weigh, or measure all inventory items. Techniques can include:
 - (i) **Manual Counting** : Directly counting items, often suitable for smaller quantities or bulk materials.
 - (ii) **Barcode Scanning** : Using scanners to quickly and accurately record inventory items.
 - (iii) **Sample Testing** : Verifying a representative sample when dealing with a large volume of items.

(4) **Verification & Condition** : Inspect items for damage or defects and ensure they match their specifications in the inventory ledgers.

(5) **Reconciliation** : Compare the physical count results with the recorded inventory in the company's accounting system.

(6) **Investigation** : Investigate any discrepancies (shortages or surpluses) and identify their causes.

(7) **Reporting & Adjustment** : Record the verified results and update inventory records to reflect the actual physical quantities, which may involve adjusting the inventory accounts.

Q.14. What is slow and non-moving inventory? What are its causes?

Ans. Slow-moving and non-moving inventory are costly, inefficient stock items that can signify deeper business issues. Slow-moving inventory sells at a slower rate than expected, while non-moving inventory (or dead stock) does not sell at all over a long period. Both tie up capital, occupy storage space, and risk becoming obsolete.

Slow-moving inventory : Slow-moving inventory consists of items with low sales velocity that take longer than a predetermined period to sell. The acceptable timeframe varies depending on the industry and specific product. For example, 90 days might be considered slow for apparel, while 12 months might be acceptable for a luxury car.

Causes of Slow-moving inventory :

- (i) **Poor forecasting**: Inaccurate predictions of customer demand lead to over-ordering.
- (ii) **Seasonal fluctuations**: A product's demand drops during the off-season.
- (iii) **Shifts in trends**: Changes in fashion or consumer preferences can reduce demand for once-popular items.
- (iv) **Ineffective marketing**: Poor product visibility on a website or on store shelves can cause items to sell slowly.

Non-moving inventory (dead stock) : Non-moving inventory includes items that have not been sold or used for an extended period, often 6 to 12 months or more. These products are considered obsolete and are unlikely to ever sell.

Causes of Non-moving inventory :

- (i) **Obsolescence**: Products become outdated due to technological advances, or perishables expire.
- (ii) **Quality issues**: A product is defective or damaged and cannot be sold.
- (iii) **End of product lifecycle**: An item has been replaced by a newer model or is no longer relevant.

Q.15. What are the key strategies for managing problematic inventory in an organization?

Ans. The key strategies for managing problematic inventory in an organization are as under:

- (1) **Identify the root cause** : Analyze your inventory data to determine why certain products are underperforming. Look at historical sales, market trends, and product lifecycles to understand whether the issue is with marketing, forecasting, or product viability.
- (2) **Manage slow-moving inventory** :
 - (i) **Offer discounts and promotions** : Clearance, flash sales, and seasonal sales can create urgency and help clear out older stock.
 - (ii) **Bundle products** : Pair a slow-moving item with a faster-selling, complementary product to make the bundle more attractive to customers.
 - (iii) **Re-market and reposition** : Improve product descriptions and photos, or feature items in new places on your website or in-store to boost visibility.
 - (iii) **Increase awareness** : Communicate with your sales team about slow-moving items and offer incentives for selling them.
- (3) **Clear out non-moving inventory** :
 - (i) **Liquidate** : Sell the stock in bulk to liquidation companies to recover some capital and free up storage space.
 - (ii) **Repurpose** : See if components or materials can be used for new products, which can save money and reduce waste.
 - (iii) **Donate** : Give products to a charity to get a tax deduction, build goodwill, and get the items off your books.
 - (iv) **Write off** : If the inventory has no value, it may be necessary to write it off as a loss.
- (4) **Prevent future issues** :
 - (i) **Improve demand forecasting** : Use inventory management software and analytics to make more accurate predictions based on historical and current data.
 - (ii) **Optimize supply chain** : Negotiate lower minimum order quantities with suppliers to avoid overstocking. This can be more cost-effective than getting bulk discounts on items that won't sell.
 - (iii) **Implement FSN or ABC analysis** : Categorize your inventory into groups based on sales velocity (Fast, Slow, Non-moving) or value (A, B, C) to guide stocking and procurement decisions.
 - (iv) **Adopt FIFO** : For perishable or time-sensitive goods, use the "First In, First Out" method to ensure older stock is sold before it expires.

Short Answer type Questions :

Q.1. What is meant by Labour Costing?

Ans. Labour Costing is the process of recording, analyzing, and controlling the cost of human labor used in production. It involves determining the total cost of wages, salaries, benefits, and related expenses. Labour cost is a major element of total production cost, so accurate accounting is essential. It helps management measure efficiency, fix wages, control idle time, and implement incentive schemes. Labour costing also supports proper product pricing and cost control. It includes both direct labor (which can be traced to specific jobs) and indirect labor (supporting activities that cannot be traced to a single job directly).

Q.2. What is Accounting and Control of Labour Cost?

Ans. Accounting and control of labour cost involves recording wages and salaries systematically and ensuring that labor is used efficiently. It starts with tracking attendance and work hours through time-keeping and time-booking systems. Accurate calculation of wages, overtime, and benefits ensures proper cost allocation. Control measures include setting labor standards, analyzing idle time, monitoring efficiency, and reducing labor turnover. Proper labor cost control helps increase productivity, reduce wastage, and maintain employee satisfaction. It also assists in fair wage distribution and better decision-making by management.

Q.3. What is Time Keeping and Time Booking?

Ans. **Time Keeping** is the process of recording the exact time when employees enter and leave the workplace. It ensures discipline and accurate calculation of wages, especially in time-based payment systems. Tools like attendance registers, punch cards, or biometric systems are used.

Time Booking records the actual time spent by workers on specific jobs or tasks. It helps in determining labor cost per job or product. Time tickets, job cards, or digital systems are commonly used. Together, time keeping and time booking form the basis of accurate wage calculation and labor cost control.

Q.4. What is Payroll in Labour Costing?

Ans. **Payroll** is the systematic record of all payments made to employees, including wages, salaries, allowances, overtime, bonuses, deductions, and benefits. Payroll preparation involves calculating gross wages based on attendance and work done, deducting statutory contributions like provident fund and taxes, and determining net pay. A well-maintained payroll ensures accuracy, transparency, and compliance with labor laws. It also provides valuable information for labor cost analysis, budgeting, and cost allocation. Payroll systems may be manual or computerized, depending on the size and nature of the organization.

Q.5. What is meant by Overtime and Idle Time?

Ans. Overtime refers to the extra hours worked by employees beyond their normal working hours. It is usually paid at a higher rate than regular wages. Overtime may increase production but also raises labor costs.

Idle Time is the time during which workers are paid but are not engaged in productive work due to machine breakdown, power failure, or lack of materials. Idle time is classified into normal (unavoidable) and abnormal (avoidable). While normal idle time is included in production cost, abnormal idle time is charged to the profit and loss account.

Q.6. What is Labour Turnover and why is it important?

Ans. Labour Turnover refers to the rate at which employees leave an organization and are replaced by new workers. It is measured by comparing the number of employees who leave during a period to the average number of employees. High labour turnover increases recruitment, training, and administrative costs, and may also affect production efficiency and quality. Low turnover indicates employee satisfaction and stability. Monitoring and controlling labour turnover is crucial to maintain productivity, reduce costs, and improve morale. Causes of labour turnover include poor working conditions, low wages, lack of motivation, and better opportunities elsewhere.

Q.7. What are Fringe Benefits?

Ans. Fringe Benefits are additional advantages or services provided to employees apart from their regular wages or salaries. These may include medical facilities, free meals, housing, transport, insurance, pension schemes, leave benefits, and recreational facilities. Fringe benefits improve employee satisfaction, reduce absenteeism, and enhance loyalty. They also help attract and retain skilled workers. From a costing perspective, fringe benefits are part of indirect labour costs and must be allocated properly to ensure accurate product costing. Providing fringe benefits also contributes to a positive organizational culture and better productivity.

Q.8. What is Employee Cost Reporting?

Ans. Employee Cost Reporting involves preparing and presenting information related to total employee costs, including wages, benefits, overtime, incentives, and welfare expenses. This reporting provides management with a clear view of the labor cost structure and helps in cost control and budgeting. It may include reports on labor utilization, idle time, absenteeism, turnover rates, and incentive payments. Proper employee cost reporting supports better decision-making, helps identify inefficiencies, and ensures compliance with statutory requirements. It also allows for performance analysis of different departments or production units.

Q.9. What are the main Methods of Wage Payment?

Ans. The two primary methods of wage payment are:

- **Time Rate System:** Wages are paid based on the time an employee works, irrespective of output. It is simple and suitable when quality is more important than quantity.
- **Piece Rate System:** Wages are paid based on the number of units produced. It motivates employees to increase output but may affect quality if not monitored.

Many companies use a combination of both methods or introduce incentive plans to balance efficiency and quality. Choosing the right wage method depends on the type of work and organizational goals.

Q.10. What are the Incentive Schemes — Halsey, Rowan and Taylor Plans?

Ans.

- **Halsey Plan:** Workers are paid for actual time worked plus a bonus (usually 50%) for time saved compared to standard time. It encourages efficiency.
- **Rowan Plan:** Workers receive a bonus based on the proportion of time saved to standard time, ensuring fair rewards and cost control.
- **Taylor's Differential Piece Rate System:** Workers are paid different rates per unit depending on whether they meet or exceed the standard output. It strongly incentivizes high productivity.

These incentive schemes aim to improve performance, reduce idle time, and align employee interests with organizational goals.

Long Answer type Questions :

Q.1. What do you understand by 'Labour Cost Control'? State briefly the purpose of labour Cost Control.

Or

What do you mean by 'Labour Cost Control'? Distinguish clearly between Direct and indirect labour.

Or

What are the expenses which are normally classified as indirect labour in a manufacturing concern? How are direct and indirect labour costs identified in a time-cum-job card.

Ans. Meaning of Labour Cost Control : Labour Cost is a significant element of Cost specially in an organisation using more manual operations. It is the cost of human endeavour in the product and requires co-ordinated efforts for its control. Labour cost control aims at the control of the labour cost per unit of production and not at the reduction of the wage rates of the workmen. Low wages do not necessarily mean low labour cost. In recent labour agreements, it has been found that substantial increase in wages has been granted against corresponding increase in productivity, thereby reducing labour cost per unit.

Labour Cost Control Factors : The following are the important labour cost control factors :

- (1) **Production Planning :** Production should be so planned as to have the maximum and rational utilisation of labour. The product and process engineering, programming, routing and direction constitute the production-planning.
- (2) **Selling up of Standards :** Work-study, time-study and motion-study help in setting up standards for product in operations.
- (3) **Use of Labour Budgets :** Labour budget is a plan for labour cost and is based on past data. Labour budget is prepared on the basis of Production budget.
- (4) **Study of the effectiveness of wage-policy :** How far two remuneration paid on the basis of incentive plan matches with increased production, is a point for study and control of cost.
- (5) **Labour Performance Report :** The labour utilisation and labour efficiency reports received periodically from the departments help the managerial control on labour and exercise labour cost control.

Departments control Labour Cost : In a large organisation, the control of labour cost involves the co-ordinated efforts of the following Departments.

(1) **Personnel Department :** This department is responsible for man-power planning and has the following functions to perform :
(i) Selection, appointment discharge etc., (ii) Training and education, (iii) Wage policy and wage administration (iv) Industrial relations (v) Labour welfare, health and safety, and (vi) Personnel services.

(2) **Industrial Engineering Department :** This department prepares plans and specifications of each job, supervises production activities, undertakes time and motion studies and performs job-analysis etc.

(3) **Time-office :** This department is primarily responsible for collection of data relating to attendance, time spent on job or process by the workmen, and providing information on attendance and leave to payroll department.

(4) **Payroll Department :** This department is responsible for computing total and net earnings of each worker, preparation of payroll and maintenance of various records relating to payroll.

(5) **Cost Department :** It collects and classifies all cost data relating to labour utilisation by departments, and allocates them to respective job or process as per available documents.

Direct and Indirect Labour : Labour Cost may be classified into Direct and Indirect Labour :

Direct Labour : Direct labour refers to the time spent in altering the construction, composition, conformation or conditions of the products manufactured. Thus the time spent by a worker, identifiable with a particular job or process or operation is a direct labour and is considered variable with the output.

Direct labour is also denoted by certain other names such as process labour, productive labour, operating labour etc.

The examples of direct labour are: Labour of machine operators and assemblers, labour of carpenter in making furniture, labour of goldsmith in making ornaments etc.

Indirect Labour : Indirect labour is one which is of general character and cannot be conveniently and economically identified with a particular job, product or process. Indirect labour does not affect the construction or the composition of finished products. It represents ancillary work done in connection with the product manufactured. It helps and facilitates production work indirectly. Examples of Indirect labour are salaries and wages paid to Inspectors, Supervisors, Maintenance Staff, Assistants in Purchase, stores and offices, security staff etc. Like indirect materials, the indirect labour costs are also collected departmentally or process-wise and then absorbed over the products on an equitable business.

Sometimes, it becomes difficult to distinguish between direct and indirect labour. A labour might be engaged in doing a particular work concerned with manufacturing a commodity, and after two hours, the same worker might be placed on a different job concerning with, say repairing or time-keeping. In such a situation, the wages paid for the first two hours should be treated as **direct** and for the rest of the period, **indirect**. The classification of labour between direct and indirect also depends upon the criteria laid down by the management for the work and nature of the industry. The distinction must be observed because direct labour is a part of Prime Cost of Production whereas Indirect labour is treated as factory overhead and, therefore, included under works or factory cost.

Direct and Indirect Labour in a time-cum job card : Time-cum-job card is that card which contains information like name of the worker, ticket number, attendance time, rate of pay, time, spent on each job or productive order. If a worker's time is looked under specific job or productive order, it is **direct labour** otherwise it will be treated as indirect. Some worker may be treated as direct or indirect labour depending upon the nature of job in which he is employed.

Q. What is the significance of Time Keeping in Labour Costing?

Or

What is the role of Time Keeping department in Labour Costing?

Ans. Like Personnel Department, this department also plays an important role in labour cost control through maintaining record of each worker's time in and time out during regular working period and reporting the time of each worker for each department, operation or production order. Thus this department is responsible for recording the attendance time of each worker accurately. This will ensure punctuality and discipline in the company and will have a positive impact on the morale of each worker. Time keeping is a statutory requirement also and therefore

accurate recording of time should be ensured. The important role of time keeping from the point of view of labour costing and control can be summarized as given below :

- (1) It shows the total number of hours worked by each workman and so the calculation of his wage becomes possible. This is applicable where the workers are paid wages as per the time rate.
- (2) Time keeping promotes punctuality and discipline amongst the workers. In the absence of the time keeping system, there will be not only indiscipline amongst them but the workers who are otherwise punctual and disciplined will be frustrated.
- (3) Certain benefits like pension, gratuity and leave with pay, provident fund, promotion, and salary scale are linked with the continuity of service. Attendance records in this regard, can be helpful in computation of these benefits.
- (4) Computation of Labour hours becomes possible through time keeping records. This will be useful in overhead apportionment and absorption, which may be made on the basis of Labour hours.
- (5) Time keeping is a statutory requirement under Labour laws.
- (6) The time keeping records can be used for further analysis like for fixation of standard time and finding out idle time as well as the efficiency of Labour. It can be used by researchers as well as by Government Authorities for various purposes.

Q.3. What are the various methods of Time Keeping?

Ans. The above-mentioned points highlight the importance of the time keeping. The question that we have to answer now is that what are the methods of time keeping? The answer to this is given in the following paragraphs. The methods of time keeping are explained below:

(1) **Attendance Records** : This is the simplest and the oldest method of marking attendance of workers. In this method, every worker signs in an attendance register against his name. Leaves taken by workers as well as late reporting is marked on the attendance register itself. The main limitation of this system is that in case there is large number of workers, there may be large queues for signing the muster. Similarly, there is little control over marking the attendance time and hence there may be irregularities in time recording.

(2) **Disc Method** : This is one of the older methods of recording time. A disc, which bears the identification number of each worker, is given to each one. When the worker comes in, he picks up his disc from the tray kept near the gate of the factory and drops in the box or hooks it on a board against his number. Same procedure is followed at the time of leaving the factory. The box is removed at starting time, and the time keeper becomes aware of late arrivals by requiring the workers concerned to report him before starting. The time keeper will record in an Attendance

Register any late arrivals and workers leaving early. He will also enter about the absentees in the register on daily basis. The main limitation of this method is that there is a possibility of marking the attendance of a worker by his friend i.e. by a proxy. Secondly if the number of workers is large, there will be a delay in recording time due to manual operation of this system.

(3) **Time Recording Clocks or Clock Cards** : This is mechanized method of time recording. Each worker punches the card given to him when he comes in and goes out. The time and date is automatically recorded in the card. Each week a new card is prepared and given to the worker so that weekly calculation of wages will be possible. If wages are paid on monthly basis, a new card may be given in each month. Due to advancement of technology, giving a new card each month is also not required as the same card continued till the worker either leaves the service or retires from the service. The only limitation of this method, [in fact it is the limitation of all the methods of time keeping] is that though the time in and time out are recorded, the records do not show the productive time of the worker, i.e. how he has spent the time in the factory. Thus if a worker comes in at 8 am and leaves at 5 pm, he has spent 9 hours in the company, which can be ascertained from the time keeping records. However, how he has spent time, is not shown by these records. For showing the productive time, separate records showing time booking are to be prepared. The time booking records can also be combined with time keeping records so that there is no need to keep dual records.

(4) **Biometric Attendance System** : According to Biometric attendance system, attendance of the employees is marked by recognising an employee based on physical and behavioural traits. An employee's unique identity like finger print, face and retina image etc. are kept in a database which is matched at the time of marking of attendance before the attendance device for this purpose. Biometric attendance system includes finger print recognition system, face recognition system, Time and attendance tracking technology etc. This system reduces the risk of time manipulation and proxy attendance. However, it may not be possible for small organisations due to cost associated with set-up and maintenance of this system.

Q.4. Name the essentials of a Good Time-Keeping System?

Ans. A good time-keeping system is supposed to have the following requirements.

- (1) System of time-keeping should be such as not to allow proxy for another employee under any circumstances.
- (2) There should be a provision of recording of time of piece employees so that regular attendance and discipline can be maintained.

- (3) Time of arrival as well as time of departure of employees should be recorded so that total time of employees can be recorded and wages can be calculated accordingly.
- (4) Method of recording of time should be mechanical as far as possible so that chances of disputes regarding time may not arise between employees and the time-keeper.
- (5) Latecomers should record late arrivals. The time-keeper should adhere to this discipline strictly.
- (6) The system should be simple, smooth and quick. Unnecessary queuing for marking attendance should be avoided.
- (7) The system should be reviewed periodically to prevent any error or loophole.

Q.5. What is Time Booking? What are its objectives?

Ans. In time keeping we have seen that the basic objective of time keeping is to mark the attendance time, i.e., time in and time out. Time keeping aims at keeping a check on the number of hours spent by a worker in the factory. However, it does not record the productive time of the workers. It means the time keeping methods do not provide information about how the time is spent by the workers in the factory. For example, the time keeping record will show that the worker has reported for duty at 8 am and left at 6 pm, thus, he has spent 10 hours in the company. But the analysis of these 10 hours is not provided by the time keeping. In view of this there is a need to have a system, which will tell about the productive time spent by the workers in the factory. The method, which supplies this information, is known as 'Time Booking Methods' and the recording the time spent by a worker in each job, process or operation is known as 'Time Booking'. The objects of time booking are as follows :

- (1) To determine the productive time spent by the worker on the job or operation. This helps in finding out the idle time and controls the same
- (2) To determine the quantity and value of work done
- (3) To determine earnings like wages and bonus
- (4) To determine the efficiency of workers

Q.6. What methods are used for Time Booking?

Ans. The following methods are used for time booking:

(1) **Daily Time Sheet** : In this method, each worker records the time spent by him on the work during the day, for which a sheet is provided to each worker. The time is recorded daily and hence accuracy is maintained. However, the main limitation of this method is lot of paper work is involved as daily sheets are maintained on daily basis by each worker.

(2) **Weekly Time Sheets** : The only difference between the daily time sheet and weekly time sheet is that these time sheets are maintained on weekly basis. This means that each worker prepares these sheets

weekly rather than daily. This helps in reducing the paper work to a great extent. The only care to be taken is that since the information is filled up on daily basis, there may be inaccuracies and hence filling the information should be done on daily basis only.

(3) **Job Ticket** : Job tickets are given to all workers where time for commencing the job is recorded as well as the time when the job is completed. The job tickets are given for each job and the recording of the time as mentioned above helps to ascertain the time taken for each job. After completing one job, the worker is given another job.

(4) **Labour Cost Card** : This card is meant for a job, which involves several operations or stages of completion. Instead of giving one card to each worker, only one card is passed on to all workers and time taken on the job is recorded by each one of them. This card shows the aggregate labour cost of the job or the product.

(5) **Time and Job Card** : This card is a combined record, which shows both, the time taken for completion of the job as well as the attendance time. Therefore, there is no need to keep separate record of both, time taken and attendance time.

Thus we may distinguish time keeping and time booking, that the time keeping is simply maintaining attendance of the workers i.e., the time of arrival and the time of departure and there by the time spent by the worker in the organization is measured, whereas time booking is not only maintaining the time spent by the workers in the organization, but also the time spent on each & every job including the idle time with reasons are recorded.

Q.7. What is Payroll Department in an organization? What are the functions of this department?

Ans. Roll of Payroll Department is of crucial importance in overall Labour Cost computation and control. The main responsibilities of this department are preparation of payroll from clock cards, job or time tickets, or time sheet. The payroll shows the number of wages payable to each worker showing the gross wages payable, the deductions and the net wages payable. For doing this calculation, they have to work in collaboration with the time office, personnel department, Cost Accounting department and with the concerned department in which the worker is working. The functions of this department are given below:-

- (1) To compute the wages of the employees
- (2) To prepare a detailed wages sheet showing the gross wages payable, various deductions and other payroll liabilities
- (3) To maintain individual employee payroll records
- (4) To prepare department wise summaries of wages
- (5) Compilation of Labour statistics for management
- (6) To install and implement an effective internal check system for preventing frauds and irregularities in payment of wages

(7) To detect and prevent ghost workers

Payroll Procedure :

Following activities and the responsibility to discharge such activities are mentioned hereunder :

S.No.	Activity	Responsibility
(1)	Attendance and time details	Time-keeping department
(2)	Preparation of list of employees and other details	Personnel/HR department
(3)	Computation of wages and other incentives	Payroll department
(4)	Payment to employees	Cost/Accounting department
(5)	Discharge of statutory liabilities	Cost/Accounting department

Q. What do you understand by 'Idle Time'? Distinguish between 'Normal' and 'Abnormal' idle time.

Or

Discuss the concept of 'Idle Time'. How idle time is treated in Cost Accounts?

Or

What is 'Idle Time'? Give its treatment in Cost Accounts. Distinguish it from Idle Capacity.

Or

How will you deal with Idle Time, Overtime, and Leave with wages in Cost Accounts?

Or

Write a short note on 'Idle time'.

Ans. Idle Time : Loss of time which takes place during the scheduled time of work is called Idle time : In other words, idle time refers to that portion of hours which are not utilised for productive purposes. Idle time may represent loss of time of labour, machines or equipment's due to lack of material, breakdown of machinery, failure of power supply etc.

Classification of Idle time : Idle time may be classified into the following two categories :

(i) Normal and Abnormal

(ii) Controllable and uncontrollable.

(i) Normal and Abnormal : Time lost in course of usual routine of activity is known as normal idle time. This loss may be regarded as unavoidable and beyond control. It is an inevitable loss. Abnormal idle time refers to that loss of time which occurs due to abnormal factors not in any way related with the normal course of activity.

(ii) Controllable and Uncontrollable : Idle time is said to be controllable when it is capable of being controlled and kept within limits. For example, idle time due to productive causes, can be controlled and set within limits. On the other hand, idle time due to administrative and economic causes is known as uncontrollable.

Causes of Idle time : Idle time may arise due to any one of the causes given below : (1) Productive causes (2) Administrative causes (3) Economic causes (4) Abnormal causes

(1) Productive Causes : Idle time due to each of the following causes is said to have arisen on account of productive causes :

(i) Machine break-down (ii) Power failures (iii) Waiting for work (iv) Waiting for tools (v) Waiting for raw material (vi) Waiting for instructions (vii) Tea-breaks (viii) Changing from one job to another etc.

Idle time due to all these reasons should be reported to the management by the foreman of the department concerned so that remedial action can be taken.

(2) Administrative Cause : Idle time may also be caused on account of administrative causes e.g., in periods of depression the factory may not be in a position to utilise its full capacity. The management may not like to terminate their trained workers. They may prefer to pay wages to them without their doing any work.

(3) Economic Causes : Some industries are of seasonal nature e.g., sugar factories work only for 4 to 7 or 8 months. For the remaining period the workers remain idle but they are paid their wages.

(4) Abnormal Causes : Idle time may arise due to strikes; lockouts; floods; fire; and major breakdown of machineries etc.

Difference between Idle time and Idle Capacity

Idle time refers to the time during which the workers do not perform any work and which happens due to several causes whereas Idle capacity means temporary non-utilisation of the capacity produce. Idle capacity is due to uncontrollable factors whereas idle time may be due to controllable factors also. Idle time is a part of idle capacity as due to idle capacity, idle time arise.

Accounting of Idle Time : Normal idle time of all workers should be collected under standing order number and charged to factory overheads. However, some of the normal idle time of direct workers, which are associated with the job or work order, such as time taken for machine setting, change over or tools setting, can be added to the product cost as direct wages inflating the hourly rate of wages. For example, if such idle time is normally 10% of total hours and wages paid for 8 hours is Rs. 72, then direct labour cost will be Rs. 72 divided by 7.2 (i.e., 8 hours less 0.8 hrs) = Rs. 10 per hour.

Abnormal idle time cost shall be collected as per standing order numbers and shall be charged to Costing Profit and Loss Account. Under no circumstances, abnormal idle time can be charged to Product cost.

Overtime : The time worked over and above the normal hours is overtime. According to Indian Factories Act, 1949, the remuneration paid for the overtime work is at double the normal rate. This Act also lays down that a worker is entitled to overtime wages when he works for more than 9 hours on any day or more than 48 hours in a week. Overtime work increases the cost of production because the hourly rate is significantly higher for overtime.

Accounting of Overtime Premium :

- (i) If overtime is paid to complete a job at the request of the customer, overtime premium is charged to the job order concerned.
- (ii) If overtime is undertaken in order to cope up with increased production, overtime premium is treated as factory overheads. Alternative method is to distribute the overhead premium over all the jobs undertaken during normal as well as overtime hours at an average rate.
- (iii) If overtime is paid for any capital order, such as, fabrication of a machine to be used internally the overtime premium shall be charged to capital work order account.
- (iv) If overtime is worked to recover production loss due to abnormal conditions such as strike, lock-out, flood etc., the premium should be charged to costing profit and Loss Account.

Leave with pay or leave with wages or Holiday with Pay

The workers are entitled to leave with pay for some days in a year as per provisions of the Factories Act. Besides it, the Casual, Privilege, Medical Leave etc., may have to be granted with pay according to the need of the worker. Leave wages cannot be allocated to cost units directly but may be allocated to the particular department or cost centres in which the workers are employed. Leave wages, therefore are treated as departmental overheads. Alternatively an inflated rate of direct wages can be applied to absorb both normal wages and an appropriate portion of leave with pay.

Q. What do you mean by 'Labour Turnover'? How is it measured? What are the reasons for it? Explain.

Or

Describe the causes of Labour Turnover.

Or

What is Labour Turnover? Enumerate the causes of such labour turnover and indicate some steps which may reduce labour turnover.

Ans. Meaning of Labour Turnover : Labour turnover is the index indicating employees leaving the concern due to various factors as well as the employees joining the concern. It indicates movement of labour force of which some factors are uncontrollable e.g., death of an employee, retirement or superannuation of employee etc. Labour turnover to some extent is preferred but significant increase in labour turnover compared to the previous period is unfavourable. Labour turnover includes the employees sacked by the concern as well as the employees who are joining due to diversification expansion programme.

Measurement of Labour Turnover : Labour turnover may be measured by any of the following methods :

(1) Separation Rate Method : This is the most commonly used method of measuring the labour turnover. The formula is :

$$\text{Labour turnover} = \frac{\text{Number of separation in a period}}{\text{Average no. of employees in the period}} \times 100$$

(2) Replacement Method : This method of measuring labour turnover takes into consideration only the actual replacement of labour irrespective of person leaving.

$$\text{Labour Turnover} = \frac{\text{Number of replacement in a period}}{\text{Average No. of worker in the period}} \times 100$$

However, new recruitment for some expansion project should not be included in the total of replacement.

(3) Flux Method : This is the combination of separation and replacement. The formula is -

$$\text{Labour Turnover} = \frac{\text{No. of Separation} + \text{No. of Replacement}}{\text{Average number of workers in the period}} \times 100$$

In the above three methods average number of employees denotes the simple average of the number of employees at the beginning and at the end of the period on pay-roll.

(4) Equipment annual Rate of Turnover : This rate is found out to know the labour turnover during a month or part thereof. The formula to find the equivalent rate is -

$$\text{Equivalent Turnover Rate} = \frac{\text{Turnover Rate} \times 365}{\text{No. of days in the relative period}}$$

Causes of Labour Turnover : Various causes of labour turnover can be broadly divided into the following three categories.

(1) Personal causes :

- (i) Dissatisfaction with job remuneration, locality or environment.
- (ii) Domestic reasons like marriage.
- (iii) Change for betterment.
- (iv) Retirement due to superannuation or ill health
- (v) Death.

In the above cases the employer can do hardly anything to prevent it.

(2) Unavoidable causes :

- (i) Redundancy due to seasonal nature of business.
- (ii) Shortage of resources like, material, power, finance etc.
- (iii) Change of Plant location.
- (iv) Drop in market demand for the product.
- (v) Discharge for disciplinary action.

In the above cases, the employer has to ask some of the employees to leave the organisation.

(c) Avoidable causes :

- (i) Dissatisfaction with job or remuneration
- (ii) Unsatisfactory working conditions.
- (iii) Bad relationship with superiors and co-workers.
- (iv) Lack of amenities like sports and recreation centres, schools etc.
- (v) Lack of career prospects.
- (vi) Lack of transport, accommodation, medical and other facilities.

In respect of the above causes, the management can take remedial action to keep such separations at the minimum.

Effect of Labour Turnover : High labour turnover reduces the productivity of labour and results in increased cost of production. A very high labour turnover has adverse effect not only on the organisation but it has repercussions on the society at large. It is a loss, which may lead to higher cost and higher prices with consequential cost effective to employ labour as before. However, a low labour turnover indicates dynamism and mobility. Specially, among young and energetic workers there will always be some movement for better future.

Q.10. What is meant by fringe benefits? Give examples

Or

Why employers offer fringe benefits? Explain.

Ans. Fringe benefits are non-wage forms of compensation that employers provide to employees in addition to their regular salary or wages, acting as valuable perks to enhance their overall compensation package. These benefits can be monetary or non-monetary, and they serve to support employees' well-being, boost job satisfaction, attract talent, and increase loyalty. Common examples include health insurance, paid time off, retirement plans, life insurance, and commuter benefits, though they can also include non-monetary items like gym memberships or educational assistance. Fringe Benefits may include :

- (1) **Non-wage compensation :** They are benefits provided beyond the employee's base salary.
- (2) **Perks or non-monetary compensation :** They are supplementary advantages, often in the form of services or goods rather than cash.
- (3) **Enhance overall compensation :** They are a significant part of an employee's total compensation package.

Examples :

- (i) **Health and wellness :** Health insurance, dental plans, mental health support.
- (ii) **Work-life balance :** Paid time off (PTO), flexible working hours, remote work options.
- (iii) **Retirement and financial security :** Retirement plans (like 401(k)s in the U.S.).

(iv) **Commuting and travel :** Commuter benefits, travel allowances, parking subsidies.

(v) **Educational and professional growth :** Tuition assistance, scholarships.

Why Employers Offer Fringe Benefits?

Employers offer Fringe benefits in order to :

- (1) **Attract and retain talent :** Competitive fringe benefits can differentiate a company and help secure top employees.
- (2) **Improve employee satisfaction and loyalty :** Offering benefits that support their needs increases job satisfaction and engagement.
- (3) **Boost productivity :** Employees who feel supported and secure are often more productive.
- (4) **Cater to diverse workforces :** A variety of benefits can appeal to employees with different needs.

Q.11. What is Employee cost reporting? What are the components of employee cost reporting?

Or

What is meant by employee cost reporting? What are the methods for calculating and reporting employee costs?

Ans. Employee cost reporting is the systematic process of identifying, measuring, analyzing, and presenting a company's total expenses related to its workforce. This includes all costs from recruitment to termination, going far beyond just salaries and wages. Reporting on these costs helps businesses with strategic financial planning, budgeting, and resource allocation.

Components of employee cost reporting :

Employee costs are typically categorized in a detailed report that provides insight into where and how an organization's money is spent on its people.

- (1) **Pre-hire expenses :** These are costs incurred before an employee's first day on the job.
 - (i) **Recruitment :** Fees for job postings, advertising, and recruitment agencies.
 - (ii) **Hiring team's time :** The time and salary allocated to recruiters and hiring managers for interviewing and assessing candidates.
 - (iii) **Assessment costs :** Expenses for background checks, skills assessments, and testing.
 - (iv) **Relocation and signing bonuses :** Payments to new hires for moving costs and as a signing incentive.
- (2) **Employee lifecycle costs :** This is the bulk of employee-related expenses, covering the period from the first day to the last.
 - (i) **Compensation :** The employee's base salary or wages, bonuses, and incentive pay.

- (ii) **Benefits** : The cost of health, life, and disability insurance, retirement contributions, and paid time off (vacation, sick leave, holidays).
 - (iii) **Taxes** : The employer's portion of payroll taxes, such as Social Security, Medicare, and unemployment taxes.
 - (iv) **Overhead** : A share of general business expenses, including office space, utilities, supplies, and equipment.
 - (v) **Training and development** : Costs associated with training programs, professional development, and certifications.
- (3) **Post-exit costs** : These are the often-overlooked costs associated with an employee's departure.

- (i) **Severance packages** : Payments made to employees who are laid off or terminated.
- (ii) **Unemployment insurance** : Payments for unemployment claims made by former employees.
- (iii) **Administrative costs** : Expenses for exit interviews, final payroll processing, and legal reviews.

Methods for calculating and reporting employee costs :

Businesses can use a variety of methods to calculate and report on employee costs, with options ranging from simple formulas to automated software.

(1) **Simple calculation** : The most straightforward method for estimating the total annual cost of an employee is to add their gross annual pay, annual payroll taxes, and additional annual expenses. A common rule of thumb is that the actual cost is 1.25 to 1.4 times the employee's base salary.

(2) **3-Stage method** : For a more comprehensive view, this method breaks costs down by the employee's lifecycle:

- (i) **Pre-hire expenses** : Total all talent acquisition costs.
- (ii) **Employee lifecycle costs** : Total the costs incurred during the period of employment.
- (iii) **Post-exit costs** : Total all costs associated with an employee's departure.

(3) **Labour cost percentage** : This metric is used to measure the efficiency of labour spending relative to revenue.

$$\text{Labour Cost Percentage} = (\text{Total Labour Cost} / \text{Total Sales}) \times 100$$

(4) **Automated expense reporting** : Many businesses use specialized software to automate the expense reporting process. This allows employees to easily track and submit expenses, while managers can track and analyze spending patterns in real-time.

Q.12. What are the best practices for effective employee cost reporting?

Ans. The best practices for effective employee cost reporting are :

- (1) **Define clear policies** : Establish comprehensive expense policies that specify which expenses are reimbursable, outline spending limits, and define documentation requirements.
- (2) **Automate where possible** : Use expense management software to automate data entry, route reports for approval, and streamline the reimbursement process. This reduces manual errors and saves time.
- (3) **Integrate systems** : Connect expense reporting software with your financial management and accounting systems to ensure seamless data transfer and a unified financial view.
- (4) **Conduct regular audits** : Perform internal audits to identify discrepancies, ensure policy compliance, and deter fraudulent activities.
- (5) **Report across departments** : In addition to company-wide totals, report on employee costs by department. This helps managers make more informed decisions about resource allocation and budgeting for their teams.

Q.13. What are the various methods of remunerating labour? Explain.

Or

Which method of wage payment will you prefer time-rate or piece rate? Give reasons for your answer.

Or

Distinguish between time rate method and piece rate method of wage payment.

Or

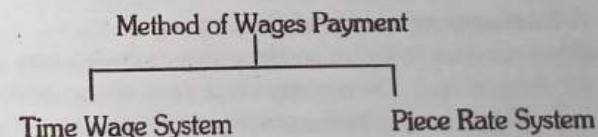
Discuss the various methods of wage payment?

Or

Give the merits and demerits of Time-Rate Method of Wage payment.

Ans. Remuneration is the reward for labour and service. Remuneration includes salaries and wages. Commission, various allowances and statutory bonus. Ordinarily wages are paid on two basis.

- (i) Payment according to time-basis
- (ii) Payment according to work basis.



(i) **Time wage system or Time Rate system** : When employees are paid as per hours worked irrespective of the quantum of output produced, the system is called 'Time wage or Time-rate' system. Payment

may be made per hour, per day, per week or per month, according to the convention and convenience of the organisation. Generally, these are used in such industries where qualifications are more valuable than the production.

Advantages or Merits of Time-wage system :

- (1) This is the easiest and oldest system and has proved the test of time due to simplicity in making calculations and recording wages.
- (2) Workers are assured of steady income of wages.
- (3) The quality of production improves as there is no hurry to the worker for completing the work. He handles it to his ability and care so the job is best executed. Hence, this system is useful for artistic and fine jobs.
- (4) It is suitable for different grades of workers where wages are paid according to different grades.

Disadvantages or Demerits of the system

- (1) Under this system, workers consume time uselessly. Hence, to make discipline, they need more control and inspection.
- (2) As the day wages can be had regardless of the quantity produced the workers do not feel encouraged to produce more. Initiative is killed for want of incentive.
- (3) There is no difference in the wages of efficient and inefficient workers, hence efficient workers work with less care hence there is less efficiency which has effect on production.
- (4) 'An empty mind is a devil's workshop'. Workers getting wages without sufficient output to their credit, find enough time to plan trade union activities, strikes etc.
- (5) The productivity is decreased, the labour cost increases, the job schedules are upset, bottlenecks in production are created, and the profits ultimately go down.

(ii) **Piece-rate System** : Under this system, the wages are paid on the basis of production or output only. Here efficiency of work is cared more. The rate is fixed per piece of work irrespective of time taken by him in completing that work. A workman is free to earn as much his ability, energy or skill would allow him to produce. This system is suitable where the production can be measured in standard units and where strict supervision is not possible.

Advantages or merits of the system :

- (1) It provides initiative and incentive to the workers to produce more.
- (2) According to the accomplished work of the workers their working ability is known hence grading and evaluation of working ability is easy.
- (3) As there is little wastage of time on the part of workers, the fixed overheads and resources like plant, machinery and space are well utilised.

- (4) Workers feel free to work, compete with fellow workers, exhibit their efficiency and earn more wages.
- (5) Workers get appropriate remuneration and producer gets enough production, mutual relations are good.

Disadvantages or Demerits of the System :

- (1) Hurried work does not produce good standard hence products have no good qualities.
- (2) As labourers do not care for quality, it needs more supervision.
- (3) Due to continuous work worker's health breaks down, this leads to absence and his work ability decreases.
- (4) To speed up production sometimes the machines, tools and equipments are not handled with the care that they require, and so the workers expose themselves to accidents, besides causing losses of breakdown to the machines, equipments etc.
- (5) The workers feel insecure of earnings during the days of ill health, holidays etc.

Ques. Distinguish clearly between Halsey and Rowan incentive plans in Wage payment.

Or

Explain different Piece Work Systems of Remuneration.

Or

Write a note on Taylor's Differential Piece Rate System.

Or

Discuss the Important Incentive Wage plans.

Or

What is Incentive Plan? Discuss with suitable example the Halsey and Rowan System of wage Payment.

Ans. Meaning of Incentive Plan : By the term 'Incentive' do we actually mean the psychological inspiration to work further more, which encourages the workers to put in more of their work.

Incentive Wage System and Bonus Plans : To encourage higher productivity a worker may be given additional remuneration for attaining and surpassing 100% efficiency. 100% efficiency may be measured in terms of time or output or in costs. Efficiency more than 100% is reflected in saving of time in done work or in exceeding output target within a time or in reduction in cost of production.

Principal incentive wage plans are as under :

(1) **Straight piece-work System** : This method is simple and the wages are calculated as follows :

$$\text{Wages} = \text{Units produced} \times \text{Piece Rate for each unit}$$

This method is simple but it does not provide incentive for increasing productivity.

(2) **Taylor's Differential Piece Rate System** : The main features of the wage plan, proposed by F.W. Taylor, are as under.

- (a) There shall be two piece-work rates, one is lower and another is higher.

- (b) The efficiency is considered with reference to standard time allowed for completing the job or with reference to standard output during the day.
- (c) If the worker finishes work within standard time (or produces more than or equal to standard output within time) he will be given higher piece rate. On the contrary, if a worker is below the standard, he shall be given lower rate of wages.

(3) Gantt Task and Bonus Scheme : It combines time, piece and bonus systems. Its main features are : (i) Day wages are guaranteed, (ii) Standard time for a task is fixed and both time wage as well as a high rate per piece are determined, (iii) A worker who cannot finish the work within the standard time is paid on time basis, (iv) if a worker reaches the standard, he will be paid time wage plus a bonus at fixed percentage (say, 20%) of normal time wage.

(4) Emerson's Efficiency System : In this system, day wages are guaranteed like Gantt Task Bonus Plan. For guaranteed day wages efficiency is set at $66\frac{1}{2}\%$ or 67%. If efficiency is 100%, then bonus of 20% is given. Efficiency above 100% is adequately rewarded by giving bonus of 1% for each increase in efficiency above 100%.

(5) Barth Premium System : This system does not guarantee day wages. In this method, the wages are calculated as follows :

$$\text{Wages} = \sqrt{\text{Standard time} \times \text{Actual time} \times \text{Daily wage Rate.}}$$

(6) Bedaux Point Premium Plan : In this, system is based on time study. Every job of a standard work is in point. One minute is counted one point. Wages of one point is equal to one minute. This system guarantees daily wages.

(7) Merrick Multiple Rate System : This is the improved form of Taylor system. It reduces the fluctuations of Taylor's System. Here, there are three rates, as under :

Efficiency	Piece rate
(a) Upto 83.33%	Wages at standard rate
(b) Above 83.33% but upto 100%	Wages at 10% above standard rate.
(c) Above 100%	Wages at 30% above standard rate.

The standard set is substantially high, but worker achieving standard gets good incentive.

(8) Halsey Premium Plan : This was propounded by Mr. F.A. Halsey. This scheme pays worker hourly wages for time taken in doing a work plus a bonus of 50% of time saved at hourly rate. A standard time is fixed for every job and the calculation of total earning is done by the following formula :

$$\text{Total earning} = (\text{Time taken} \times \text{Rate per hour}) + \frac{1}{2} (\text{Time saved} \times \text{Rate per hour})$$

Where Time saved = Standard time - Time taken

$$\text{In short, } E = T \times R + \frac{1}{2} (S - T) R$$

A modified form of this scheme is Halsey-wage plan which gives the bonus equal to wages for $\frac{1}{3}$ of time saved.

(9) Rowan Premium Plan : This system is similar to Halsey Plan, only there is difference in the reckoning of dividends. Here also time is standardised and work is prescribed. A minimum wages are also prescribed and the bonus on unscientific basis but here we find the ratio of standardised time from saved time on the basis of percentage of bonus. The formula is given herein :

$$\text{Dividend} = \frac{\text{Saved Time}}{\text{Standardised time}} \times \text{Utilised Time} \times \text{Rate of wages per hour.}$$

Q.15. Standard time for finishing work is 16 Hours. wages rate is Rs. 5 per hour. According to Halsey Plan bonus is 50% of time saved. Calculate bonus and total wages according to Halsey and Rowan Plans in the following cases when the work is actually finished in (i) 14 Hours (ii) 12 Hours (iii) 8 Hours (iv) 7 Hours. and (v) 3 Hours.

Solution : **Halsey Plan**

(i) Work finished in 14 Hours :

$$\text{Wages} = 14 \times 5 = \text{Rs. } 70$$

$$\begin{aligned} \text{Bonus} &= \frac{\text{T.S.} \times 50}{100} \times \text{Rate of Wages} \\ &= \frac{2 \times 50}{100} \times 5 = \text{Rs. } 5 \end{aligned}$$

$$\text{Total Wages Rs. } 70 + 5 = \text{Rs. } 75$$

(ii) Work finished in 12 Hours :

$$\text{Wages} = 12 \times 5 = \text{Rs. } 60$$

$$\begin{aligned} \text{Bonus} &= \frac{\text{T.S.} \times 50}{100} \times \text{Rate of Wages} \\ &= \frac{4 \times 50}{100} \times 5 = \text{Rs. } 10 \end{aligned}$$

$$\text{Total Wages} = \text{Rs. } 60 + \text{Rs. } 10 = \text{Rs. } 70$$

(iii) Work finished in 8 Hours :

$$\text{Wages} = 8 \times 5 = \text{Rs. } 40$$

$$\begin{aligned} \text{Bonus} &= \frac{\text{T.S.} \times 50}{100} \times \text{Rate of Wages} \\ &= \frac{8 \times 50}{100} \times 5 = \text{Rs. } 20 \end{aligned}$$

$$\text{Total Wages} = \text{Rs. } 40 + \text{Rs. } 20 = \text{Rs. } 60$$

(iv) Work Finished in 7 Hours :

$$\text{Wages} = 7 \times 5 = \text{Rs. } 35$$

$$\text{Bonus} = \frac{\text{T.S. 9 Hours} \times 50}{100} \times 5 = \text{Rs. } 22.50$$

$$\text{Total wages} = \text{Rs. } 35 + \text{Rs. } 22.50 = \text{Rs. } 57.50$$

(v) Work finished in 3 Hours :

$$\text{Wages} = 3 \times 5 = \text{Rs. } 15$$

$$\text{Bonus} = \frac{\text{T.S. 13 Hours} \times 50}{100} \times 5 = \text{Rs. } 32.50$$

$$\text{Total Wages} = \text{Rs. } 15 + 32.50 = \text{Rs. } 47.50$$

Rowan Plan

(i) Work Finished in 14 Hours :

$$\text{Wages} = 14 \times 5 = \text{Rs. } 14 \text{ Hours}$$

$$\text{Bonus} = 70 \times \frac{2}{16} = \text{Rs. } 8.75$$

$$\text{Total Wages} = \text{Rs. } 70 + \text{Rs. } 8.75 = \text{Rs. } 78.75$$

(ii) Work finished in 12 Hours :

$$\text{Wages} = 12 \times 5 = \text{Rs. } 60$$

$$\text{Bonus} = 60 \times \frac{4}{16} = \text{Rs. } 15$$

$$\text{Total Wages} = \text{Rs. } 60 + \text{Rs. } 15 = \text{Rs. } 75$$

(iii) Work finished in 8 Hours :

$$\text{Wages} = 8 \times 5 = \text{Rs. } 40$$

$$\text{Bonus} = \frac{40 \times 8}{16} = \text{Rs. } 20$$

$$\text{Total wages} = \text{Rs. } 40 + \text{Rs. } 20 = \text{Rs. } 60$$

(iv) Work finished in 7 Hours :

$$\text{Wages} = 7 \times 5 = \text{Rs. } 35$$

$$\text{Bonus} = \frac{35 \times 9}{16} = \text{Rs. } 19.69$$

$$\text{Total wages} = \text{Rs. } 35 + 19.69 = \text{Rs. } 54.69$$

(v) Work finished in 3 Hours :

$$\text{Wages} = 3 \times 5 = \text{Rs. } 15$$

$$\text{Bonus} = \frac{15 \times 13}{16} = \text{Rs. } 12.19$$

$$\text{Total wages} = \text{Rs. } 15 + 12.19 = \text{Rs. } 27.19$$

Conclusion : From the above it is clear that (i) when the time saved is less than half, the amount of bonus under Rowan Plan in comparison to Halsey Plan is more. (ii) When time saved is 50%, the bonus is equalling both the Plans (iii) When the time saved is more than 50%, the bonus under Halsey Plan is more in comparison to Rowan Plan. (iv) Bonus Under Halsey Plan may be more than 100% of the wages if the time saved is more than Half.

Q.16. A worker a day rate of Rs. 12 per week of 48 hours and cost of living bonus of 10 paise per hour worked. He has been given an 8 hours work to perform which he completes in 6 Hours. He is allowed as premium bonus @ 30% of the time saved. Calculate his remuneration under Halsey Premium Plan and Rowan Premium Plan and hourly rate of earnings.

Solution : Under Halsey Premium Plan :

$$\text{TT} \times \text{R} + (\text{ST} - \text{TT}) \times \% \times \text{R}$$

Where, TT = Time Taken; ST = Standard Time

R = Rate

$$= 6 \times .25 + (8-6) \times \frac{30}{100} \times .25$$

$$= 1.50 + 2 \times \frac{30}{100} \times .25$$

$$= 1.50 + .15 = \text{Rs. } 1.65$$

Cost of living bonus :

$$10 \text{ Paise Per hour} = \frac{.60}{2.25}$$

$$\text{Hourly Rate of Earnings} = \frac{2.25}{6} = \text{Rs. } 0.375$$

Under Rowan Plan :

$$\text{TT} \times \text{R} + \left(\frac{\text{ST} - \text{TT}}{\text{ST}} \right) \times \text{TT} \times \text{R}$$

$$= 6 \times .25 + \left(\frac{8-6}{8} \right) \times 6 \times .25$$

$$= 1.50 + \frac{2}{8} \times 1.50$$

$$= 1.50 + .375 = 1.875$$

Cost of Living Bonus :

$$10 \text{ Paise per hour} = \frac{.60}{2.475}$$

$$\text{Hourly Rate of Earnings} = \frac{2.475}{6} = \text{Rs. } .4125$$

Q.17. A worker under the Halsey method of remuneration has a day rate of Rs. 120 per week of 48 hours, plus a cost of living bonus of Re. 1 per hour worked. He is given 8 hours' task to perform, which he accomplished in 6 hours. He is allowed 30% of the time saved as premium bonus. What would be his total hourly rate of earnings, and what difference would be his total hourly rate of earnings, and what difference would it make if he was paid under the Rowan Method?

Solution : (i) **Under Halsey Plan :**
 Standard Time = 8 Hours
 Less : Actual Time taken = 6 hours
 Time saved = 2 hours

Standard Wages

Day rate = Rs. 120 for 48 hours or Rs. 2.50 per hour

Cost of living Bonus = Re. 1 per hour

Premium bonus = 30% of time saved.

Therefore, wages for 6 hours @ Rs. 2.50 per hour = Rs. 15.00
 + Cost of living bonus for 6 hours @ Re 1 per hour = Rs. 6.00
 + Premium : 30% of time saved i.e., 30% of 2 hours × Rs. 2.50 = Rs. 1.50
 Earning for 6 hours = Rs. 22.50

Hourly rate = Rs. $\frac{22.50}{6}$ = Rs. 3.75

(ii) Under Rowan Method :

Wages for 6 hours @ Rs. 2.50 per hour = 15.00
 Cost of living bonus for 6 hours @ Re. 1 per hour = 6.00

Premium :

Proportion of time saved to the standard time =
 $\frac{2}{8}$ of the wages of the time worked i.e., $\frac{2}{8}$ of Rs. 15 = Rs. 3.75
 Earning for 6 hours = Rs. 24.75

Hourly rate = Rs. $\frac{24.75}{6}$ = Rs. 4.125

	Rowan Method	Halsey Method	Difference
	Rs.	Rs.	Rs.
Difference of Earning for 6 hours	24.75	22.50	2.25
Difference in hourly rate	4.125	3.75	0.375

Q.18. In a factory one work order was completed in 26 hours. The standard time for this work was 40 hours and the worker was paid at Rs. 10 per hour. Factory overhead charges were 80% of the standard time. Find out worker's total earnings, employer's savings and effective earnings rate per hour under (i) Halsey Plan, and (ii) Rowan Scheme.

Solution :

Halsey plan :

Wages for 26 hours @ Rs. 10 per hour = Rs. 260
 Premium : 14 hours × Rs. 10 × 50% = (+) 70

Total for 26 hours Rs. 330

Employer's Savings :

40 hours × Rs. 10 = 400
 Wages paid = (-) 330

Savings Rs. 70

Effective Rate

Rs. 330 ÷ 26 hours = Rs. 12.60

Rowan Plan :

Wages for 26 hours @ Rs. 10 = Rs. 260.00

Premium: $\frac{260 \times 14}{40}$ = Rs. 91.00

Total for 26 Hrs. Rs. 351.00

Employer's savings Rs. 400 – Rs. 351 = Rs. 49.00

Effective Rate : Rs. 351 ÷ 26 hrs. Rs. 13.50

Employer's Saving on factory overheads :

Predetermined factory overheads :
 80% of Rs. 400 = Rs. 320.00

Presently :

Halsey 80% of Rs. 330 = Rs. (-) 264.00

Savings = Rs. 56.00

Rowan: 320 – 80% of Rs. 351

Savings: 320 – 280.80 = Rs. 39.20

UNIT-4**Short Answer type Questions :****Q.1. What are Overhead Expenses in Cost Accounting?**

Ans. Overhead expenses are indirect costs that cannot be directly attributed to a specific product, job, or service but are necessary for the overall operation of the business. They include expenses such as rent, insurance, electricity, factory supervision, depreciation, and office expenses. Overheads are usually classified into three categories — factory overheads, administrative overheads, and selling & distribution overheads. Unlike direct costs, overheads are allocated and absorbed into the cost of production using suitable methods. Proper accounting of overhead expenses ensures accurate product costing, better budgeting, and improved cost control.

Q.2. What is Classification of Overhead Expenses?

Ans. Overhead expenses are classified mainly into three categories:

- **Factory/Production Overheads:** Indirect costs related to manufacturing (e.g., factory rent, power, machine depreciation).

- **Administration Overheads:** Costs for managing and controlling business operations (e.g., salaries of administrative staff, office rent).
- **Selling and Distribution Overheads:** Expenses incurred in promoting and delivering products (e.g., advertising, packaging, transportation).

They can also be classified by **behavior** as fixed, variable, or semi-variable overheads. Classification helps in proper allocation and control of overhead costs, leading to more accurate product costing.

Q.3. What is Allocation and Absorption of Overhead Expenses?

Ans. **Allocation of overhead** means charging a particular overhead cost directly to a specific cost center or department when it can be clearly identified.

Absorption of overhead refers to the process of distributing allocated overhead costs to different units of production using predetermined overhead rates (e.g., machine hour rate, labor hour rate). This ensures each unit bears a fair share of the total overhead. Accurate allocation and absorption help determine the true cost of production and pricing decisions, and also avoid under or over costing.

Q.4. What is Under and Over Absorption of Overheads?

Ans. **Under absorption** occurs when the amount of overhead absorbed is **less than** the actual overhead incurred. This may happen due to low production or incorrect estimates.

Over absorption occurs when the amount absorbed is **more than** the actual overhead. It may result from higher production or overestimation.

Both situations need adjustments at the end of the period — either by transferring to the Costing Profit and Loss Account or by revising rates. Controlling under and over absorption ensures that costs reflect actual production conditions and remain accurate.

Q.5. What is meant by Capacity Level of Cost?

Ans. **Capacity Level of Cost** refers to the relationship between overhead costs and the level of production capacity utilized. Fixed overheads remain constant regardless of output, but their per-unit cost decreases with higher capacity utilization. When capacity is underutilized, per-unit costs increase, affecting profitability. Businesses must analyze **practical capacity**, **normal capacity**, and **actual capacity** to calculate accurate overhead absorption rates. Maintaining optimal capacity utilization helps in better cost control, improved efficiency, and competitive pricing of products or services.

Q.6. How are Interest on Capital, Packaging Expenses, Bad Debts, and R&D treated in Costing?

Ans.

- **Interest on Capital:** Usually not included in cost accounts, as it is considered a financial cost. However, it may be included in notional costs for managerial decisions.
- **Packaging Expenses:** If packaging is essential for production (primary), it is included in factory overheads. If for marketing (secondary), it is included in selling and distribution overheads.
- **Bad Debts:** Generally treated as a selling and distribution overhead.
- **Research & Development Expenses:** If related to production, included in factory overheads; if for product development, may be treated as deferred revenue expenditure.

Correct treatment ensures accurate costing and fair pricing.

Q.7. What is Unit Costing and when is it used?

Ans. **Unit Costing** is a costing method used to determine the cost of producing a single unit of output. It is particularly suitable for industries producing homogeneous products in continuous processes (e.g., cement, bricks, sugar, paper). Total cost is divided by the total number of units produced to arrive at the **cost per unit**. It includes direct materials, direct labor, direct expenses, and overheads. Unit costing helps in pricing decisions, cost control, and profitability analysis. It is simple to apply and provides valuable cost information for mass production industries.

Q.8. What is a Cost Sheet and why is it prepared?

Ans. A **Cost Sheet** is a statement showing the detailed breakdown of total cost for a given period or job. It includes prime cost (direct materials, labor, and expenses), works cost, cost of production, and total cost. It also shows profit or loss if selling price is given. The main purposes of preparing a cost sheet are to determine unit cost, assist in cost control, help fix selling prices, and provide a basis for comparison over time. It serves as an important tool for managerial planning and decision-making.

Q.9. What is Job Costing and how does it work?

Ans. **Job Costing** is a costing method used when products are manufactured or services are rendered according to specific customer orders. Each job is treated as a separate cost unit. Direct materials, direct labor, and overheads are recorded and assigned to each job individually. This method is suitable for industries like construction, printing, shipbuilding, or repair works. Job costing helps in determining the cost and profitability of each job, controlling costs effectively, and preparing customized quotations for clients. It requires accurate record keeping and monitoring of each order.

Q.10. How is Unit Cost Calculated in Cost Accounting?

Ans. Unit cost is calculated by dividing the **total cost** of production by the **number of units** produced in a period.

Unit Cost = Total Cost / Total Units Produced

Total cost includes prime cost, factory overheads, administrative overheads, and selling & distribution overheads. Accurate calculation of unit cost helps in setting selling prices, controlling expenses, and evaluating operational efficiency. This method is widely used in process industries producing standardized goods. Regular monitoring ensures competitive pricing and better profitability.

Long and Short Answer type Questions :

Q.1 What is 'Unit Costing'? Discuss the procedure for treatment of costs under unit costing.

Or

What is Output costing? In which industries is it applicable?

Ans. Unit or Output Costing : This system is applied in those undertakings which produce a single commodity. Under this method, cost per unit of output is ascertained and the amount of each element constituting such cost is determined. The relevant cost data are collected from various records e.g., Material Abstract, Wage Abstract, Cost Ledger etc. The per unit cost can be arrived at after dividing the total expenditure by the quantity produced. A cost sheet or statement of cost is made out which throws light on every aspect of cost. Comparative figures of a preceding period are also provided in the cost statement so as to enable the management to assess the progress of the business.

The total cost is analysed into Prime Cost, Factory Cost, Office Cost or Total Cost of production.

This method of costing is employed in case of industries (i) Where the production is uniform and continued affair, (ii) where the units of production are identical and (iii) where the cost units are physical and natural.

This method is most commonly used in case of industries, mines, quarries, collieries, breweries, dairies, steel works, paper mills, rubber factories, brick making concerns.

Q.2 What do you understand by Overhead. Explain the various methods for allocation of overheads?

Or

Define Overhead. Give its classification. Explain in detail the meaning of fixed, variable and semi-variable overheads.

Or

Define Overhead. Explain the various methods of classification of Overheads.

Or

What do you mean by Overheads? Give examples of different classes of overheads.

Or

What are 'Overheads'? How are they classified?

Or

Explain classification of Overhead Costs.

Ans. Meaning of Overhead : Overhead may be defined as 'the aggregate of indirect material cost, indirect wages and indirect expense.' Amounts spent by a concern in carrying on day to day work, except those on prime cost, may be termed as overheads. It comprises the following :

(i) Indirect labour cost (ii) Indirect wages (iii) Indirect expenses.

Overheads cannot conveniently be identified with the commodities produced or services rendered. They constitute an element of cost as they have been incurred for manufacturing a commodity or making it ready for sale. They may happen either inside or outside the factory. The salary of a foreman is an overhead charge occurring within the factory while the salary of an office supervisor is an overhead charge occurring outside the factory.

Definitions : ICMA defines overhead as, 'total cost of indirect materials, wages and expenses.'

According to **Whelden**, 'Overhead may be defined as the cost of indirect materials, indirect labour and such other expenses including services as cannot conveniently be charged to a specific unit. Alternatively, overheads are all expenses other than direct expenses.'

While ascertaining cost of anything overheads must not be ignored.

Classification of Overheads : Overheads are classified into the following categories :

(1) **According to Nature :** According to nature, overheads may be classified into the following heads :

- (i) **Indirect Material :** Indirect material is that material which is used for purposes ancillary to the business and which cannot be conveniently assigned to specific physical units.
- (ii) **Indirect Labour :** Indirect labour is that labour which is employed for the purpose of carrying out tasks incidental to goods or services provided.
- (iii) **Indirect Expenses :** Indirect expenses are those expenses which cannot be directly, conveniently and wholly allocated to cost centres or cost units.

(2) **According to Normality :** According to Normality, overheads may be classified into the following two heads :

- (i) **Normal Overheads :** Normal Overheads are those overheads which are expected to be incurred in attaining a given output. They are included in production costs and are unavailable.
- (ii) **Abnormal Overheads :** These are the overheads which are not expected to be incurred in attaining a given output e.g., cost of abnormal idle time. Such costs are charged from costing profit and has account.

(3) **According to Controllability** : According to controllability, overheads are classified as under :

(i) **Controllable Overheads** : The overheads which can be controlled by executive action at the point of their incurrence. Variable overheads are generally of controllable nature.

(ii) **Uncontrollable Overheads** : Such overheads are not influenced by the action of a managerial personnel. Such expenses are generally of fixed nature and are beyond the control of a managerial personnel.

(4) **According to Variability** : According to variability, the classification of overhead is as under :

(i) **Fixed Overheads** : Fixed overhead refers to those expenses which do not vary with the volume of production. The examples of such costs are rent, office or factory manager's salary, insurance of buildings etc.

(ii) **Variable Overheads** : These overheads refers to those expenses which do not vary with the volume of production. The examples of such costs are power, selling commission etc.

(iii) **Semi-variable Overheads** : Semi-variable overheads are those which are neither fixed nor variable in relation to the output. They do not change in the same ratio in which the output changes. The examples are depreciation, cost of repairs and maintenance etc.

(5) **According to Function** : According to function the overheads are classified into the following three heads :

(i) **Factory Overheads** : Factory overheads are those expenses which are incurred in connection with production over and above the prime cost.

(ii) **Administrative Overheads** : It includes cost of general management e.g. office rent, rates, salaries, stationary, legal expenses, directors fees etc.

(iii) **Selling and Distribution Overheads** : It comprises marketing, selling, publicity and distribution cost.

Q.3. Explain briefly the principles of the overhead Apportionment.

Or

Write short note on 'Departmentalisation of overheads'.

Or

Why do you consider departmentalisation of overheads necessary?

Or

Describe the various methods of allocation of Indirect Expenses?

Or

What is the difference between the following :

- (i) Allocation of overheads.
- (ii) Apportionment of overheads.
- (iii) Absorption of overheads.

Or

Discuss Allocation of Cost and Apportionment of Cost. Mention the principles of overhead Apportionment.

Or

What do you mean by allocation, apportionment, and absorption of overheads?

Ans. Departmentalisation of Overheads : After collecting and classifying overheads under suitable headings, it is necessary to assign them to departments or cost centres on a suitable basis.

This involves the following stages :

- (1) Allocation of overheads
- (2) Apportionment of overheads

Allocation of overheads is the process of charging the whole amount of overhead costs to a particular cost centre. Whereas **Apportionment of Overheads** is the process of splitting up an item of overheads cost and charging it to the cost centre on an equitable basis.

Departmentalisation of overheads is necessary for the following reasons :

- (1) It helps in Control of Costs.
- (2) It helps in the Correct Valuation of Work-in-Progress.
- (3) It helps in the true and correct valuation of different cost centres.
- (4) It helps in knowing cost of operating different service departments.

Cost Allocation : The term allocation implies relating overheads directly to the various departments. The estimated amounts of various items of manufacturing overheads should be allocated to various cost centres or departments. The salary of the work manager cannot be directly allocated to any one department since he looks after the whole factory. It is, therefore, obvious that many overhead items will remain unallocated after this step.

According to ICNA terminology; 'The allotment of whole items of cost to cost centres or cost units' is known as allocation.

Cost Apportionment : The allotment to two or more cost centres of proportions of common items of cost on the estimated basis of benefit received' is known as apportionment.

At this stage, those items of estimated overheads (like the salary of the works manager) which cannot be directly allocated to the various departments and cost, centres are apportioned. Apportionment implies 'the allotment of proportions of items of cost to cost centres or departments'. It implies that the unallocable expenses are to be spread over the various departments or cost centres on an equitable basis. After this stage, all the overhead cost would have been either allocated to or apportioned over the various departments

Absorption of Cost : It is defined as the process of absorbing all overhead costs allocated or apportioned over particular cost centre or production department by the units produced. Allocation of cost involves

the process of charging total expenditure to cost centres or cost units while the apportionment of overheads involves the process of charging expenditure to cost centres or cost units in the specified proportions.

Absorption of overheads takes place only after the allocation and apportionment of overhead expenses. In other words, the overheads cost are either allocated or apportioned over different cost centres or cost units and afterwards they are absorbed on equitable basis by the output of the same cost centres.

Principles of the overhead Apportionment :

(1) **Service or Utility Principle** : It is based on the concept that the share of overheads borne by a cost centre should be in proportion to the service received by it, or the utility derived by it. In other words, if the service has been availed by some special department, the expenditure incurred upon that service must be recovered from that very department.

(2) **Ability to Pay Principle** : This is based on the principle 'What the traffic may bear'. The departments which contribute greater profits share more of the overheads, and the departments which are weak or are new, share less of them. The products of a new department may still have to find out a market and if the products are to bear full overheads, the cost will increase and the products may not compete well.

(3) **Theory of Survey or observation** : The service or utility received by the different departments may possibly be different from one period to the other. In such a condition, in order to test the real impact of the expenditure, there must be some survey so that on the basis of that survey, the expenses could be distributed among various departments. For instance, a works manager devotes his time to different production centres, a survey may be made to find out the time devoted by him to each department so that his salary may be apportioned to different centres on the basis of time given.

(4) **Efficiency or Incentive Principle** : Under this principle, production budget is set and the overheads are also budgeted to achieve the production target. If the production is more than the budgeted one, the rate of overhead per unit of production will be less than the budgeted rate. It speaks of better efficiency. So the apportionment is made on the basis of standards set for each centre.

Difference between Allocation and Apportionment :

- (1) Under the process of allocation, the amount of overheads is allocated in some or one particular department while under the apportionment, the total amount of the overheads is distributed among various departments on some fair basis.
- (2) When some expenditure is allocated, adequate basis is not essential, while for the apportionment of expenses suitable basis is necessary.

- (3) Really speaking, when the overheads are concerned with some particular department, these are allocated, and when these are related with various departments, it is the apportionment of the expenses.

Difference between Apportionment and Absorption :

- (1) According to **Wheldon**, distributing the total overheads of the factory among the production departments or cost centres is known as Apportionment. But distributing the overheads of a production department or cost centre on the basis of units manufactured in that department, is known as the Absorption.
- (2) The process of apportionment of overheads is performed before the absorption of the overheads, i.e. only after the apportionment, could the absorption of the overheads be made.
- (3) The overheads are apportioned on reasonable basis whereas for the absorption the percentage rates are taken into use.
- (4) Within the apportionment of overheads, these are apportioned between the production-centres and the service centres, but in the absorption, the overheads are distributed among the various units.

Q.4. Explain different overhead absorption rates.

Ans. The overhead absorption rates may be (1) Actual (2) Pre-determined or Standard (3) Blanket (4) Multiple.

(1) **Actual Absorption rate** : The rate is based on actual overhead expenses and suitable actual base related to production. The rate is arrived at as follows :

$$\text{Overhead Rate} = \frac{\text{Actual overhead Expenses}}{\text{Actual quantity (or value) of suitable base}}$$

The actual rate is available after the period, and therefore, it results in a delay in determining the cost of jobs, processes or products. The actual rate may be different from period to period and therefore, comparison of costs from period to period is difficult.

(2) **Pre-determined or Standard Rate** : The rate is based on budgeted overhead expenses (not actual) and suitable quantity (or value) of budgeted base. The rate is arrived at as follows :

$$\text{Overhead rate} = \frac{\text{Budgeted overhead expenses}}{\text{Budgeted quantity (or value) of suitable base}}$$

The pre-determined rate is available in advance, and therefore, cost of jobs, processes, or products is determined promptly. There is no delay in determination of cost (similar to actual overhead absorption rate). The predetermined rates are normally uniform and therefore, comparison and analysis of costs compared to the previous period become easier.

(3) **Blanket (Or Single) Overhead Absorption Rate** : A blanket or single overhead absorption rate is calculated. Such rate is applicable for entire factory.

The rate is calculated as follows :

$$\text{Overhead Rate} = \frac{\text{Overhead expenses (for entire factory)}}{\text{Total quantity (or value) of suitable base}}$$

Application of blanket rate is not practicable, because the benefit received by each production cost centre is not taken into account. Determination of cost becomes arbitrary, if blanket overhead absorption rate is used.

(4) Multiple Overhead Absorption Rate : Multiple or different overhead absorption rates are calculated for service cost centres, production cost centres etc. Determination of multiple overhead absorption rates according to the suitable base results into ascertainment of proper costs of jobs, processes and products. Further comparison of costs with the previous period becomes more meaningful from control point of view.

Q.5. Explain different methods for determination of overhead rates.

Or
Name the different methods of absorption of factory overheads.

Ans. The different methods for determination of overhead rates are as follows:

(1) Output : The overhead absorption rate is calculated with reference to budgeted output or budgeted production. Overhead rate is arrived at as follows :

$$\text{Overhead Rate} = \frac{\text{Budgeted Overhead Expenses}}{\text{Budgeted Output}}$$

This rate is normally applied in mining industry, brick laying, shoe industry etc.

(2) Direct Wages Cost : The overhead absorption rate is calculated with reference to budgeted direct wages. Overhead rate is arrived at as follows :

$$\text{Overhead Rate} = \frac{\text{Budgeted Overhead Expenses}}{\text{Budget Direct Wages (Cost)}}$$

This rate is applied in industries in which direct wages is the main factor of production.

(3) Direct wages Hours (or Production hours) : Instead of Costs, the rate is determined on the basis of budgeted direct wages (or labour) hours. Overhead rate is arrived at as follows :

$$\text{Overhead rate} = \frac{\text{Budgeted Overhead Expenses}}{\text{Budgeted Direct Wages Hours}}$$

This rate is also applied in industries in which direct labour hours is the main factor of production.

(4) Machine Hour Rate : The overhead absorption rate is calculated with reference to budgeted machine hours. This rate is arrived at as follows :

$$\text{Overhead Rate} = \frac{\text{Budgeted Overhead Expenses}}{\text{Budgeted Machine Hours}}$$

This rate is applied in industries in which machine hour is not important. However, sometimes supplementary rate is also used to include all other overhead costs not included in the rate.

(5) Direct Material Cost : Budgeted Material Cost is the basis for determination of overhead absorption rate. It is applied in industries in which direct material cost is most important.

The rate is calculated as follows :

$$\text{Overhead Rate} = \frac{\text{Budgeted Overhead Expenses}}{\text{Budgeted Direct Material Cost}}$$

(6) Prime Cost : The base used for determination of overhead absorption rate prime cost. The rate is calculated as follows :

$$\text{Overhead Rate} = \frac{\text{Budgeted Overhead Expenses}}{\text{Budgeted Prime Cost}}$$

(7) Sale Price : Instead of Costs, the sale price is considered as a base to determine overhead absorption rate. The rate is calculated as follows :

$$\text{Overhead Rate} = \frac{\text{Budgeted overhead Expenses}}{\text{Sale Price of Units of Production}}$$

The overhead absorption rates determined according to budgeted overhead expenses result into under absorption or over absorption of overheads because the actual expenditure is normally different from the budgeted expenditure. If budgeted expenditure is less than the actual expenditure, then there is under absorption of expenses. If budgeted expenditure is more than the actual expenditure then there is over-absorption. Such under absorption or over absorption may be adjusted through supplements overhead absorption rates or it may be written off to the costing Profit and Loss Account. Sometimes, such amounts are adjusted in the subsequent years.

Q.6. Explain the Concept of 'Under-absorption' and 'Over-absorption' of factory overheads.

Or

What do you mean by Over-absorption and Under-absorption of expenses? Explain the causes and the method of dealing with them.

Or

How would you treat in Cost Accounts over or under absorbed expenses?

Ans. Over and Under Absorption of Overheads : In case the amount of overheads recovered from production is more than the actual

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overheads, there is said to be 'Over absorption of overheads'. In a reverse case it will be termed as 'under absorption of overheads'.

Causes of Over or Under-Absorption : Over and under absorption of overheads may arise due to one or more of the following reasons :

- (1) Error in estimating overhead expenses.
- (2) Error in estimating output or hours to be worked.
- (3) Seasonal fluctuations
- (4) Unanticipated change in production activity.
- (5) Changes in the methods or techniques of production.

Disposal of Over or under-Absorption of Overheads : Any of the following methods may be adopted for disposal of over or under absorption of overheads, depending upon the circumstances :

(1) **Carrying Over of Overheads :** The amount of over or under-absorbed may be carried over to the next year's accounts for absorption in the following circumstances :

- (i) In case of new project,
- (ii) In case of a seasonal factory.
- (iii) In case the normal business cycle extends over more than one year and overhead rates are determined on a long-term basis.

This method is criticised on the ground that costs of one period should be absorbed in the products of that period and it is not proper to transfer the costs of one period to another.

(2) **Use of Supplementary Rate :** In case it is observed that the actual and absorbed overheads cannot coincide because of serious error in estimating overheads and the difference is quite large, a supplementary rate may be calculated. By applying this supplementary rate, the cost of each job, order or process may be adjusted. The formula for calculating the rate is as follows :

$$\text{Supplementary Rate} = \frac{\text{Amount of over or under absorbed overhead}}{\text{Actual base}}$$

(3) **Transfer to Costing Profit and Loss Account :** In case the difference between actual and absorbed overheads is not large, such difference should be transferred at the end of the accounting year to the Costing Profit & Loss Account.

Q.7. The following annual charges are incurred in respect of a machine where manual labour is almost nil and where the work is done by means of five machines of exactly similar type and specifications :

	Rs.
(1) Rent and Rates (Proportional to the floor space occupied) for the shop	48,000
(2) Depreciation on each machine	5,000
(3) Repairs and maintenance for 5 machines	10,000

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- (4) Power (as per metre) @ Rs. 10 per 16 units consumed for the shop 37,500
- (5) Electric charges for light in the shop 5,400
- (6) Attendants : There are two attendants for the five machines and they are each paid Rs. 600 p.m.
- (7) Supervisors : For the five machines in the shop, there is one supervisor whose emoluments are Rs. 2,500 per months
- (8) Sundry Supplies, such as lubricants, jute and cotton waste, etc. for the shop 4,950
- (9) Hire Purchase Instalment payable for the machines (including Rs. 3,000 as interest) 12,000
- (10) The machine uses 10 units of Power per hour.

Calculate the Machine Hour Rate for the year.

Solution: Computation of Machine Hour Rate

Machine No.....	Rs.	Per Hour Rs.
Standing Charges		
Rent & Rates Per machine (Rs. 48,000÷5)	9,600	
Light in workshop per machine (Rs. 5,400÷5)	1,080	
Salary of Attendants per machine (Rs. 600×2×12)÷5	2,880	
Supervision per machine (Rs. 2,500×12)÷5	6,000	
Sundry supplied per machine (Rs. 4,950÷5)	990	
Hire Purchase Instalment Interest Per machine (Rs. 3,000÷5)	600	
Total Standing Charges	Rs. 21,150	
Hourly Rate for standing charges	Rs. 21,150	
	Hours 1,200	17.63

Variable Charges :

- (1) Hourly Rate of Depreciation Rs. 5000÷1200 Hrs. 4.16
- (2) Hourly Rate of Repairs & Maintenance Rs. Maintenance Rs. $\frac{10,000}{5} \div 1,200$ Hrs. 1.67
- (3) Power (10 units per hour @ Rs. 10 per 16 units 6.25 12.08

Machine Hour Rate

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Q.8. What is the method of finding Machine Hour Rate? To find the rate what points will you consider? Explain.

Or

What is Machine Hour Rate? For what purpose is it computed?

Or

What do you understand by Machine Hour Rate? How is it calculated? Describe its area of application.

Or

What principle factors are to be considered while computing Machine Hour Rate?

Ans. Meaning of Machine Hour Rate : This is the most scientific method of absorption of overheads. It refers to the computation of absorption rate on the basis of the number of hours a machine has been utilised in completing a particular job. By machine hour rate, we mean the cost incurred in running a machine for one hour.

It is on the basis of this rate, that a charge is made to the jobs for the overheads depending upon the number of hours for which a machine has worked on that job. This method is applicable where work is carried on mostly by the machines because the overheads in such a case are more related to the machines.

Method or Procedure adopted to determine rate :

- (1) The factory overheads of the factory are first apportioned to the different machines or group of machines. It should be well understood that under this method, a machine or a group of machines is a cost centre, and all the overhead related to the machines are allocated or apportioned to the machines. The individual items of factory overheads are not to be apportioned to production department and service departments, but are to be apportioned to the machines including the machines installed in the service departments. The overhead share falling to machines of service departments are again apportioned to machines of production department on the basis of utility derived.
- (2) The working hours of a machine are calculated for the period for which the machine is to run.
- (3) If the overheads of a machine cost centre are divided by the effective machines hours, we get the machine hour rate pertaining to the machine or the group of machines as the case may be.
- (4) Where a job is completed by a single machine, the hours spent by the job on the machine are multiplied by the machine hour rate to determine the overheads chargeable to the job. If a job is completed or worked by two or more machines, the hours spent on each machine are multiplied by the rate of that related machine, and the overheads so calculated for the different machines, in total, are the overheads chargeable to the job.

- (5) The estimated hours forming the base for calculation should often be compared, with the actual hours worked, and necessary adjustment affected.

It can be calculated by dividing the amount of factory overheads incurred for a machine during a particular period by the number of machine hours during the same period.

The formula is :

$$\text{Machine Hour Rate} = \frac{\text{Amount of Factory overheads}}{\text{Total machine hours}}$$

Advantages :

- (1) This is the most accurate method of allocating overheads to each jobs.
- (2) This method involves the use of an overhead costing method which is scientific, logic, theoretically sound and practical in its use.
- (3) It provides a basis for the measurement of the monthly cost of the idle machines.
- (4) It provides an ideal method for estimating the cost of a job on a specification and route sheet with a high degree of accuracy.
- (5) It makes it possible for sales manager to quote more accurate estimated selling prices for jobs.
- (6) It gives more equitable results.
- (7) It uses time as a base in applying overheads.

Disadvantages :

- (1) It is not universally applicable.
- (2) This increases the cost of accounting procedures.
- (3) It is quite difficult to estimate total production hours in advance.

Q.9. A Company has three Production departments and two Service Departments. The summary of distribution of overheads is as under :

Production Departments	Service Departments
X Rs. 9,000	R Rs. 702
Y Rs. 6,000	N Rs. 900
Z Rs. 3,000	

Expenses of Service departments are charged on the basis of following percentage :

	X	Y	Z	R	N
R	20%	40%	30%	-	10%
N	40%	20%	20%	20%	-

Apportion the expenses of Service departments by repeated distribution method.

Solution : Apportionment of overheads of Service Departments

	Total	Production Departments			Service Dept.	
		X	Y	Z	R	N
Total overheads	19,602	9,000	6,000	3,000	702	900
Apportionment of overheads of N Deptt.	-	360	180	180	180	-
	19,602	9,360	6,180	3,180	882	900
Apportionment of overheads of R Deptt.	-	176	353	265	-882	88
	19,602	9,536	6,533	3,445	-	988
Apportionment of overheads of N Deptt.	-	35	17	18	18	-988
	19,602	9,571	6,550	3,463	18	-
Apportionment of overheads of R Deptt.	-	4	7	5	-18	2
	19,602	9,575	6,557	3,468	-	2
Apportionment of overheads of N Deptt.	-	1	.50	.50	-	-2
	19,602	9,576	6,557.50	3,468.50	-	-

Q.10. On the job No. 100 materials used were Rs. 1,000 and Labour charged directly to job was Rs. 800. Other informations are as follows:

Machine No. 10 used for 50 hours (Machine Hour Rate Rs. 4.00) Machine No. 20 used for 30 hours (Machine Hour Rate Rs. 3.00) 8 Welders worked on the job for 5 days of 8 hours each (Direct Labour Hour Rate for welders is 50 paise). Expenses not allocated for calculating machine hour rate on Direct Labour hour rate totalled Rs. 3,000.

The total direct wages for the period were Rs. 30,000. Ascertain works costs of a job No. 100.

Solution : Statement of Cost of Job No. 100

Materials			Rs.	1,000
Labour				800
	Prime Cost	Rs.		1,800
Works overheads :				
Machine No. 10-50 Hrs. × Rs. 4		200		
Machine No. 20-30 Hrs. × Rs. 3		90		
Direct Labour Hour Rate				
Welders (8 × 5 - 8) = 320 Hrs. × 0.50 =		160		
General overheads : 10% of Direct Wages		80		530
	Works' Cost	Rs.		2,330

Q.11. Why do you consider departmentalisation of Overheads necessary?

Ans. After collecting and classifying overheads under suitable headings, it is necessary to assign them to departments or cost centres on a suitable basis. This involves the following stages :

- Allocation of overheads
- Apportionment of overheads.

Allocation of overheads is the process of charging the whole amount of overheads costs to a particular cost centre. Whereas, Apportionment of overheads is the process of splitting up an item of overhead cost and charging it to the cost centres on an equitable basis.

Departmentalisation of overheads is necessary for the following reasons :

- (1) It helps in Control of Costs.
- (2) It helps in the correct valuation of work-in-progress.
- (3) It helps in the true and correct valuation of different cost centres.
- (4) It helps in knowing cost of operating different service departments.

Q.12. What are the requisites of a good method of absorption.

Or

What factors would you consider before formulating overhead items?

Ans. While formulating overhead rates and choosing a method of absorption, the following factors should be taken into account.

- (1) **Adequacy** : The method should be such that the total amount of overhead should be more or less equal to the actual amount of overhead incurred.
- (2) **Convenience** : The method of determining overhead rate should be simple and convenient in its application. It should not require any unnecessary clerical labour.
- (3) **Skill** : The method of absorption of overhead should make distinction between the work done by skilled and unskilled workers while deciding about the overhead rate.
- (4) **Time** : Where the overheads vary with the time e.g., salaries, wages, rent, electricity etc., these should be absorbed after taking into account the time taken by various jobs for completion.
- (5) **Manual and Machine work** : A distinction should be made between the work done by manual labour and work done on machine. The method of absorption of overheads should make such distinction.
- (6) **Separate overhead rates** : The nature of work done in different departments or cost centres may be different. In such a case, it is advisable that separate overheads rates should be worked out for each department and cost centres.

(7) **Frequent changes in overhead rates** : Overhead rates should not be changed from time to time due to changes in the volume of output.

Q.13. From the following information, compute the Direct Labour Hour Rate of a concern :

Number of workers in the factory - 100

Working days in a month 28 and working hours per day is 8.

Idle hours per worker 30 (40% of which may be treated as normal)

Factory Overheads for the month Rs. 10,600.

Solution : (i) Calculation of Labour Hours :

$$\begin{array}{rcl} \text{No. of workers} \times \text{Working days} \times \text{Working hours} & & \\ 100 \times 28 \times 8 & = & 22,400 \\ \text{Less: Idle hours} = 40\% \text{ of } (30 \times 100) & = & -1,200 \\ \hline & & 21,200 \end{array}$$

(ii) Direct Labour Hour Rate

$$\begin{aligned} &= \frac{\text{Factory overhead}}{\text{Direct Labour Hours}} \\ &= \frac{10,600}{21,200} = 50 \text{ paise} \end{aligned}$$

Q.14. Explain Direct Materials Cost Method of Absorption of overheads.

Ans. Direct materials Cost Method or Percentage of Direct Materials Cost : Under this method the basis of charging overheads is the cost of direct materials used in production. The overhead rate is expressed as a percentage of direct materials cost. It can be obtained by dividing the total factory overhead cost by the cost of direct materials used during a period and multiplying the ratio by 100.

The formula for calculating this rate is as follows :

$$\% \text{ of direct material cost} = \frac{\text{Amount of Factory overhead cost}}{\text{Cost of direct materials used}} \times 100$$

If the amount of factory overhead cost is Rs. 25,000 and the cost of direct materials is Rs. 50,000, then the percentage rate will be -

$$\frac{25,000}{50,000} \times 100 = 50\%$$

A Job which consumes Rs. 8,000 worth of materials will be charged -

$$\text{Rs. } 4,000 \left(8,000 \times \frac{50}{100} \right)$$

This method of absorbing overhead is applied in manufacturing units where :

- (i) Different quantities of raw materials are used.
- (ii) The prices of these raw-materials vary widely, and

(iii) The material cost form a very high percentage of the total cost (say about 80% to 90%).

Advantages/Merits :

- (1) This method is simple and easy to operate.
- (2) This method gives a fairly accurate result if the prices and grades of materials do not fluctuate widely from time to time and the cost of materials used remains uniform.

Disadvantages/Demerits/Drawbacks/Limitations :

- (1) There is no logical relationship between manufacturing cost and the cost of the raw-materials used.
- (2) It ignores the time element while many of the overheads vary with time.
- (3) It assumes that factory overhead expenses are related to the value of materials used. This is not a realistic assumption.
- (4) It does not make any distinction between the jobs done by skilled workers and those by unskilled workers.

Q.15. What is meant by capacity level of cost? What is their types and importance?

Ans. In cost accounting, capacity levels refer to different potential rates of production or service delivery, classified into installed, practical, normal, and actual capacity. These levels are used to assess resource utilization, allocate costs accurately, and inform management decisions by providing benchmarks for operations and helping to identify opportunities for cost reduction.

Types of Capacity Levels :

- (1) **Installed Capacity** : The maximum output achievable under ideal, theoretical conditions, assuming the best possible use of all available resources without any downtime.
- (2) **Practical Capacity** : The maximum output achievable by realistically accounting for factors like regular breakdowns, maintenance, and operational inefficiencies.
- (3) **Normal Capacity** : The expected average output over a period, considering seasonality and planned maintenance.
- (4) **Actual Capacity** : The actual level of output achieved or the actual utilization of resources in a specific period.

Importance of Capacity Levels :

- (1) **Cost Allocation** : Understanding capacity levels helps allocate fixed costs more accurately by associating them with the level of output they support.
- (2) **Performance Measurement** : Comparing actual capacity utilization against planned or theoretical capacity helps measure efficiency and identify areas of under- or over-utilization.

(4) **Cost Control and Reduction** : Analyzing the difference between actual and ideal capacity utilization can highlight wasteful expenses and lead to strategies for improving efficiency and reducing costs.

(5) **Management Decisions** : Capacity level data provides crucial insights for strategic planning, budgeting, and pricing decisions.

Q.16. How would you treat Interest on Capital in costing?

Ans. In cost accounting, the treatment of certain financial items can vary, especially items like interest on capital. The guiding principle is to distinguish between costs related to production and financial expenses.

The treatment of interest on capital depends on whether the capital is owned or borrowed.

Interest on owned capital : This refers to a hypothetical or imputed cost representing the return on a business owner's or partner's own investment.

(1) **Costing treatment** : A significant amount of controversy surrounds the inclusion of interest on owned capital in cost accounts. The general opinion is that it should be excluded from the cost of production and treated as an appropriation of profit. Reasons for exclusion include:

- (i) It is not a true cost of production, unlike wages or material expenses.
- (ii) Including it can make cost comparisons between different companies or periods misleading.

(2) **Financial accounting treatment** : In financial accounting for a partnership, interest on capital is typically shown on the debit side of the Profit and Loss Appropriation Account. It is considered an expense to the business and increases the owner's capital account on the balance sheet.

Interest on borrowed capital : This is the interest paid on borrowed funds, such as loans or debentures, that are used for the business.

(1) **Costing treatment** : In most cases, interest on borrowed capital is a financial expense and is excluded from the cost sheet. It is recorded separately in the financial Profit and Loss account.

(2) **Exception for qualifying assets** : Interest on funds borrowed for the acquisition, construction, or production of a "qualifying asset" (one that takes a substantial time to prepare for use) may be capitalized. The interest is added to the cost of the asset, not expensed immediately.

Q.17. How would you treat Packaging Expenses in costing?

Ans. The treatment of packaging expenses in costing depends on its purpose and type. Packaging can be categorized as either primary or secondary, and each is accounted for differently.

(1) **Primary packaging** : Primary packaging is the material that directly holds and preserves the product for the customer. It is considered an essential part of the finished product and is treated as a direct material.

(i) **Examples** : The bottle for a beverage, the wrapper for a candy bar, or the box holding a box of cereal.

(ii) **Costing treatment** : Primary packaging costs are included in the Cost of Goods Sold (COGS).

(2) **Secondary and shipping packaging** : Secondary and shipping packaging is used for storage, transport, promotion, or to group items together, rather than being an integral part of the product itself. These expenses are treated as an indirect cost.

(i) **Examples** : Outer cardboard shipping boxes, packing peanuts, bubble wrap, and pallets.

(ii) **Costing treatment** : Secondary packaging costs are typically classified as distribution or selling overheads.

(iii) **Accounting** : They are generally expensed as an operating or selling expense.

(3) **Reusable packaging** : For containers or materials that are used repeatedly, such as industrial gas cylinders or crates, the cost is assigned differently.

(i) **Costing treatment** : The cost is spread over the number of times it is used or depreciated over its useful life. The depreciation or amortized cost is then included in the packaging cost.

Q.18. How one would treat Bad debts in cost accounting?

Ans. In cost accounting, the treatment of bad debts depends on whether they are considered a normal business expense or an abnormal loss. The core principle is that financial losses should be excluded from cost calculations so as not to distort the cost of products or services.

Normal vs. Abnormal bad debts : The distinction between normal and abnormal bad debts determines their treatment in cost accounting:

(1) **Normal bad debts** : Bad debts that are a regular and expected part of doing business on credit are considered a selling and distribution overhead.

(i) **Reasoning** : Extending credit carries an inherent risk of non-payment. This is a normal and predictable cost of generating revenue, similar to advertising or sales commissions.

(ii) **Accounting treatment** : They are included in the overall selling and distribution costs, which are then absorbed into the cost of production or treated as a period cost.

(2) **Abnormal bad debts** : These are exceptionally large or unusual bad debts caused by unexpected events, such as the bankruptcy of a major customer.

(i) **Reasoning** : Such losses are unusual and not directly related to the efficient operation of the business. Including them in the cost of goods would lead to a misleading and inflated cost figure.

(ii) **Accounting treatment** : Abnormal bad debts are excluded from cost accounts. Instead, they are charged to the Profit and Loss Account (or Costing Profit and Loss Account) as a financial loss.

Impact of Bad debts on cost sheets : The treatment of bad debts directly affects the cost sheet, which is a document used to determine the total cost of a product.

(1) For normal bad debts :

- (i) They are added as part of the selling and distribution expenses.
- (ii) This increases the "Cost of Sales."

(2) For abnormal bad debts:

- (i) They are not included in the cost sheet.
- (ii) This ensures the calculated cost per unit remains stable and reflects the cost of production and normal operations.

Q.19. How would you treat Research and Development Expenses in costing?

Ans. In cost accounting, Research and Development (R&D) expenses are treated differently depending on the specific cost and whether it provides an alternative future use for the company. While most R&D costs are expensed immediately, certain assets used in the process can be capitalized.

Overview of Research and Development cost treatment:

(1) **Expensing** : The majority of R&D costs, including personnel, materials used for a single project, and most services, are expensed to the income statement in the period they are incurred. This is due to the high uncertainty of future benefits from R&D activities.

(2) **Capitalizing** : Only items with a clear alternative future use can be capitalized and recorded as an asset on the balance sheet. This includes facilities, equipment, and certain software that can be used for purposes other than the R&D project they were originally purchased for.

(3) **Cost allocation** : R&D is a type of overhead or indirect cost for production. In management accounting, these costs must be allocated to products and services to properly determine their full cost.

Cost accounting methods for Research and Development:

For management accounting purposes, a company must decide how to treat R&D costs. The two main approaches are to expense the costs as they occur or to capitalize and amortize them over time. The method

chosen affects financial metrics like earnings before interest, taxes, depreciation, and amortization (EBITDA) and net income.

(1) **Expensing Research and Development costs** : This method treats R&D as a period cost and is the more conservative accounting approach.

(a) Pros :

(i) **Simplicity** : It avoids the subjectivity and complexity involved in estimating an R&D project's future benefits.

(ii) **Transparency** : It provides a clearer picture of the company's current spending on innovation.

(b) Cons:

(i) **Distorted profitability** : It can understate current-period profitability because significant investments are immediately recorded as expenses, while the revenue benefits may not be realized for years.

(ii) **Reduced earnings** : It can lead to lower reported net income and EBITDA in periods of high R&D spending.

(2) **Capitalizing and amortizing Research and Development costs** : Under this method, qualifying R&D costs are treated as an asset and gradually written off over their estimated useful life. This is only allowed for financial reporting under certain rules, such as International Financial Reporting Standards (IFRS)

(a) Pros :

(i) **Improved financial metrics** : It can boost reported net income and EBITDA by spreading the expense over time.

(ii) **Better matching principle** : It aligns R&D expenses with the future revenues the investment is expected to generate, providing a more accurate reflection of long-term profitability.

(c) Cons :

(i) **Subjectivity** : It requires management to make judgments about future commercial viability and useful life, which can be prone to manipulation.

(ii) **Complexity** : It adds complexity to the accounting process, requiring detailed tracking and clear criteria for what can be capitalized.

Cost classification of Research and Development expenses :

Regardless of whether costs are expensed or capitalized, R&D expenses must be identified and classified.

(1) **Direct costs** : These are costs directly related to a specific R&D project, such as :

- (i) Salaries of R&D personnel
- (ii) Materials and supplies consumed in research
- (iii) Fees for outsourced research services

(2) **Indirect costs** : These are overhead costs that must be allocated to R&D activities, such as :

- (i) Utilities and administrative costs for research facilities
- (ii) Depreciation of buildings or equipment used for R&D

(3) **Capitalizable costs** : These are expenditures that have a use outside of a specific R&D project. Their depreciation or amortization is included as an R&D cost during the project.

- (i) **Equipment and facilities** with an alternative future use.
- (ii) **Purchased intangible assets**, like patents, that have a use beyond the initial project.
- (iii) **Software development costs**, which can be capitalized at certain stages.

The cost allocation process : In cost accounting, companies must allocate the total R&D costs to the products that benefit from the research, particularly for internal decision-making. Common allocation methods include:

(1) **Direct-cost basis** : R&D costs are allocated based on a percentage of a product's direct costs, such as labour and materials. This is a simple method but may distort the true cost of each product.

(2) **Activity-Based Costing (ABC)** : This method assigns R&D costs based on specific activities that drive overhead. For example, the cost of testing equipment could be allocated based on the time each product or project uses it.

(3) **Payroll-based allocation** : This method is effective for R&D-intensive businesses where labour is the primary driver of cost. Costs are allocated proportionally to the salaries of the key personnel involved in each project.

Q.86. What do you understand by Cost Sheet? What are its objectives? Also give a specimen of Cost Sheet.

Or

What is meant by Cost Sheet? Give its merits and a specimen of Cost Sheet.

Or

What do you understand by Cost? What are the various contents of Cost?

Or

Give the meaning of Cost Sheet. Mention their advantages also.

Or

What is a Cost Sheet and why is it prepared? Prepare a cost sheet with imaginary figures.

Or

What do you mean by Cost Sheet? Discuss briefly its objectives and advantages.

Or

Define Cost Sheet. Give the characteristics and advantages of it.

Or

What is Unit Cost. In which industries it is used?

Or

Discuss the objectives and merits of Cost Sheet.

Ans. Meaning of Cost-Sheet : Cost sheet is a statement which provides for the assembly of the detailed cost of a centre a cost unit. Cost sheets are prepared for the use of management to fix the selling price, to compare the costs of any two periods and to ascertain the efficiency in the production. According to **I.C.M.A. England**, 'Cost Sheet is a document which provides for the assembly of the estimated detailed cost in respect of a cost centre or a cost unit.'

The total cost is analysed into Prime cost, Factory/work cost, office cost or cost of production. The cost sheet presents the total as well as cost per unit. The cost sheet can be drawn with as much details regarding costs.

Characteristics : The main characteristics of a cost sheet are as under :

1. Cost sheet shows total cost as well as per unit cost of any product.
2. It is prepared on periodical basis, say weekly, monthly, half yearly & yearly etc.
3. It explains every stage of cost such as prime cost, factory cost, total cost etc.
4. It shows the relation and percentage of various types of cost with the total cost.
5. It indicates the comparative statement of various years.
6. It shows a comparative statement of one type of product having different varieties, such as Fan, Cooler, Air conditioner etc.
7. It explains the changes made in cost on account of increase in production.
8. It analyses the profit and loss of production.

Objectives of Cost Sheet : The chief objectives of preparing Cost Sheet are as follows :

- (1) To show the classification of costs.
- (2) To prepare estimate of costs for future use.
- (3) To compare that actual cost with the standard cost and to take remedial action.

Advantages : The main advantages of cost sheet are as under :

1. **Fixation of Sales Price** : Cost sheet fixes the sales price of a product in advance, which is an essential part in a competitive economy.

2. **Control on Expenses** : Cost sheet presents the comparative statement of different periods, hence it may control the expenses whenever it shows a rising trend.
3. **Fixation of Cost** : With the help of cost sheet, total cost as well as per unit cost of any product can be ascertained easily.
4. **Tender Price** : On the basis of cost sheet, tender prices can be fixed in advance.
5. **Other Advantages** : It includes the following advantages :
- (i) It is easier to have a comparative study of cost of various years.
 - (ii) It gives the information regarding various components of total cost.
 - (iii) It indicates the ratio and percentage of each aspect of cost on its total cost.

**Specimen of Cost Sheet with imaginary figures
units produced 4,000 :**

Particulars	Total Cost Rs.	Cost per unit Rs.
Raw Material	1,60,000	40.00
Direct Wages	80,000	20.00
Prime Cost	2,40,000	60.00
Factory Overhead	16,000	4.00
Factory Cost	2,56,000	64.00
Office overhead: (10% of factory cost)	25,600	6.40
Cost of Production	2,81,600	70.40
Less : Units in stock (Unsold)	28,160	—
Cost of goods sold	2,53,440	70.40
Selling overhead	12,000	3.33
Total Cost	2,65,440	73.73
Profit	94,560	26.27
Sales	3,60,000	100.00

Q.21. Mention the difference between Cost Account and Cost Sheet.

Or

What is Cost Sheet? Give its merits? Differentiate between Cost Account and Cost Sheet?

Or

What is Cost Account? How does it differs from Cost Sheet?

Or

What is the difference between Cost Account and Cost Sheet?

Ans. Difference between Cost Account and Cost Sheet

Point of Difference	Cost Account	Cost Sheet
1. Double Entry System	It is based on Double Entry System and has Debit and Credit sides.	It is a Statement and not based on double Entry Principles.
2. Preparation	Cost accounts are used to prepare Production A/c, Process A/s, Contract or Job A/c, Ledger A/c, etc. These accounts show the costs but only after the end of the year or period when they are closed.	Cost Sheets are prepared during the currency of production weakly, monthly etc. The cost sheets render cost information related to that period for which they are prepared.
3. Helpful to know Comparative Costs.	Cost accounts are not helpful to know comparative costs etc.	Cost Sheets are helpful to know comparative costs, to exercise cost control and costs reduction effectively, and to fix selling prices and quote articles.
4. Analytical Presentation	Cost accounts do not show cost per unit in a detailed or analytical way.	Cost Sheets ascertain cost per unit in a detailed or analytical way.
5. Comparison with Financial Accounts	The cost accounts are useful in reconciling the profits of financial books with cost books.	The cost sheet are not so useful.

Q.22. A manufacturer finds that an advance in the cost of production has taken place and that whereas formerly his goods cost as : Raw materials 30%, Wages 20%, Rent and Rates 5%, Fuel and Power 10% and General Expenses 15%. Now there has been an increase of 50% in Fuel, 30% in materials, 25% in wages and 20% in rent. He consults you as to what percentage he must add to the selling price in order to obtain the same profit as before.

Solution : Former Components of Costs :

Materials	30%
Wages	20%
Rent & Rates	5%
Fuel & Power	10%
General Expenses	15%
	<u>80%</u>

Profit 20%
Selling Price 100%

Profit is 20% on sales or 25% on Total Cost.

Increased Cost of Production:		Rs.
Materials	30% + 30% increase =	39
Wages	20% + 25% increase =	25
Rent & Rates	5% + 20% increase =	6
Fuel & Power	10% + 50% increase =	15
General Expenses	15% (No increase) =	15
Total Cost		100
Profit		25
Selling Price		125

In order to obtain the same profit Selling Price should be increased by 25% then only he will get 20% profit on the selling price or 25% on the cost price.

Q.23. A furniture company prepares Stools, Chairs and Tables. The following is the statement of cost :

	Stool Rs.	Chair Rs.	Table Rs.
Materials	36	60	440
Labour	48	40	120
Production	600	300	60

Total factory overhead Rs. 30,000

Office Expenses Rs. 12,000

Selling Expenses Rs. 24,000

Prepare a cost sheet showing cost of each item separately. Assuming that one table is equal to 4 stools and 2 chairs to one table.

Ans. Statement of Cost

Particulars	Stools 600 Rs.	Chairs 300 Rs.	Tables 60 Rs.
Materials	21,600	18,000	26,400
Labour	28,800	12,000	7,200
Prime Cost	50,400	30,000	33,600
Add : Factory Exp. (5 : 5 : 2)	12,500	12,500	5,000
Works Cost	62,900	42,500	38,600
Add : Office Exp. (5 : 5 : 2)	5,000	5,000	2,000
Cost of Production	67,900	47,500	40,600
Add : Selling Exp. (5 : 5 : 2)	10,000	10,000	4,000
Total Cost	77,900	57,500	44,600

Notes :

Since 2 Chairs = 1 Table

Hence 300 Chairs = 150 Table = 600 Stools (150 × 4)
1 Table = 4 Stool, ∴ 60 Tables = 240 Stools
= 600 : 600 : 240 or 5 : 5 : 2

Q.24. From the following particulars, prepare a statement of cost. Calculate factory cost at 20% of Prime cost and office on cost at 80% of factory on cost :

Opening stock of Raw Material	72,000
Purchase of Raw Material	4,32,000
Closing stock of Raw material	1,08,000
Wages	1,80,000

Solution : Statement of Cost

Opening Stock of Raw Materials	72,000
Add : Purchase	4,32,000
	5,04,000
Less : Closing Stock of Raw Materials	1,08,000
Raw Material Consumed	3,96,000
Add : Wages	1,80,000
Prime Cost	5,76,000
Add : Factory on Cost : 20% of Prime Cost	1,15,200
Work Cost	6,91,200
Add : Office on Loss : 80% on Factory Cost	92,160
Total Cost	7,83,360

Q.25. A factory produces a standard product. The following information is given to you from which you are required to prepare 'Cost Sheet' for the period ended 31st July 2026 :

Consumable Materials :	Rs.
Opening Stock	10,000
Purchases	85,000
Closing Stock	4,000
Direct wages	20,000
Other Direct expenses	10,000
Factory overheads	100% of Direct Labour
Office overheads	10% of works cost
Selling and Distribution expenses	2 per unit sold
Units of Finished products.	

In hand at the beginning of the period 1,000

(valued at Rs. 16)

Produced during the period 10,000

In hand at the end of the period 2,000

Also, find out selling price per unit on the basis that profit mark up is uniformly made to yield a profit of 20% of the selling price. There was no work-in-progress either at the beginning or at the end of the period.

Solution : Cost Sheet for the period ended 31st July, 2026

Particulars	Units	Rs.	Rs./ Unit
Opening Stock of Consumable Materials		10,000	
Add: Purchases		85,000	
Less : Closing Stock of Consumable materials		95,000	
Cost of Materials Consumed	10,000	4,000	
Add: Direct Wages	—	91,000	9.10
Add : Other direct expenses	—	20,000	2.00
Prime Cost	10,000	10,000	1.00
Add: Factory Overheads (100% of Direct wages).		121,000	12.10
Works Cost	10,000	20,000	2.00
Add: Office overheads (10% of works cost)		14,100	1.41
Cost of Production	10,000	1,41,000	14.10
Add: Opening Stock of Finished Products	1,000	16,000	15.51
Less : Closing Stock of Finished Products (F.I.F.O. Basis)	11,000	1,71,100	
Cost of Goods Sold	2,000	31,020	
Add: Selling and Distribution Expenses	9,000	1,40,080	15.56
Cost of Sales	9,000	18,000	2.00
Add: Profit (20% on selling Price i.e., 25% on Cost of Sales)		1,58,080	17.56
Sales	9,000	39,520	4.39
		1,97,600	21.95

Q.26. The following particulars relate to a Slate manufacturing company for Jan, 2026.

Slate quarried 2000 tons on which Royalty @ Rs. 2 per ton paid, wages Rs. 80,000, Coal Rs. 20,000, Internal transport Rs. 40,000. Works overhead 20% on Direct charges.

Office overhead 10% on works cost.

Cost of Removal of overburden 1 paisa per Slate manufactured.

Opening Stock 20,000 Slates.

Manufactured Slates purchased 40,000 @ Re. 1 per Slate.

Slate sold 2,00,000 @ Rs. 2 per Slate

Closing stock 60,000 Slates

Selling expenses 10 paisa per Slate sold.

Prepare a Cost Sheet of Slate Company

Solution :

Calculation of Output :

	No. of Slates
Slates Sold	2,00,000
Add : Closing stock	60,000
	2,60,000
Less : Opening Stock	20,000
Purchase of Manufactured Slates	40,000
	60,000
Output	2,00,000

Particulars	Total	Per unit
Royalty on Slates quarried		
20,000 tons @ Rs 2 per ton	40,000	0.20
Wages	80,000	0.40
Coal	20,000	0.10
Internal transport	40,000	0.20
Prime Cost	1,80,000	0.90
Add : Work Overhead :		
20% on Direct wages	36,000	0.18
Cost of removal of over burden	2,000	0.01
Works Cost	2,18,000	1.09
Add : Office Overhead :		
10% on Works Cost	21,800	0.109
Cost of Production	2,39,800	1.119

Statement of Profit

	States Nos.	Total Amount Rs.
Opening Stock	20,000	23,980
Add : Cost of Production	2,00,000	2,39,980
Add : Purchase of Manufactured Slates	40,000	40,000
	2,60,000	3,03,960
Less : Closing Stock	60,000	71,940
Cost of Goods Sold	2,00,000	2,32,020
Add : Seeling Expenses @ 10 paisa per Slate sold	—	20,000
Cost of Sales	2,00,000	2,52,020
Profit	—	1,47,980
		4,00,000
Selling Price @ Rs. 2/- per Slate		

Q.27. From the following particulars prepare a cost sheet showing per unit cost. Also prepare a statement of profit to show profit of the year.

	Rs.		Rs.
Depreciation on Plant & Machinery	2,600	Carriage on Raw Material	782
Opening stock on 30th June, 2023 :		Office Salary	1,880
Finished goods (175 Units)	1,948	Carriage on Sales	466
Raw Material (250 Units)	600	General Expenses	434
Cash discount on sales	748	Factory Rent & Rates	4,542
Printing & Stationery	186	Manufacturing Wages	20,000
Purchases :		Travelling Expenses	558
Raw Material (5000 Units)	17,452	Sales	59,684
Finished goods (255 Units)	2,548	Stock on 31st July 2026 :	
Plant Repairs	500	Finished goods (535 Units)	2,794
Office Rent & Rates	1,300	Raw Material (150 Units)	200
Coal	1,158		

Solution : Cost Sheet for the Month of July, 2026
(Output 5,100 Units)

Particulars	Units	Rs.	Total Cost (Rs.)	Cost per Unit (Rs.)
Opening Stock of Raw Materials	250	600		
Add : Purchases	5,000	17,452		
Add : Carriage	-	782		
	5,250	18,834		
Less : Closing Stock of Raw Material	150	400		
	5,100	18,434	18,434	3.61
Add : Manufacturing Wages			20,000	3.92
Prime Cost			38,434	7.53
Add : Factory Expenses :				
Depreciation on Plant		2,600		
Repairs to Plant		500		
Coal		1,158		
Factory Rent & Rates		4,542		
Manufacturing Salaries		2,058	10,858	2.13
Factory Cost			49,292	9.66
Add : Office & Administrative Ex.				

Printing & Stationery	186		
Office Rent & Rates	1,300		
Office Salaries	1,880		
General Expenses	434	3,800	0.75
Cost of Production (5,100 Units)	53,092		10.41

Statement of Profit during the Month of July, 2026

Particulars	Units	Rs.
Opening Stock finished goods on 30-6-2023	175	1,948
Add : Cost of Production of Goods during July, 2023	5,100	53,092
	5,275	55,040
Purchase of Manufactured Goods	255	2,548
	5,530	57,588
Less : Closing Stock of Manufactured Goods (31st July, 2023)	535	5,588
Cost of Goods Sold	4,995	52,000
Add : Selling Expenses :		
Carriage on Sales		466
Travelling Expenses		558
	4,995	53,024
Net Profit		6,660
Sales	4,995	59,684

Notes : (i) Cash discount on sales Rs. 748 has been ignored because this is an item of financial nature. (ii) Where there is no wastage in production, output will be equal to the units of materials used i.e., units of materials consumed shows output.

Q.28. From the following information, Prepare a monthly cost sheet of Radha Bricks works showing cost and profit (M = 1,000 bricks)

Material Used :

Lime	:	895 tons @ Rs. 500 per ton
Coal	:	820 tons @ Rs. 300 per ton
Sand	:	Rs. 15 per 'M' Bricks
Stone	:	Rs. 1,46,500
Labour	:	Sand digging and running Rs. 1,00,000
	:	Brick making Rs. 4,00,000

Factory on cost: 25% of direct charges

Office on cost : 10% of direct charges.

Bricks sold : 3,500 'M' @ Rs. 550 per 'M' bricks

Stock of Bricks at the beginning of the month 100 'M'

Stock of Bricks at the end of the month 600 'M'

Solution :

Cost Sheet for a Month
(Bricks Manufactured 40,00,000)

Particulars	Rate per 'M' (1,000) Bricks Rs.	Total Amount Rs.
Materials :		
Lime		4,47,500
Coal		2,46,000
Sand		60,000
Stone		1,46,500
	225.00	9,00,000
Labour : Digging Sand Rs. 1,00,000		
Brick Making Rs.	125.00	5,00,000
<u>4,00,000</u>		
Prime Cost	350.00	14,00,000
Works Overhead (25% of direct wages)	87.50	350,000
Works Cost	437.50	17,50,000
Office overhead (10% of direct charges)	35.00	1,40,000
Cost of Production	472.50	18,90,000

Selling Price 550.00

Less : Cost Price 472.50

Profit per 1,000 bricks or per 'M' Bricks 77.50

* Number of bricks manufactured has been ascertained as follows :

Stock of Bricks at the end of Month 600 'M'

Add : Sold bricks during the Month 3,500 'M'

4,100 'M'

Less : Stock of bricks at the beginning of the months 100 'M'

Bricks Manufactured during the month 4,000 'M'

Or $4,000 \times 1,000 = 40,00,000$ Bricks

Q.2. What is meant by Job Cost? What are its objectives? Describe the procedure of Job Costing.

Or

What do you understand by Job Costing? Explain its features, Objects, pre-conditions and advantages.

Or

What is Job Costing or Job-order Costing? In which industries is it used?

Or

Explain the purpose and procedure of Job costing.

Ans. Meaning of Job Costing : Job Costing or Job order costing refers to that form of specific order costing which applies where work is undertaken to customer's special requirements and each order is of comparatively short duration. The work is usually carried out within a factory or workshop and moves through processes or operations as a continuously identifiable units.

Features of Job Costing : Main features of Job costing are :

- (1) Production is undertaken after obtaining customer's order.
- (2) Each order is different and suited to the requirement of the customer.
- (3) Identity of each order is retained throughout the manufacturing process from start to finish.
- (4) Production cycle is usually short, but a large order may extend beyond one year.
- (5) Cost information is collected by job order.
- (6) Generally, the Jobs are performed according to the customers orders and therefore, there is no need to maintain stock of finished or completed jobs.

Industries in which Job Costing is used : Job costing is generally applied in the following industries :

- (i) Ship-building (ii) Motor repairs and garages (iii) Engineering concerns (iv) Printing Press (v) Text Book Publications (vi) Machine Making, (vii) Building concerns (viii) Wood working shops (ix) Ready-made garments concerns etc.

Objectives of Job-Costing :

- (1) To bring together all the costs incurred for completing a job.
- (2) To ensure that there is proper control over costs incurred.
- (3) To ascertain the profit (or loss) of the production or the execution of the product or order or job.

Advantages of Job Costing :

- (1) It helps management to detect which jobs are profitable and which are not.
- (2) On the basis of accurate job costs, estimates of cost for similar work in the future may be conveniently made.
- (3) Efficiency of operation can be controlled by comparing actual costs against the estimates made while submitting tender for jobs or giving quotations.
- (4) Job costing provides accurate cost data.

Limitation : The practical limitation of job costing is that it involves a great deal of clerical and paper work in connection with estimation of costs, designing and scheduling of production.

Procedure Essential for Job-Costing : The following procedure is usually adopted in Job-Costing :

- (1) **Estimating :** In job costing the procedure starts with the receiving of enquiry from the customer or submission of quotations against tenders floated by customers. In either case, cost estimates are prepared in detail, considering cost of material, labour, factory and administrative and selling overheads. Price quotation will depend on the estimated cost, profit loading and policy guidelines in pricing. Thus, estimating plays a vital role in job costing.
- (2) **Job order Number :** When firm order is received, the commercial department allots a job order Number with a clear title or description of the job. The number continues till the completion of production process.
- (3) **Production or Job order :** It is a written order to the foreman to proceed with the job. It contains all the relevant information regarding production such as, the quantity or units to be manufactured, delivery date, details of materials required as per Bill of materials, the operations and departments concerned.
- (4) **Bill of Materials :** It is a list of all materials and parts required for a job. It is prepared by Production Planning Department. The foreman receives a copy of the bill of materials alongwith the production order. Thereby he is authorised to call for and receive the materials.
- (5) **Job-cost card or Job-cost sheet :** The cost accounting department opens a job-cost card on receipt of the Production order, wherein all cost information is recorded.
- (6) When Job is completed, and transferred to finished goods store, entries in the Job Cost card is verified with reference to the estimates, and then final cost of the order and profit or loss amounts are computed.

UNIT-5

Short Answer type Questions :

Q.1. What is Contract Costing?

Ans. Contract Costing is a method of costing used to determine the cost of specific contracts, usually in construction and large-scale engineering projects. Each contract is treated as a separate cost unit, and costs are recorded individually. All direct costs like materials, labor, and expenses are charged to the contract, while overheads are allocated appropriately. Contract costing is suitable for industries where work is carried out on-site, often over a long period. It helps in tracking cost, profit, and progress of each contract accurately. Common examples include building roads, bridges, or large infrastructure projects.

Q.2. What is Sub-Contract Costing?

Ans. Sub-Contract Costing involves assigning a part of the main contract to an external party, called a subcontractor, while the main contractor retains overall responsibility. The subcontractor's cost becomes or additional work efficiently without overloading the main contractor. Sub-contract costs are recorded separately for clarity and control. Proper monitoring ensures the subcontractor's performance aligns with quality, cost, and time requirements. Sub-contracting helps reduce workload and enables better resource utilization in contract-based industries.

Q.3. How are Contract Costs Recorded and Controlled?

Ans. In contract costing, a separate **contract account** is opened for each contract. All direct costs (materials, labor, direct expenses) are charged directly to the contract. Overheads are allocated on a fair basis. Any materials returned to stores or plant transferred out are deducted. Payments received from contractees are shown separately from work certified and uncertified. Cost control is maintained through regular site reports, budget comparison, and monitoring progress. This ensures contracts are completed within budget and schedule, preventing cost overruns and improving profitability.

~~Q.4~~ What is Work Certified and Work Uncertified in Contract Costing?

Ans.

- **Work Certified** refers to the portion of work completed on a contract that has been **inspected and approved** by the contractee or an architect/engineer. It is valued at an agreed rate and used as a basis for interim payment.
- **Work Uncertified** refers to work completed but not yet approved or certified. It is valued at cost and not at contract price.

This distinction helps in accurate profit calculation and financial reporting during the course of a contract. Work certified indicates progress and payment due, while work uncertified reflects pending completion.

Q.5. What is Escalation Clause in Contract Costing?

Ans. An **Escalation Clause** is a provision in a contract that allows the contractor to revise the contract price if there are significant increases in material costs, labor rates, or other inputs during the contract period. This protects contractors from losses caused by inflation or unexpected cost rises. The amount of escalation is calculated based on the agreed formula between contractor and contractee. Escalation clauses are commonly included in long-term contracts to ensure fairness to both parties and maintain profitability despite changing cost conditions.

Q.6. What is Process Costing and when is it used?

Ans. Process Costing is a method of costing used in industries where production is continuous and the products are homogeneous. Costs

are accumulated for each process or department over a specific period and then divided by the total units produced to find the unit cost. It is best suited for industries like cement, sugar, chemicals, oil refining, and textiles. In process costing, materials, labor, and overheads are charged to each process separately. It simplifies cost determination, enables cost control at each stage, and helps in setting appropriate selling prices for mass-produced goods.

Q.7. How does Process Costing differ from Job or Contract Costing?

Ans. Process costing applies to mass production of standardized products, while job and contract costing are used for customized, one-off jobs. In process costing, costs are accumulated by process or department, and unit cost is calculated by averaging total costs over all units produced. In job and contract costing, costs are tracked individually for each job or contract. Process costing is continuous and systematic, while job and contract costing are specific and project-based. This makes process costing suitable for industries like paper or oil, and job/contract costing suitable for construction or printing industries.

Long Answer type Questions :

Q.8. What do you mean by Contract Costing? Discuss the specific aspects of Contract Costing.

Or

Define Contract Costing. What important points should be borne in mind, in its preparation?

Ans. Meaning of Contract Costing : Contract Costing is type of job costing in which a contract constitutes a unit of cost. Contract costing is defined by the I.C.W.A. Terminology as, 'That form of specific order costing which applies where work is undertaken to customer's special requirements and each order is of long duration. The work is usually of a constructional nature.' Under this system, costs are accumulated for each contract which is regarded as a separate activity for cost determination and control.

Contract costing is adopted usually in the case of companies engaged in : (i) Building construction ; (ii) Road constructions (iii) Ship Building, (iv) Civil Engineering and similar activities.

Specific Aspects of Contract Costing : The Specific aspects of contract costing are as under :

(1) **Material :** Materials purchased and directly supplied to a contract are debited to the concerned contract account. Material returned to stores are valued at cost and credited to the Contract Account. Similarly, the cost of materials transferred from one contract to another is credited to the former contract and debited to the latter contract. However, such transfers

should be duly authorised by a responsible officer through a 'Material Transfer Note'.

If materials, plant or equipment at work-site are damaged, destroyed or lost on account of theft, accident, earthquake etc., the cost of such losses should be determined and transferred to P and L Account because such losses are 'abnormal'. Where loss could be recovered from insurance company or from sale of scrap, the proceeds may be credited to the contract and net loss (i.e., the difference between the cost of goods damaged and the amount realised) is transferred to P and L account.

When the loss of material is negligible, it should be treated as 'Normal loss'. Normal loss is entirely neglected so that it is automatically borne by the contracts.

(2) **Labour :** All the labour engaged for the completion of contract is direct labour and is treated as such. For each work Wages Analysis Sheet is prepared and the total are debited to the Contract Account. In case of transfer of workers from one contract to another or supervisory staff working on a number of contracts, detailed time sheets should be maintained and wages may be allocated on the basis of time spent on each contract.

(3) **Overheads :** So far overhead costs are concerned most of them can be directly related to particular contracts and therefore charged entirely to the contract. The direct expenses include architects fees, hire charge of special equipment, cost of sub-contract etc. However, costs common to more than one contract such as head office expenses should be apportioned to the respective contracts or some suitable bases, e.g., direct labour hours or value of contracts.

(4) **Plant & Machinery :** The expenditure incurred on plant and machinery used at contract site may be accounted for in any of the following ways :

- The cost of operation of each plant may be determined and an hourly-rate may be fixed in advance. The cost of plant and machinery may be charged to the different contracts in accordance with the number of hours of use during the period.
- If machinery is used at the Site of contract for a long period of time, the Contract Account may be debited with the value of the plant at the commencement of use (i.e., original cost or written down value) and credited with the depreciated value when it is returned to Head office. If a portion of the plant is returned to the store, the Contract Account should be credited with the appreciated value of that part of the plant.

(5) **Work-in-Progress :** In Contract Account, the amount of Work-in-Progress consist of :

- Cost of work completed both certified and uncertified.
- Cost of work uncompleted.
- Amount of profit taken as credit.

(6) **Escalation Clause** : It is a clause in the agreement of a contract which provides for revision of contract price to accommodate variation in the cost of material and labour due to market fluctuations. This clause is generally included in case of long period contracts. It protects the interests of both the parties from unfavourable changes in the prices of materials and labour.

(7) **Surveyor's Certificate** : In case of large contracts, the part payment is made to the contractor from time to time on the basis of surveyor's or architect's certificate.

Q.2. What do you understand by Cost Plus method of Contracts? In what cases is this method usually employed?

Or

Explain Cost Plus Contract? Discuss its advantages and disadvantages.

Or

What is Cost Plus Contract? Explain its main Advantages.

Ans. Meaning of 'Cost Plus' Contract : Cost plus contract is one wherein the contract agrees to pay to the contractor the cost price of the work done on the contract plus an agreed percentage margin to the costs allowed on the contract. In such contracts, the items of cost that may be allowed and the percentage of profit to be added are provided in the agreement. Such contracts are undertaken when it is not possible to fix the contract price in advance.

Cost plus contracts are advantageous where the contract supplies the material and labour. Secondly, it is convenient and safe to enter into a new or a special contract on cost plus basis because in that case contractor as well as contract are protected from arbitrary fixation of contract price. Thirdly, the contractor is assured of his profit and relieved from the risk of uncertainty. Lastly, in case of cost plus contracts quality of work do not suffer, which is a matter of satisfaction to contract. Therefore, where the quality of the work is of utmost importance, cost plus contract is desirable.

A demerit of these contracts is that there is no incentive to economise the cost of the contract, because paradoxically enough, in case of cost plus contracts, higher the cost, higher will be the profit, since profit is allowed as a percentage of cost. Cost plus contracts have the following Advantages and Disadvantages :

Advantages :

- (i) The Contractor is assumed of a fixed percentage of profit. There is no risk of incurring any loss on the contract.
- (ii) It is useful, specially, when the work to be done is not definitely fixed at the time of making the estimate.
- (iii) Contractee can ensure himself about 'the cost of the Contract', as he is empowered to examine the books and documents of the contractor to ascertain the variety of the cost of the Contract.

Disadvantages :

- (i) Since contractor is assured of a profit margin, there may not be any incentive for cost reduction by avoiding wastage and effective economies in production.
- (ii) The Customer has to pay not only the resultant high cost but also the resultant high profit and thus, customer may have to pay substantially for lack of proper attitude (towards cost and efficiency) on the part of contractor. The final price to be paid by customer cannot be exactly ascertained up to the last and may create difficulty for the customer in preparing purchase budget.

Q.3. What is Sub-contract costing? How does it works?

Or

What are the key components of Sub-contract costing?

Ans. Subcontract costing is the accounting and management process of tracking the costs associated with hiring a third-party specialist to perform specific tasks on a larger project. Rather than being a stand-alone costing method, it is a component of a larger project's total cost. It is particularly common in industries like construction, manufacturing, and IT.

Key components of subcontract costing : For the main contractor, the total cost of the subcontract includes several components:

- (1) **Subcontractor's fee** : The direct payment to the subcontractor for the work they perform, which can be based on a fixed price, hourly rates, or per-unit pricing.
- (2) **Cost of materials** : The cost of any materials the main contractor provides to the subcontractor for the work.
- (3) **Administrative costs** : The expenses incurred by the main contractor for managing and coordinating the subcontractor, such as processing invoices and overseeing the work.
- (4) **Contractual risks** : Financial considerations for potential risks, such as performance bonds, retainage (a portion of the payment withheld until project completion), and potential liquidated damages for missed deadlines.

How subcontract costing works?

The process of managing and accounting for subcontract costs follows several steps :

- (1) **Define the scope of work** : A clear, detailed contract specifies the tasks and deliverables the subcontractor is responsible for, along with the agreed-upon payment terms.
- (2) **Assign costs** : In the accounting system, the costs associated with the subcontract, such as the direct fee and any materials provided, are allocated directly to the specific project or "contract account".

(3) **Track progress and costs** : The main contractor monitors the subcontractor's progress and tracks incurred costs. Invoices from the subcontractor must be carefully documented and verified against the work completed.

(4) **Manage work-in-progress (WIP)** : For projects spanning multiple accounting periods, the value of uncompleted work (WIP) on the contract must be reported. This includes the value of certified work (approved by the client) and uncertified work.

(5) **Recognize revenue and profit** : The main contractor recognizes revenue and profit from the subcontract based on the percentage of completion. For uncompleted long-term contracts, only a portion of the total profit is recognized in the current period, with the rest held in reserve.

(6) **Finalize the contract** : Upon completion, the contract account is credited with the final contract price, and the total profit or loss is transferred to the company's profit and loss account.

Q.4. What are the benefits of using subcontractors?

Ans. Using subcontractors can be a strategic way for businesses to manage their resources and finances.

(1) **Access to expertise** : A company can hire specialists for tasks that require unique skills not available in-house.

(2) **Cost savings** : Subcontracting can eliminate the costs of hiring and training permanent employees, as well as the need to purchase specialized equipment for one-off tasks.

(3) **Increased flexibility** : Companies can scale their workforce up or down based on project needs without the long-term commitment of hiring full-time staff.

(4) **Reduced risk** : Some risks, such as legal compliance and financial liabilities related to the work, are shifted to the subcontractor.

Q.5. What points are taken into consideration at the time of Work-in-progress? How it is shown in Balance Sheet?

Or

What is Work-in-Progress and how is it calculated? How it is shown in Balance Sheet?

Ans. Work in progress is an incomplete work. Work in progress A/c is prepared and used in finding out profit or loss on incomplete contracts. The debit and credit side of work in progress A/c shows the following items:

Debit Side : (a) Work certified value, (b) Cost of work certified, (c) Material at site.

Credit Side : A portion of profit in the form of reserve is shown on the credit side of this account.

Work-in-Progress A/c			
Dr.		Cr.	
	Rs.		Rs.
To Contract A/c - Work Certified		By Contract A/c -	
Work uncertified		By Profit Reserve	
		By Balance c/d	

Work-in-Progress A/c and Balance Sheet

Generally the balance of work in Progress A/c is debit, and it is shown on the asset side of Balance Sheet. The account of Contract A/c is credited by the amount received in cash from time to time. The credit balance of contract A/c is deducted out of the assets amount shown in Balance sheet.

It is shown in Balance sheet as under :

Balance Sheet		
Liabilities	Assets	Rs.
	Work-in-Progress :	
	Works Certified	
	Works Uncertified	
	Total	
	Less : Profit Reserve	
	Balance	
	Less : Cash Received	
	From contract	

Q.6. State whether or not profit should be determined on incomplete Contract also. How would you determine profit on incomplete Contract?

Or

What are the various methods of calculating profits on incomplete Contracts? Explain with illustration.

Or

What are the principles of ascertaining profits on incomplete Contracts?

Or

How profit is calculated on incomplete Contracts? Mention the method of calculating such profits.

Or

Comment on the method of ascertaining in the following types of contract-

(i) When contract is completed (ii) When contract is incomplete, and (iii) When contract is near completion.

Ans. Contract account is prepared to find out the profit and loss on a contract. After the completion of contract, the total of debit side and credit side of contract account is made and balance is known. If the credit side is larger, there will be profit and on the contrary if debit side is bigger, it will

show the loss. If the contract is incomplete, the profit on it will not be treated as profit. The actual profit can be known after the completion of the contract.

Calculation of Profits on Incomplete Contracts

The following rules are applied for calculating profits on the incomplete contracts:

1. If certified work is more than 1/4 but less than 1/2 of Contract Price: In such a case, only 1/3 of the profit received should be transferred to P/L A/c and the balance to work in progress A/c. This profit should be in proportion to the cash received to work certified. The formula is given as under:

$$\text{Profit} = \text{Total Profit} \times \frac{1}{3} \times \frac{\text{Cost Received}}{\text{Work Certified}}$$

2. If certified work is less than 1/4 of Contract Price: In such case, no profit is ascertained and the whole balance is transferred to work in progress A/c.

3. If certified work is more than 1/2 of Contract Price: In this case 2/3 part of the total profit on cash received basis is transferred to profit and loss account and the balance is transferred to work in progress account. The formula is as under:

$$\text{Profit} = \text{Total Profit} \times \frac{2}{3} \times \frac{\text{Cost Received}}{\text{Work Certified}}$$

4. Profit on Certified Work: The contractor tries to complete the contract and gets the certificate of work completed from the architect. It is this certified work, on which the profit is calculated.

5. On Completion of Contract: If the contract has been completed, the following formula is used for calculating the profit on such contract:

$$(i) \text{ Profit} = \text{Total Profit} \times \frac{\text{Work Certified}}{\text{Contract Price}} \text{ or}$$

$$(ii) \text{ Profit} = \text{Total Profit} \times \frac{\text{Cash Received}}{\text{Contract Price}}$$

Example 1. Total profit on contract Rs. 9,600, Cash received Rs. 32,000 and the contract has retained 20% of work. More than 1/3 work has been completed. Find out the profit to be transferred to P/L A/c.

Solution:

$$\text{Profit} = \text{Total Profit} \times \frac{1}{3} \times \frac{\text{Cash Received}}{\text{Work Certified}}$$

$$\text{Rs. } 9,600 \times \frac{1}{3} \times \frac{80}{100} = \text{Rs. } 2,560$$

$$\text{Work-in-Progress Rs. } 7,040 (9,600 - 2,560)$$

Example 2. A contractor received cash Rs. 64,000 against Rs. 80,000 as work certified. The total profit on incomplete contract is Rs. 21,000. Find out profit to be transferred to P/L A/c.

Solution:

$$\begin{aligned} \text{Profit} &= \text{Total profit} \times \frac{2}{3} \times \frac{\text{Cash Received}}{\text{Work Certified}} \\ &= 21,000 \times \frac{2}{3} \times \frac{64,000}{80,000} = \text{Rs. } 11,200 \end{aligned}$$

$$\text{Work-in-Progress : Rs. } 21,000 - 11,200 = \text{Rs. } 9,800.$$

Q.7. Mr. X started contract on 1st January, 2025 and incurred the following expenditure on the contract during the year 2025. Materials Rs. 15,000, Wages Rs. 25,000, Plant Rs. 10,000; overhead charges Rs. 3,000.

At the end of the year works to the value of Rs. 1,50,000 has been certified of which 75% had been received in cash; work completed but not certified was estimated Rs. 1,500 and materials valued at Rs. 2,000 were on hand at the site. Of the plant and materials charged to the contract, plant which cost Rs. 2,000 and materials which cost Rs. 1,200 were lost. Plant which cost Rs. 1,000 was returned to stores.

After allowing depreciation at the rate of 10% per annum on plant, prepare contract account and point out how much profit is to be taken to Profit & Loss account?

Solution:

Contract Account

(For the year ended 31st Dec, 2025)

Date	Particulars	Amt. (Rs.)	Date	Particulars	Amt. (Rs.)
2025 Jan/ Dec.	To Materials	15,000	2025 Jan/ Dec.	By Material on site	2,000
	To Wages	25,000		By Plant on Site	6,300
	To Plant	10,000		By P & L A/c:	
	To Overhead charges	3,000		Plant Lost	2,000
	To Balance c/d	1,10,900		Material Lost	1,200
					3,200
				By Plant returned to Stores:	
				Cost	1,000
				(-) Dep. 10%	100
					900

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			By Work-in-progress : Work certified 1,50,000
			(+)Uncertified 1,500
		1,63,900	1,51,500
	To P & L A/c	55,450	1,63,900
	To Work-in-progress A/c (Profit in reserve)	55,450	1,10,900
		1,10,900	1,10,900

Note : Plant sent on Contract		Rs.
(-) Plant lost	2,000	10,000
(-) Plant Returned	1,000	3,000
Remaining Plant on Contract		7,000
(-) 10% Depreciation		700
Plant on Site		6,300

Q.8. A contractor took a contract for Rs. 15 lakh and spends the following.

	Rs.
Material issued	10,000
Materials Purchased	3,00,000
Labour	2,50,000
Plant	3,50,000
Indirect expenses	90,000
Materials returned to stores	20,000
Material lost by fire	5,000
Materials at site	15,000
Plant at site	3,00,000
Works certified	7,00,000

Cash received from contractee Rs. 5,60,000, being 80% of works certified. Prepare contract A/c, work in progress A/c and Balance Sheet.

Ans.

Contract Account

Date	Particulars	Amt.(Rs.)	Date	Particulars	Amt.(Rs.)
Jan./	To Materials		Jan./	By Materials	
Dec.	Purchased	3,00,000	Dec.	Returned to Store	20,000
	To Materials issued			By Profit & Loss	
	From Stores	10,000		A/c	5,000
	To Labour	2,50,000		(Materials	
				lost by Fire)	

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	To Plant	3,50,000		By Materials at site c/d	15,000
	To Indirect Exp.	90,000		By Plant at site	3,00,000
	To Balance c/d	40,000		By Work-in-Progress (Work Certified)	7,00,000
		10,40,000			10,40,000
" 31	To Profit & Loss A/c	21,333	Dec.31	By Balance b/d	40,000
	($40,000 \times \frac{2}{3} = 5,60,000$ $\times 7,00,000$)				
	To Work-in-Progress (Profit in Reserve)	18,667			
		40,000			
Jan.1	To Work in Progress b/d	6,81,333			40,000
	(7,00,000-18,667)				
	To Materials at Site b/d	15,000			
	To Plant at Site b/d	3,00,000			
Dr.					

Dr. Work-in-Progress A/c		Cr.	
To Contract A/c (Work Certified)	7,00,000	By Contract A/c (Reserve)	18,667
		By Balance c/d	6,81,333
	7,00,000		7,00,000

Balance Sheet

Assets		
Work-in-Progress A/c :		
Balance		6,81,333
Less : Cash Received		5,60,000
		1,21,333
Material at Site		15,000
Plant at Site		3,00,000

Q.9. The Contract ledger of a Company showed the following expenditure on account of a contract at 31st December, 2025.

Materials	1,20,000
Plant	20,000
Wages	1,64,400
Establishment Charges	8,600
	3,13,000

The contract was commenced on 1st January, 2025 and the contract price was Rs. 6,00,000. Cash received on account to

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date was Rs. 2,40,000 representing 80% of the work certified, the remaining 20% being retained until completion. The value of materials on hand was Rs. 4,000 and the cost of work finished but not certified at 31st December, 2025 was Rs. 6,000.

Prepare an account in respect of the contract, showing the profit to date, assuming depreciation on plant at 10% per annum and state the proportion of profit company would be justified in taking to the credit of its profit and loss account.

Solution : Contract Account

Date	Particulars	Amt. (Rs.)	Date	Particulars	Amt. (Rs.)
2025 Dec.31	To Material	1,20,000	2025 Dec. 31	By Work-in-Progress	
	To Wages	1,64,400		Work certified	3,00,000
	To Plant	20,000		Uncertified	6,000
	To Establishment Charges	8,600		To Material on Hand	4,000
	To Balance being Profit c/d	15,000		By Plant on Hand	20,000
		3,28,000		(-) 10% Dep.	2,000
	To Profit & Loss A/c	8,000			3,28,000
	To Work-in-Progress	7,000		By Balance b/d	15,000
	c/d (Profit in Reserve)	15,000			15,000
2026 Jan 1	To Work-in-Progress b/d	2,99,000			
	To Plant on Hand b/d	18,000			
	To Material on Hand	4,000			

* Cash received from Contractee is Rs. 2,40,000 which is 80% of the work certified, so the value of work certified will be Rs. $2,40,000 \times \frac{100}{80} =$ Rs. 3,00,000

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Q.10. The Expenses on a Contract no. 521 were as under :

	Rs.
Materials Purchased	2,80,000
Materials transferred from Contract No. 408	20,000
Wages Paid	3,50,000
Wages Accrued	10,000
Indirect Expenses	14,000
Inspection Fees	6,000
Stores Issued	40,000

An old plant of Rs. 34,000 was purchased on 1st January and Rs. 6,000 was spent on its overhauling. This plant was transferred to job No. 450 on 30th June and on 30th September an additional plant of Rs. 80,000 was purchased. The contract price was Rs. 20,00,000.

On 31st December the value of work certified was Rs. 10,00,000 and 80% of it received in cash. The cost of uncertified work was Rs. 30,000. The stores at site was Rs. 8,000.

Charge 10% depreciation per annum on plant and transfer 1/4 profit earned to reserve. Prepare Contract A/c, Work in progress A/c and Balance Sheet.

Ans. Contract Account

Particulars	Amt. (Rs.)	Particulars	Amt. (Rs.)
To Materials	2,80,000	By-Work-in-Progress	
To Transferred from No. 408	20,000	Works certified:	10,00,000
		Work uncertified	30,000
			10,30,000
To Wages	3,50,000	By Plant transferred to other job	38,000
Add Accrued	10,000		
	3,60,000	By Plant at site	78,000
To Indirect Exp.	14,000	By Stores on hand	8,000
To Inspection fees	6,000		
To Stores	40,000		
To Plant	34,000		
+ overhauling	6,000		
	40,000		
To Additional Plant on 30th Sept.	80,000		
To Balance	3,14,000		

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	11,54,000	By Balance Profit	11,54,000
To P/L A/c 3/4	2,35,500		3,14,000
To Reserve	78,500		
	3,14,000		3,14,000

Notes : (1) Value of Plant transferred to Job No. 450 :

Cost of Plant	Rs. 34,000
Add : Overhauling	6,000
	40,000
Less : Depreciation up to 30th June	
$\frac{40,000}{100} \times \frac{10}{1} \times \frac{6}{12}$	2,000
	38,000

(2) Value of Plant at site :

Cost on 30th Sept.	Rs. 80,000
Less : Depreciation for 3 months @ 10% per annum	2,000
	78,000

Work-in-Progress A/c

To Contract A/c	10,30,000	By Balance c/d	10,30,000
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Balance Sheet

Assets		Rs.
Work in Progress	10,30,000	
Less : Reserve	78,500	
	9,51,500	
Less : Cash Received	8,00,000	1,51,500

Q.11. The following particulars relate to two houses which a firm of builders had in course of construction under contract :

	House A	House B
	Rs.	Rs.
Work in Progress on 1 st Jan. 2025 (Excluding Rs. 800 estimated profit which was taken to P & L A/c in 2025)	1,40,000	-
Materials Purchased	2,30,000	1,66,000
Wages	2,00,000	1,40,000
Electrical Services & fittings	1,40,000	3,000
Road making charges	80,000	-
Contract price including road making	6,00,000	4,00,000
Cash received on 31 st Dec, 2024	6,00,000	2,40,000
Percentage of Cash received to work certified	100%	66 2/3%
Value of materials in hand on 31 st Dec, 2024	4,000	5,400
Completed work not certified	-	25,000

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Value of Plant used on sites	1,20,000	60,000
Period of Plant remained on sites during the year	10 months	8 months

The total establishment expenses increased during the year 2025 amounted to Rs. 12,240. These are to be charged to two contracts in proportion to wages. Depreciation on plant is to be taken into account at the rate of 10% per annum.

Prepare two Contract Accounts (in columnar form) showing the profit and loss on each house for the year 2025 and the sums which you consider appropriately transferable to the profit & Loss Account.

Solution :

Particulars	House 'A' (Rs.)	House 'B' (Rs.)	Particulars	House 'A' (Rs.)	House 'B' (Rs.)
To Work-in-progress	1,48,000	-	By Work certified	-	3,60,000
„ Materials	2,30,000	1,66,000	„ Contractee's A/c	6,00,000	-
„ Wages	2,00,000	1,40,000	„ Work uncertified	-	25,000
„ Electric filling charges	14,000	3,000	„ Plant Less Dep.	1,10,000	56,000
„ Road making charges	80,000	-	„ P & L A/c (Loss)	1,50,000	-
„ Plant	1,20,000	60,000			
„ Administration charges	72,000	50,400			
„ Balance c/d	-	27,000			
	8,64,000	4,46,400			
To P/L Account	-	12,000	By Balance b/d	8,64,000	4,46,000
„ Balance c/d	-	15,000		-	27,000
	-	27,000		-	27,000

Q.12. Building Contractors Ltd., undertakes contracts. On 31st October, 2025, when the actual accounts were prepared the position of contract No. 101, commenced on 1st January, 2025, was as under :

Rs.	
Materials Purchased	37,500
Materials in hand on 31-10-2025	1,500
Wages paid	43,750
Wages outstanding	615
Proportionate Share of indirect expenses	875
Cost of Plant	6,260
Overhead expenses accrued due on 31-10-2025	1,000

The value of work certified was Rs. 90,000, of which Rs. 67,500 had been received. Work completed, but uncertified was valued at Rs. 2,500. The contract price was Rs. 1,50,000. The plant on the site valued at Rs. 5,000 on 31st October, 2025.

Proper contract No. 101 after taking credit for profit which you think reasonable.

Solution: Contract No. 101 Account
For the period ended at 31st October, 2025

Dr.	Rs.	Particulars	Cr.	Rs.
To Materials purchased	37,500	By Work in progress:		
To Wages paid	43,750	Work certified	90,000	
To Plant	6,260	By Work completed		
To Proportionate share of indirect expenses	875	but	2,500	
To Outstanding wages	615	uncertified	1,500	
To Overhead expenses accrued	1,000	By Material in hand		
To Balance being profit c/d	9,000	By Plant (depreciated value)	5,000	
	99,000		99,000	
To P & L A/c	4,500*	By Balance b/d		9,000
To Work-in-Progress	4,500			
	9,000			9,000

$$* \text{Balance} \times \frac{2}{3} \times \frac{\text{Cash Received}}{\text{Work Certified}}$$

$$\text{Rs. } 9,000 \times \frac{2}{3} \times \frac{67,500}{90,000} = \text{Rs. } 4,500$$

Q.13. A firm of building contractors began to trade on 1st April, 2025. The following was the expenditure on the contract for Rs. 3,00,000.

	Rs.
Material	51,000
Plant	15,000
Wages	81,000
Other expenses	5,000

Cash received on account to 31st March, 2025 amounted to Rs. 1,28,000 being 80% of the work certified of the plant and materials charged to the contract plant which cost Rs. 3,000 and material which cost Rs. 2,500 were lost. On 31st March, 2025 Plant which cost Rs. 2,000 was returned to stores, the cost of work done but uncertified was Rs. 1000 and materials costing Rs. 2,300 were in hand at site. Charge 15% depreciation on plant; reserve 1/3 profit received. Prepare a contract account from the above particulars.

Solution : Contract Account
(as on 31st March, 2025)

Particulars	Amt.(Rs.)	Particulars	Amt.(Rs.)
To Material issued	51,000	By Work-in-Progress	
To Wages	81,000	Work certified	1,60,000
To Plant	15,000	(1,28,000 × $\frac{100}{80}$)	
To Other Expenses	5,000	By Work uncertified	1,000
To Balance c/d	27,000	By P/L A/c:	
		Plant lost	3,000
		Material lost	2,500
		By Plant Returned	2,000
		(-) Dep.	300
		By Material at site	2,300
		By Plant in hand	8,500
	1,79,000		1,79,000
To P/L A/c	14,400	By Balance b/d	27,000
(27,000 × $\frac{2}{3} \times \frac{80}{100}$)			
To Reserve	7,200		
To General Reserve	5,400		
	27,000		27,000

Q.14. The following figures are extracted from the books of a contractor for the year ending 31st Dec. 2025.

	Rs.	Rs.
Work-in-Progress on 31 st Dec. 2024	1,70,000	
Less : Advances	1,10,000	60,000
Transactions during 2025 :		
Material supplied to contract direct		1,20,000
Materials issued from stores		21,000
Wages		17,000
Working Expenses		30,000
Administration Expenses (From which Rs. 500 is chargeable to P/L A/c)		2,500
Plant issued		5,000
Materials returned from contracts direct to suppliers		900
Materials returned to stores		1,100
Contract finished		45,000
Work certified		30,000
Profit Taken from contracts		23,000
Advance from contractee's		80,000

Prepare the Contract Account

Solution :

Contract Account		Contract Account	
	Rs.		Rs.
To Work-in-Progress	1,70,000	By Material Returned to suppliers	900
To Material	12,000	By Material Returned to stores	1,100
To Issued from stores	21,000	By Contract finished	45,000
To Wages	17,000	By Work Certified	30,000
To Expenses	3,000	By Work uncertified	1,76,000
To Administrative Exp.	2,000	(Balancing figure)	
To Plant issued	5,000		
To Profit	23,000		
	2,53,000		2,53,000

Q.15. Define Process Costing and explain its working. For which kinds of business is process costing suitable?

Or

What do you mean by Process Costing? What are its objects, its features and general principles? Explain.

Or

What is Process Cost Accounts? Explain the objects and principles of Process Cost Accounts.

Or

What is Process Costing? Explain its various objectives.

Or

Mention the main characteristics of Process Cost System. Where this system can be made applicable?

Or

What do you mean by Process Cost Accounts? Discuss its objectives.

Or

What is Process Costing? Write the salient features of the system. For which kind of business process costing suitable?

Or

What do you understand by 'Process Costing'? In which industries is it used?

Or

What do you mean by 'Process Costing'? What are its features, general principles, advantages and disadvantages.

Ans. Meaning of Process Costing : There are certain commodities which have to undergo several distinct stages of manufacture before completion. These stages are called 'Processes'. In such case it is desirable to find out the cost of production at the end of each process. So that the expenditure incurred at different stages of production may be ascertained.

Thus, Process costing is a system of costing in which elements of cost are collected and summarised separately for each process. This system of costing is applied when the production is continuous and standardised (or identical or homogeneous) goods are produced. The cost of each process is accumulated or collected according to the accounting period and the cost per unit are ascertained at the end of the accounting period. The output of one process is transferred to the next process and it becomes input of the next process. The output of the last process is the finished product. Generally each process and the sequence for processing each product are predetermined.

Objects of Process Costing : The main objects of process costing are as under :

- 1. Cost of each Process :** This method is followed so that the cost of each process may be ascertained easily.
- 2. Price of By-Product :** There are certain products, whose production becomes automatic on account of its main product and that is known as bye products, for example molasses in the production of sugar, oil cake during the production of oil. The valuation of these bye product is made in process account.
- 3. Profit on each Process :** In order to get the profits at each stage, process costing system is followed.
- 4. Treatment of Wastage :** During manufacture, there are certain wastages in each process. In order to find out the cost of wastage, the process accounts are prepared.

Main Features of Process Costing : The main features of process costing are as under :

1. The whole production activities are divided into various processes, which are limited to a certain operation.
2. Production records of each process are to be maintained so that it may show the quantity of output in each process.
3. The cost of normal loss in any process is included in the cost of the total units produced. In this way, the average cost per unit of that process is increased.
4. The work-in-progress in each process at the end of accounting period is calculated in terms of equivalent units of costing.
5. The items of cost can be identified with specific processes, all these are accumulated at regular intervals.
6. The total cost of each process is divided by the total units produced at the end of each period and the cost per unit of output is calculated.
7. The finished product of one process becomes the raw material to the next process and it continues, till the final products are completed. Thus the accumulated cost of one process is transferred to the next process at initial costs.

Application of Process Costing : This costing method is applied to the following industries :

1. **Manufacturing Industries :** Iron and steel, cement, paper, rubber, ceramics, automobile plants, ice, paints industries etc.
2. **Mining Industries :** Mineral oil, coal, sulphur, gold, iron, zinc, gas etc.
3. **Chemical Industries :** Chemicals, perfumery, soap, oil, cosmetics, medicines etc.
4. **Public Utility Concerns :** Electricity generation and distribution, water supply, gas supply etc.

General Principles of Process Costing : The general principles are as follows :

- (1) Both direct and indirect expenses are accumulated and classified according to the process.
- (2) The quantity or physical units of output in each process or department are recorded in the respective process accounts.
- (3) To determine the cost per unit of output the total cost of each process is divided by the total production at the end of each period.
- (4) The cost of one process is transferred to the next process and charged as an initial cost.
- (5) The computation of production and inventories is made in terms of complete units.
- (6) In the event of spoilage or loss in any process the units produced in the process bear the loss and the average cost per unit of that process is increased.

Main Advantages of Process Costing :

- (1) It helps the determination of cost in each process and of the final product at short intervals.
- (2) When the methods of production are standardised the average cost of manufacturing uniform products can be easily determined.
- (3) It involves a minimum of clerical effort and expense for cost records.

Disadvantages of Process Costing : The disadvantages/ demerits/ drawbacks/ limitations of process costing are as follows :

- (1) Average costs do not always show the true costs.
- (2) It is usually based on historical costs, hence, cost data are not available until the end of a period which implies delay and difficulty in cost control.

Q 16. Define the following and state how they should be dealt with in process costing?

(i) Joint-Product, and (ii) By-Product.

Or

What do you mean by the term 'Joint Products' and 'By-Products'? Write a brief note on them.

Or

Define and explain the term 'Joint Products' and 'By-Products'.

Ans. Meaning & Definition of Joint Products : In certain industries, two or more than two products of equal significance and value are produced simultaneously in a process. Such products are called 'Joint-products'. Motor, Spirit, Kerosene oil, Fuel oil, Lubricating oil, Wax, Tar and Asphalt are the examples of joint products produced from crude petroleum. Such products are common in agricultural product industries, extractive industries and chemical industries.

In 'Cost Accountants' Handbook' (edited by T. Lang), Joint Products have been defined as 'Two or More Products separated in the course of the same processing, operations, usually requiring further processing, each product being in such proportion that no single product can be designated as a major product.'

As such joint products are those which :

(a) are produced in the same process, with the same materials, (b) are separated in the course of same process, (c) are comparatively of the same importance and value, and (d) require further processing to finish them into more useful and valuable products.

By-Products : While producing, by various industries, of different commodities, alongwith the manufacturing of the products, several other products too are suddenly produced. These are known as the subsidiary or By-products. For instance, in the oil industry, the remains after crushing, the oilseeds become the chemical manures. In the dairy industry, alongwith butter, the cheese and other items are also produced.

'Cost Accountants' Handbook' defines by-products as Products recovered from material discarded in a main process, or from the production of some major product, where the material value is to be considered at the time of severance from the main product.'

As such, by-products are those which :

(a) Come forth as a result of the production of the main product, (b) are produced from the discarded material or scrap of the main product, and (d) possess a value which can be obtained by the sale thereof.

Apportionment of Joint Costs : The main point in the accounting for costs in case of joint products and by products is the apportionment of joint costs. There are common costs incurred in both the cases upto the point of separation, and after this split off point the independent costs are debited to each of them, more so, to the joint products where further processing is required in the form of independent products.

In relation to the apportionment of joint costs, there may be three possibilities :

- (a) Where two or more products are simultaneously produced and there is no by-product.
 (b) Where there are both joint products as well as by-products, and
 (c) Where there is a main product and a by-product.

The various methods of the distribution of the joint expenses are used to apportion the joint costs :

- (1) **Standard Cost Method** : In it, the joint cost is distributed on the basis of the prescribed standards.
- (2) **Average unit Cost Method** : Under this method the total joint cost is distributed on the basis of the number of total produced commodities.
- (3) **Market Price method** : Under this method, the pre-separation cost is apportioned on the basis of market price of the products. The joint cost is divided in the ratio of the market price of the products.
- (4) **Sale Value Method** : Under this method, the joint cost is apportioned in proportion to the total sale price of each product.

Q.17. What is Abnormal Gain? Explain with an example.

Ans. There are a number of exceptional cases when the actual process loss is comparatively less than the expected loss. It is called abnormal gain. It is also known as effectiveness.

The reporting of abnormal gains has to be done separately so that the management has knowledge of the gains. Basically, the treatment of abnormal gain is similar to abnormal loss. It is credited to the abnormal gain account. Further, the abnormal gain account is shown at the debit side of the concerned process account.

Calculation of Units of Abnormal Gain :

Units of Abnormal Gain = Actual Output - Expected Output

Expected Output = Input - Normal Loss

Example :

Using the given information compute:

- Normal Loss in units
- Abnormal Loss in units
- Abnormal gain in units

Input 4000 units and normal loss is 10% of the input.

Actual Output is :

- (1) Case A : 3400 units
- (2) Case B : 3800 units

Case A : Actual Output is 3400 units

Normal Loss = 10% of 4000 units = 400 units

Abnormal Loss = Expected Output - Actual Output

Expected Output = Input - Normal

Loss = 4000 - 400 = 3600 units

Abnormal Loss = 3600 - 3400 = 200 units

Case B : Actual Output is 3800 units

Normal Loss = 10% of 4000 units = 400 units

Abnormal Gain = Actual Output - Expected

Output = 3800 - 3600 = 200 units

Q.18. The following details are extracted from the costing records of an oil mill for the year ended 31st March, 2026.

Purchase 600 Tons of Coconuts : Rs. 2,40,000

	Crushing	Refining	Finishing
Cost of Labour	3,000	1,200	1,600
Electric Power	700	400	300
Sundry Materials	200	2,000	-
Repairs to Machinery	800	500	600
Steam	300	440	220
Factory Expenses	1,400	700	240

400 tons of crude oil were produced, 350 tons of oil were produced by Refining process, 350 tons of refined oil were finished for delivery.

Rs.
 Coconut sacks sold 500
 180 tons of coconut residue sold 12,000

Loss in weight in crushing : 20 tons ; 40 tons of by-product obtained from Refining process Rs. 7,200.

You are required to show the accounts in respect of each of the following stages of manufacture, for the purpose of arriving at the cost per ton of each process and the total cost per ton of the finished oil : (a) Coconut Crushing Process, (b) Refining Process, and (c) Finishing Process including Casking.

Solution :

Crushing Process A/c

Particulars	Tons	Amt. (Rs.)	Particulars	Tons	Amt. (Rs.)
To Khopra Purchased	600	2,40,000	By Loss in Weight	20	
To Cost of Labour		3,000	By Khopra reduced sold	180	12,000
To Electric Power		700	By Coconut sacks Sold	-	500
To Sundry Materials		200	Refining Process		
To Steam		800	(Crude Oil @ 584.75 per ton)	400	2,33,900
To Repairs to Machinery		300			
To Factory Expenses		1,400			
	600	2,46,400		600	2,46,400

Refining Process A/c

To Crushing Process %	400	2,33,900	By Loss in Weight	10	-
To Cost of Labour		1,200	By Bye-Product	40	7,200
To Electric power		400	By Finishing process A/c	350	2,31,940

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To Sundry Material	2,000	(Refined Oil @		
To Steam	500	Rs. 662.69 per ton)		
To Repairs of Machinery	440			
To Factory Expenses	700			
	400	2,39,140	400	2,39,140

Finishing Process A/c (including Casking)

To Refining Process A/c	350	2,31,940	By Loss in Weight	5	-
To Cost of Labour		1,600	By Finishing Stock	345	2,43,100
To Electric Power		300	(Finished Oil Stored		
			in Casks @ Rs. 704.64		
			per tons)		
To Steam		600			
To Repairs of Machinery		220			
To Factory Expenses		240			
To Cost of Casks		8,200			
	350	2,43,100		350	2,43,100

Q.19. A product passes through two distinct process 'A' and 'B' and then to finished stock. The output of 'A' passes direct to 'B' and that of 'B' passes to finished stock. From the following information you are required to prepare the process accounts :

	Process 'A'	Process 'B'
	Rs.	Rs.
Materials consumed	24,000	12,000
Direct labour	28,000	16,000
Manufacturing expenses	8,000	8,566
Input in Process 'A' (Unit)	10,000	-
Input in Process (value)	20,000	-
Output (units)	9,400	8,500
Normal wastage (% of input)	5%	10%
Value of normal wastage	40	50
(Per 100 unit)		

No opening or closing stock is held in process.

Solution: Process 'A' Account

Particulars	Units	Amt(Rs.)	Particulars	Units	Amt(Rs.)
To Units Introduced	10,000	20,000	By Normal Wastage	500	200
To Materials Consumed	-	24,000	By Abnormal Wastage	100	840
To Direct Labour	-	28,000	By Process 'B'	9,400	78,960
To Manufacturing Expenses	-	8,000			
	10,000	80,000		10,000	80,000

$$\text{Normal cost of Production per Unit} = \frac{80,000 - 230}{9,500} = \text{Rs. 8.40}$$

$$\text{Cost of Abnormal Wastage} = 100 \times 8.40 = \text{Rs. 840}$$

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Process 'B' Account

Particulars	Units	Rs.	Particulars	Units	Rs.
To Process 'A'	9,400	78,960	By Normal Wastage	940	470
To Materials Consumed	-	12,000	By Finished Stock	8,500	1,15,600
To Direct Labour	-	16,000			
To Manufacturing Exp.	-	8,566			
To Abnormal Effective	40	544			
	9,440	1,16,070		9,440	1,16,070

Calculation of Cost of Abnormal Effective :

$$\begin{aligned} \text{Normal Cost of Production} &= 1,15,526 - 470 = \text{Rs. 1,15,056} \\ \text{Expected Normal Production} &= 8,460 \text{ Units} \end{aligned}$$

$$\text{Cost per Unit} = \frac{1,15,056}{8,460} = \text{Rs. 13.60}$$

$$\text{Cost of Abnormal Effective} = 13.60 \times 40 = \text{Rs. 544}$$

Q.20. A particular brand of Phenyl passed through three important processes. During the month ended 31st January 2026, 600 Gross bottles were produced. The Cost books show the following informations:

	Process I	Process II	Process III
	Rs.	Rs.	Rs.
Materials	4,000	2,000	1,500
Labour	3,000	2,500	2,300
Direct Exp.	600	200	500
Cost of Bottles	-	2,030	-
Cost of Corks	-	-	325

The indirect expenses for the period were Rs. 1,600. The by-products of Process No. II were sold for Rs. 240. The residue of Process III sold for Rs. 125.50.

Prepare the account in respect of each of the process showing its cost and cost of production of the finished product per gross bottles.

Solution :

Process I

(Output : 600 Gross of Bottles)

Particulars	Rs.	Particulars	Rs.
To Materials	4,000.00	By Transfer to Process	
To Labour	3,000.00	No. II (Cost per gross of	
To Direct Expenses	600.00	bottles Rs.13.69	8215.38
		Approx.)	
To Indirect Expenses	615.35		
	8215.35		8215.38

Process No. II

Particulars	Rs.	Particulars	Rs.
To Transfer from Process No. I	8,215.38	By Sale of By-Product	240.00
To Materials	2,000.00	By Transfer to Process No. III (cost per gross of bottles Rs. 25.36 Approximately)	15,218.20
To Labour	2,500.00		
To Direct Expenses	200.00		
To Indirect Expenses	512.82		
To Cost of Bottles	2,030.00		
	15,458.20		15,458.20

Process No. III

Particulars	Rs.	Particulars	Rs.
To Transfer from Process No. II	15,218.20	By Sale of residue	125.50
To Materials	1,500.00	By Finished Products A/c	20,189.50
To Labour	2,300.00	(Cost per gross of bottles Rs. 33.65)	
To Direct Expenses	500.00		
To Indirect Expenses	471.80		
To Cost of corks	325.00		
	20,315.00		20,315.00

Note : Indirect expenses have been charged to three process in the labour ratio of 30:25:23.

Q.21. A product passes through two distinct processes 'A' and 'B' and then to finished stock. A's production passes to 'B' and B's production passes to finished stock. From the following information prepare the Process Account showing the cost per unit of each item.

	Process A (Rs.)	Process B (Rs.)
Materials	3,000	1,500
Wages	3,500	2,000
Manufacturing Exp.	1,000	1,000
Entering into Process 'A' (units)	5,000	
Entering into Process 'A' (costing)	2,500	
Production (Units)	4,700	4,150
	(Units)	(Units)
Normal wastage of the units entering the process	5%	10%
Scrap value of the wastage	Rs. 8	Rs. 10
There is no opening or closing stock in processes.		

Solution :

Dr.

Process 'A'

Particulars	Units	Rs.	Particulars	Units	Cr. Rs.
To Units Introduced	5,000	2,500	By Normal Wastage	250	20
To Materials		3,000	By Abnormal Wastage*	50	105
To Wages		3,500	To Transfer to Process B	4,700	9,875
To Manufacturing expenses		1,000			
	5,000	10,000		5,000	10,000

$$* \frac{9,980 \times 50}{4,750} = \text{Rs. } 105$$

Dr.

Process 'B'

Particulars	Units	Rs.	Particulars	Units	Cr. Rs.
To Trasfer from Process A	4,700	9,875	By Normal Wastage	470	47
To Materials		1,500	By Abnormal Wastage*	80	271
To Wages		2,000	By Transfer to Finished Stock	4,150	14,057
To Manufacturing expenses		1,000			
	4,700	14,375		4,700	14,375

$$* \frac{14,328 \times 80}{4,230} = \text{Rs. } 271$$

Q.22. In a manufacturing concern in a certain period A yields by products B and C. The joint expenses of the manufacture are : Materials Rs. 10,200; Labour Rs. 11,500 and on Cost Rs. 7,500. Subsequent expenses are as under :

	A (Rs.)	B (Rs.)	C (Rs.)
Materials	2,500	1,200	1,400
Labour	1,900	1,600	2,000
On Cost	1,500	900	1,050
	5,900	3,700	4,450

Selling Prices are - A = Rs. 30,000; B = Rs. 20,000 C = Rs. 15,000.

Estimated Profits on turnover are :

A = 40%, B = 30%, C = 25%

Show how would you apportion the joint expenses of manufacture and prepare necessary accounts.

Solution : Apportionment of Joint Expenses

	A (Rs.)	B (Rs.)	C (Rs.)
Selling price	30,000	20,000	15,000
Less: Estimate Profit	12,000	6,000	3,750
Total Costs	18,000	14,000	11,250
Less: Total Separate Expenses	5,900	3,700	4,450
Share in joint Expenses	12,100	10,300	6,800
(Rs. 29,200)			

A (Main Product) Account

Particulars	Rs.	Particulars	Rs.
To Materials : Joint 10,200 Separate 2,500	12,700	By By-product B (Share in joint exps.)	10,300
To Labour : Joint 11,500 Separate 1,900	13,400	By By-product C (Share in joint exps)	6,800
To On Cost : Joint 7,500 Separate 1,500	9,000	By Cost of Production c/d	18,000
	35,100		
To Cost of Production b/d	18,000	By Sales	30,000
To Profit	12,000		
	30,000		30,000

B (By-Product) Account

Particulars	Rs.	Particulars	Rs.
To A (Main Product) A/c	10,300	By Cost of Production c/d	14,000
To Materials	1,200		
To Labour	1,600		
To On cost	900		
	14,000		14,000
To Cost of Production b/d	14,000	By sales	20,000
To Profit	6,000		
	20,000		20,000

C (By-Product) Account

Particulars	Rs.	Particulars	Rs.
To A (Main Product) A/c	6,800	By Cost of Production c/d	11,250
To Materials	1,400		
To Labour	2,000		
To On cost	1,050		
	11,250		11,250
To Cost of Production b/d	11,250	By Sales	15,000
To Profit	3,750		
	15,000		15,000

Q.23. State in brief the objects and main features of Process Costing.

Ans. Objects of Process Costing : The main objects of process costing are as under :

1. **Cost of each Process :** This method is followed so that the cost of each process may be ascertained easily.

2. **Price of By-Product :** There are certain products, whose production becomes automatic on account of its main product and that is known as by-products, for example molasses in the production of sugar, oil cake during the production of oil. The valuation of these by-product is made in process account.

3. **Profit on each Process :** In order to get the profits at each stage, process costing system is followed.

4. **Treatment of Wastage :** During manufacture, there are certain wastages in each process. In order to find out the cost of wastage, the process accounts are prepared.

Main Features of Process Costing : The main features of process costing are as under :

1. The whole production activities are divided into various processes, which are limited to a certain operation.
2. Production records of each process are to be maintained so that it may show the quantity of output in each process.
3. The cost of normal loss in any process is included in the cost of the total units produced. In this way, the average cost per unit of that process is increased.
4. The work-in-progress in each process at the end of accounting period is calculated in terms of equivalent units of costing.
5. The items of cost can be identified with specific processes, all these are accumulated at regular intervals.

Q.24. What do you mean by 'Abnormal Wastage' and 'Abnormal Effectiveness'? Explain in brief.

Ans. Abnormal Wastage : This wastage is higher than the normal wastage and is called abnormal wastage. It may be on account of carelessness of management, defects in plants and machinery and natural calamities etc. This loss is not on account of normal process hence it can not be treated as loss. It does not affect the cost of production and it is shown in process A/c on cost basis. The following formula is applied for calculating abnormal wastages :

$$\frac{\text{Normal Cost}}{\text{Normal Output}} \times \text{Abnormal wastage in units}$$

Normal cost is arrived the normal wastages out of the total expenses. Normal output is arrived by deducting normal wastages in units. Units of Abnormal wastages are arrived by deducting the actual production out of the normal production.

Abnormal Effectiveness : If actual wastage is less than the normal wastages and this difference is known as abnormal effectiveness. The accounts of abnormal effectiveness is shown in the manner of abnormal wastages. It is shown on the debit side of the process account. The formula is as under :

Cost of Abnormal Effectiveness = $\frac{\text{Normal Cost}}{\text{Normal Output}} \times \text{Abnormal effective units}$

Q.25. Explain in brief the Principles of Process Costing and name the different types of Process Costing accounts.

Ans. General Principles of Process Costing : The general principles are as follows :

- (1) Both direct and indirect expenses are accumulated and classified according to the process.
- (2) The quantity or physical units of output in each process or department are recorded in the respective process accounts.
- (3) To determine the cost per unit of output the total cost of each process is divided by the total production at the end of each period.
- (4) The cost of one process is transferred to the next process and charged as an initial cost.
- (5) The computation of production and inventories is made in terms of complete units.
- (6) In the event of spoilage or loss in any process the units produced in the process bear the loss and the average cost per unit of that process is increased.

Different Types of Process Costing : The following may be the types of Process Costing :

- (1) Simple Process Accounts
- (2) Process Accounts in case of loss in weight and wastage.
- (3) Process Accounts in case of By-Products or Residue.
- (4) In each process keeping a portion of output in stock/warehouse for sale.
- (5) In case of normal wastage and Abnormal wastage/Effective.
- (6) In case of joint or By-Products.
- (7) Oil Refining Process.

UNIT-6

Short Answer type Questions :

Q.1. What is Operating Cost Accounting?

Ans. Operating Cost Accounting is a method of costing used for organizations that provide services rather than tangible goods. It involves determining and controlling the cost of operating a service, such as transportation, hotels, power supply, or hospitals. The total cost of providing the service is calculated and then divided by an appropriate cost unit (e.g., per passenger-km or per room-night). This method helps in pricing decisions, cost control, and profitability analysis. Since services are intangible, operating costing focuses on efficiency, capacity utilization, and overhead allocation for accurate cost determination.

Q.2. What is Service Costing and where is it applied?

Ans. Service Costing (a form of operating costing) is used to calculate the cost of services provided by organizations. Unlike manufacturing, there is no tangible output. Costs are classified as fixed and variable, and the total operating cost is divided by suitable cost units. Common areas where service costing is applied include transportation (roadways, airlines), hotels, canteens, power supply, hospitals, and educational institutions. It helps management fix appropriate service charges or fares, control operational costs, and ensure financial sustainability of service-based businesses or public utilities.

Q.3. What is Transportation Costing?

Ans. Transportation Costing is the application of operating costing to transportation services such as buses, trucks, airlines, and railways. The most common cost unit used is **per passenger-kilometer** or **per tonne-kilometer**. Costs are classified into fixed (e.g., insurance, depreciation), variable (e.g., fuel, wages), and semi-variable. Total cost is divided by total cost units to determine the cost per unit. Transportation costing helps in fare fixation, route planning, cost control, and profitability measurement. It is widely used in both public and private transport operations.

Q.4. What is Hotel Costing?

Ans. Hotel Costing refers to the process of determining the operating cost of providing accommodation and related services in hotels, lodges, and guest houses. The cost unit is usually **per room per night**. Costs are classified as fixed (e.g., salaries, depreciation, rent) and variable (e.g., laundry, electricity, food). Proper allocation of costs among different departments like rooms, restaurant, and housekeeping ensures accuracy. Hotel costing helps in deciding room tariffs, improving cost control, and increasing profitability through efficient utilization of capacity.

Q.5. What is Integral Cost Accounting?

Ans. Integral Cost Accounting is a system where both **financial accounts and cost accounts** are integrated into a single set of books. All transactions are recorded in one accounting system, avoiding duplication of work. This ensures that cost and financial data are reconciled automatically. It provides a complete and real-time picture of costs, revenues, and profits. Integral systems are efficient for organizations that want better coordination between costing and financial reporting, and it simplifies the preparation of financial statements and cost reports.

Q.6. What is Non-Integral Cost Accounting?

Ans. Non-Integral Cost Accounting is a system in which cost accounts and financial accounts are **maintained separately**. The cost accounting system records only cost-related transactions, while financial accounts record all financial transactions. At the end of the period, both sets of accounts are reconciled to ensure accuracy. This system allows for

detailed cost analysis, control, and decision-making, independent of financial reporting. It is commonly used in industries where cost control and product costing are highly emphasized. Although it requires more effort to maintain, it provides flexibility and detailed cost information.

Q.7. What is the importance of Operating Cost Accounting?

Ans. Operating Cost Accounting is important because it:

- Helps determine the **cost of providing services** accurately.
- Assists in **fixing fair prices or tariffs** for services.
- Enables **cost control** by identifying and managing fixed and variable costs.
- Supports **capacity utilization analysis**, ensuring efficient use of resources.
- Provides data for **decision-making**, budgeting, and performance evaluation.
- Helps in maintaining **financial sustainability** of service organizations.

It is particularly valuable in transportation, hospitality, healthcare, education, and other service sectors.

Long Answer type Questions :

Q.1 What is Operating costing? Discuss the nature of operating costing?

Ans. Operating or Service Costing : Cost Accounting has been traditionally associated with manufacturing companies. However, in the modern competitive market, cost accounting has been increasingly applied in service industries like banks, insurance companies, transportation organizations, electricity generating companies, hospitals, passenger transport and railways, hotels, road maintenance, educational institutions, road lighting, canteens, port trusts and several other service organizations. The costing method applied in these industries is known as 'Operating Costing'.

According to CIMA [London] operating costing is, 'that form of operating costing which applies where standardized services are provided either by an undertaking or by a service cost center within an undertaking'.

Nature of Operating Costing : The main objective of operating costing is to compute the cost of the services offered by the organization. For doing this, it is necessary to decide the unit of cost in such cases. The cost units vary from industry to industry. For example, in goods transport industry, cost per ton kilometer is to be ascertained while in case of passenger transport, cost per passenger kilometer is to be computed. Cost units used in different service units are explained in detail later in chapter. The next step is to collect and identify various costs under different headings. The headings used are :

- (i) Fixed or standing charges

- (ii) Semi-fixed or maintenance charges
- (iii) Variable or running charges.

One of the important features of operating costing is that mostly such costs are fixed in nature. For example, in case of passenger transport organization, most of the costs are fixed while few costs like diesel and oil are variable and dependent on the kilometers run.

Because of the diverse nature of activities carried out in service undertakings, the cost system used is obviously different from that of manufacturing concern.

Q.2. How costing is done in Transport industry? How the classification of costs is made in transport organisation?

Or

How costing is done in transport organisation? What is the significance of operating costs in transport organisation?

Ans. Costing in a transport industry consists of determining the operating cost of each vehicle and applying this cost to find out the cost per unit of service rendered by a vehicle. The cost unit is selected with proper care keeping in view the needs of each concern, the weight, bulk, volume and type of goods carried and distance covered in each trip. Transport undertakings include goods transport organizations as well as passenger transport organisations. The cost unit is either ton kilometer or passenger kilometer. The meaning is cost of carrying one ton over a distance of one kilometer or cost of carrying one passenger for a distance of one kilometer.

Collection of Costs : A log book is maintained for each vehicle to record details of trips made by the vehicle during a specified period of time. Log book is maintained usually on a daily basis. The details shown in the log book enables the management to make suitable allocation of vehicles, to avoid the duplicate trips, or to avoid idle running capacity. The log book also provides the information relating to the fuel consumed, distance travelled, no of hours travelled, chargeable kilo meters. The log book provides the data for proper allocation of cost and in this respect, these may be compared with the production details available in a manufacturing concern.

Classification of Costs : The costs of a transport organisation can be classified and accumulated under the following heads:-

(1) Fixed or stand-by costs : These costs which include garage charges, insurance, taxes, license, depreciation, wages of drivers, cleaner's salary, establishment cost of workshop and office. Out of the above some of the costs are directly identifiable for each vehicle such as license fee and some are apportioned such as office expenses.

(2) Maintenance Charges : These costs are in the nature of semi-variable nature includes expenditure on maintenance, repairs, tyres, tubes and other charges.

(3) **Operating and Running costs** : These costs are variable in nature, includes fuel, lubricating oil, wages of drivers / cleaners (if paid on per trip / kilometer). These costs can be easily identifiable with each of the vehicle.

Significance of Operating or Running Costs :

- Control of operating and running cost and avoidance of waste of fuel and other consumable material.
- Cost of running own vehicles may be compared with the hired or other forms of transport.
- Facilitates quotation of hiring rates to outside parties who ask for the transport service.
- If transport service is treated as a separate department or service cost center, the costs to be charged to the requesting department may be easily determined.
- Suitable information is obtained for efficient routing of vehicles.
- Cost of idle vehicles and lost running time is easily obtained.

Q.3. There are two warehouses for storing finished goods produced in a factory. Warehouse 'A' is at a distance of 10 kms. and Warehouse 'B' is at a distance of 15 kms from the factory. A fleet of 5 tonne lorries is engaged in transporting the finished goods from the factory. The records show that the lorries average a speed of 30 kms. per hour when running and regularly take 40 minutes to load at the factory. At warehouse 'A' unloading takes 30 minutes per load while at warehouse 'B' it takes 20 minutes per load.

Drivers' Wages, depreciation, insurance and taxes amount to Rs. 18 per hour operated. Fuel oil, tyres, repairs and maintenance cost Rs. 2.40 per kilometer. You are required to draw up a statement showing the cost per tonne kilometer of carrying the finished goods to the two warehouses.

Solution:

Statement showing computation of total cost and cost per tonne kilometer of carrying finished goods to warehouses:

Particulars		A	B
Time for travelling		40 Min	60 Min
Time for loading		40 Min	40 Min
Time for unloading		30 Min	20 Min
		110 Min	110 Min
Cost of Insurance, wages, tax, etc.	$[(110/60) \times 18]$	Rs. 33.00	
	$[(120/60) \times 18]$	Rs.	36.00
Fuel & oil etc.	$(20 \times 2.4) (30 \times 2.4)$	Rs. 48.00	72.00
Total Cost		Rs. 81.00	108.00
Tonne Kilometers	$(5 \times 10) (5 \times 15)$	Rs. 50.00	75.00
Cost per tonne KM		Rs. 1.62	1.44

Q.4. A transport service company is running 4 buses between two towns which are 50 miles apart. Seating capacity of each bus is 40 passengers. The following particulars were obtained from their books for April, 2026.

	Amount (Rs.)
Wages of Drivers, Conductors and Cleaners	2,400
Salaries of Office and Supervisory Staff	1,000
Diesel and oil and other oil	4,000
Repairs and Maintenance	800
Taxation, Insurance, etc.	1,600
Depreciation	2,600
Interest and Other Charges	2,000
	14,400

Actual passengers carried were 75% of the seating capacity. All the four buses ran on all days of the month. Each bus made one round trip per day. Find out the cost per passenger mile.

Solution: Computation of Cost per Passenger Mile :

Passenger miles = No. of buses $\times$ Distance $\times$ Round trip $\times$ No. of Passengers $\times$ No. of days in month $\times$ Capacity.

$$= 4 \times 50 \times 2 \times 40 \times 30 \times 75\%$$

$$= 3,60,000 \text{ miles}$$

$$\text{Cost per passenger mile} = 14,400 / 3,60,000$$

$$= \text{Rs. } 0.04$$

Q.5. Mr. Sohan Singh has started transport business with a fleet of 10 taxies. The various expenses incurred by him are given below:

- Cost of each taxi Rs. 75,000
- Salary of office Staff Rs. 1,500 p.m.
- Salary of Garage's Supervisor Rs. 2,000 p.m.
- Rent of Garage Rs. 1,000 p.m
- Drivers Salary (per taxi) Rs. 400 pm.
- Road Tax and Repairs per taxi Rs. 2,160 p.a.
- Insurance premium @ 4% of cost p.a.

The life of a taxi is 3,00,000 km. and at the end of which it is estimated to be sold at Rs. 15,000. A taxi runs on an average 4,000 Km. per month of which 20% it runs empty, petrol consumption 9 Km. per litre of petrol costing Rs. 6.30 per litre. Oil and other sundry expenses amount to Rs. 10 per 100 Km.

Calculate the effective cost of running a taxi per kilometre. If the hire charge is Rs. 1.80 per Kilometre, find out the profit that Mr. Sohan may expect to make in the first year of operation.

Solution: Statement Showing Computation of Effective Cost and Profit for the Year

Particulars	Amount (Rs.)	Amount (Rs.)
Fixed expenses:		
Salary of staff	1,500	
Salary of garage supervisor	2,000	
Rent of garage	1,000	
Driver Salary (10 × 400)	4,000	
Road tax and repairs (2,160 × 10/12)	1,800	12,800
Insurance premium (75,000 × 4 % × 10/12)	2,500	
Fixed cost of 10 taxi's per month		
Cost per taxi = Rs. 12,800/10 = Rs. 1,280		0.32
Cost per km = 1280/4,000 = 0.32		
Running Costs:		
Depreciation [(75,000-15,000) / 3,00,000]		0.20
Petrol (6.3/9)		0.70
Oil & sundry expenses (10/100)		0.10
		1.32
Effective cost per Km = 1.32 × (100/80)		1.65

$$\text{Profit for year} = (1.80 - 1.65) \times 10 \times 3,200 \times 12$$

$$= \text{Rs. } 57,600$$

Q.6. Janata Transport Co. has been given a route 20 km. long for running buses. The company has a fleet of 10 buses each costing Rs. 50,000 and having a life of 5 years without any scrap value.

From the following estimated expenditure and other details calculate the bus fare to be charged from each passenger.

	3 % p.a.
(i) Insurance charges	Rs. 1,000
(ii) Annual tax for each bus	Rs. 1,000
(iii) Total garage charges	Rs. 150 p.m.
(iv) Drivers' salary for each bus	Rs. 100 p.m.
(v) Conductor's salary for each bus	Rs. 1,000
(vi) Annual repairs to each bus	
(vii) Commission to be shared by the driver and conductor equally : 10% of the takings	
(viii) Cost of stationary	Rs. 500 p.m.
(ix) Manager's salary	Rs. 2,000 p.m.
(x) Accountant's salary	Rs. 1,500 p.m.
(xii) Petrol and oil	Rs. 25 per 100 km

Each bus will make 3 round trips carrying on an average 40 passengers on each trip. The bus will run on an average for 25 days in a month. Assuming 15% profit on takings, calculate, the bus fare to be charged from each passenger.

Solution:

Particulars	Amount (Rs.)
Insurance (50,000 × 3% × 10/12)	1,250
Tax (1,000 × 10/12)	833.33
Garage charges	1,000
Drivers salary (150 × 10)	1,500
Conductor salary (100 × 10)	1,000
Repairs (1,000 × 10/12)	833.33
Cost of stationary	500
Managers salary	2,000
Accountant salary	1,500
Depreciation (50,000 × 10/5 × 1/12)	833.33
Petrol * (30,000/100) × 25	7,500
Commission of conductor & driver 35,000 × (10/100)	3,500
	29,750
(+) Profit @ 15% on takings (35,000 × 15/100)	5,250
	35,000

$$* 10 \times 20 \times 3 \times 2 \times 25 = 30,000$$

Let 'X' be the takings

$$X = 26,250 + (10/100 X) + (15/100 X)$$

$$100 X = 26,25,000 + 25 X$$

$$\Rightarrow X = 35,000$$

$$\text{Fare per passenger Km} = 35,000 / (30,000 \times 40)$$

$$= 0.0292 = \text{Rs. } 0.03$$

Q.7. Union Transport Company supplies the following details in respect of a truck of 5 tonne capacity:

Cost of truck	Rs. 90,000
Estimated life	10 years
Diesel, oil, grease	Rs. 15 per trip each way
Repairs and maintenance	Rs. 500 p.m.
Driver's wages	Rs. 500 p.m.
Cleaner's wages	Rs. 250 p.m.
Insurance	Rs. 4,800 per year
Tax	Rs. 2,400 per year
General supervision charges	Rs. 4,800 per year

The truck carries goods to and from the city covering a distance of 50 kms. each way.

On outward trip freight is available to the extent of full capacity and on return 20% of capacity.

Assuming that the truck runs on an average 25 days a month, work out:

(a) Operating cost tonne-km.

(b) Rate for tonne per trip that the company should charge if a profit of 50% on freight is to be earned.

Solution:

Particulars	Amount (Rs.)
Repairs & Maintenance	500
Drivers wages	500
Cleaners wages	250
Insurance	400
Tax	200
Supervision charge	400
Depreciation $[(90,000/10) \times (1/12)]$	750
Diesel, oil, grease $(15 \times 2 \times 25)$	750
	3,750
(+) 50% profit on freight (100% on cost)	3,750
	7,500

$$\text{Tonne Kms} = 25 [(50 \times 5) + (20/100 \times 50 \times 5)] = 7,500$$

$$\text{Cost per tonne km} = 3,750 / 7,500 = 0.50$$

$$(+) \text{ Profit @ 50\% on freight} = 1.00$$

HOTELS

Service costing is an effective tool in respect of hotel industry which run on commercial basis. Hence it is necessary to compute the cost in order to determine the price of various services provided by the hotel and to find out the profit or loss at the end of a particular period.

In this case, the costs associated with different services offered may be identified and cost per unit may be worked out. The cost unit may be Guest-day or room-day. For calculation of cost per guest day or room day, estimated occupancy rate at different point of time are taken into account, for example, peak season or lean season.

Q.8. XYZ Ltd. runs a holiday home. For this purpose, it has hired a building at a rent of Rs.10,000 per month along with 5% of total taking. It has three types of suites for its customers, viz., single room, double rooms and triple rooms.

Following information is available:

Type of suite	Number	Percentage of occupancy
Single room	100	100%
Double rooms	50	80%
Triple rooms	30	60%

The rent of double rooms suite is to be fixed at 2.5 times of the single room suite and that of triple rooms suite as twice of the double rooms suite.

The other expenses for the year 2025 are as follows:

Particulars	Rs.
Staff salaries	14,25,000
Room attendants' wages	4,50,000
Lighting, heating and power	2,15,000
Repairs and renovation	1,23,500
Laundry charges	80,500
Interior decoration	74,000
Sundries	1,53,000

Provide profit @ 20% on total taking and assume 360 days in a year.

Calculate the rent to be charged for each type of suite.

Solution:

Working Notes : Total equivalent single room suites

Nature of suite	Occupancy (Room-days)	Equivalent single room suite (Room-days)
Single room suites	36,000 (100 rooms $\times$ 360 days $\times$ 100%)	36,000 (36,000 $\times$ 1)
Double rooms suites	14,400 (50 rooms $\times$ 360 days $\times$ 80%)	36,000 (14,400 $\times$ 2.5)
Triple rooms suites	6,480 (30 rooms $\times$ 360 days $\times$ 60%)	32,400 (6,480 $\times$ 5)
		1,04,400

Statement of total cost

Particulars	Rs.
Staff salaries	14,25,000
Room attendants' wages	4,50,000
Lighting, heating and power	2,15,000
Repairs and renovation	1,23,500
Laundry charges	80,500
Interior decoration	74,000
Sundries	1,53,000
	25,21,000
Building rent {(Rs.10,000 $\times$ 12 months) + 5% on total taking}	1,20,000 + 5% on total takings
Total cost	26,41,000 + 5% on total takings

Profit is 20% of total takings

$\therefore$ Total takings = Rs. 26,41,000 + 25% (5% + 20%) of total takings

Let R be rent for single room suite

Then $1,04,400 R = 26,41,000 + (0.25 \times 1,04,400 R)$

Or, $1,04,400 R = 26,41,000 + 26,100 R$

Or, $78,300 R = 26,41,000$

Or, $R = \text{Rs. } 33.73$

Rent to be charged:

Rent to be charged for single room suite = Rs. 33.73

Rent for double rooms suites = $\text{Rs. } 33.73 \times 2.5 = \text{Rs. } 84.33$

Rent for triple room suites = $\text{Rs. } 33.73 \times 5 = \text{Rs. } 168.65$

Q.9. Angel Holiday Home runs in a small hill station with 100 single rooms. The home offers concessional rates during six off season months in a year. During this period, half of the full room rent is charged. The management's profit margin is targeted at 20% of the room rent. The following are the cost estimates and other details for the year ending on 31st March 2018 [Assume a month as 30 days].

(i) Occupancy during the season is 80% while in the off-season it is 40% only.

(ii) Total investment in the home is Rs. 200 lakhs of which 80% relate to buildings and balance for furniture and equipment.

(iii) Expenses:

Particulars	Rs.
Staff salary [Excluding room attendants]	5,50,000
Repairs to building	2,61,000
Laundry charges	80,000
Interior	1,75,000
Miscellaneous expenses	1,90,800

Annual depreciation is to be provided for buildings @ 5% and on furniture and equipment @ 15% on straight-line basis.

(v) Room attendants are paid Rs. 10 per room day on the basis of occupancy of the rooms in a month.

(vi) Monthly lighting charges are Rs. 120 per room, except in four months in winter when it is Rs. 30 per room and this cost is on the basis of full occupancy for a month.

You are required to work out the room rent chargeable per day both during the season and the off-season months on the basis of the foregoing information.

Solution:

Working Notes

Total Room days in a year

Season	Occupancy (Room-days)	Equivalent (Full Room charge days)
Season - 80% occupancy	100 Rooms $\times$ 80% $\times$ 6 months $\times$ 30 days in a month = 14,400 Room Days	14,400 Room Days $\times$ 100% = 14,400
Off-season - 40% occupancy	100 Rooms $\times$ 40% $\times$ 6 months $\times$ 30 days in a month	7,200 Room Days $\times$ 50% = 3,600

Total Room days	= 7,200 Room Days	
	14,400 + 7,200 = 21,600	
Room Days		18,000 Full Room days

Lighting Charges : It is given in the question that lighting charges for 8 months is Rs. 120 per month and during winter season of 4 months it is Rs. 30 per month. Further it is also given that peak season is 6 months and off season is 6 months.

Being Hill station, winter season is to be considered as part of off-season. Hence, the non-winter season of 8 months include: peak season of 6 months and off-season of 2 months.

Hence, the lighting charges are calculated as follows:

Season	Occupancy (Room-days)
Season & Non-winter - 80% Occupancy	100 Rooms $\times$ 80% $\times$ 6 months $\times$ Rs. 120 per month = Rs. 57,600
Off-season & Non-winter - 40% Occupancy (8 - 6 months)	100 Rooms $\times$ 40% $\times$ 2 months $\times$ Rs. 120 per month = Rs. 9,600
Off-season & winter - 40% Occupancy months	100 Rooms $\times$ 40% $\times$ 4 months $\times$ Rs. 30 per month = Rs. 4,800
Total Lighting charges	Rs. 57,600 + 9,600 + 4,800 = Rs. 72,000

Statement of total cost

Particulars	Rs.
Staff salary	5,50,000
Repairs to building	2,61,000
Laundry & Linen	80,000
Interior	1,75,000
Sundries Expenses	1,90,800
Depreciation on Building (Rs. 200 Lakhs $\times$ 80% $\times$ 5%)	8,00,000
Depreciation on Furniture & Equipment (Rs. 200 Lakhs $\times$ 20% $\times$ 15%)	6,00,000
Room attendant's wages (Rs. 10 per Room Day for 10,800 Room Days)	1,08,000
Lighting charges	72,000
Total cost	28,36,800
Add: Profit Margin (20% on Room rent or 25% on Cost)	7,09,200
Total Rent to be charged	35,46,000

Calculation of Room Rent per day:

Total Cost/ Equivalent Full Room days = $\text{Rs. } 35,46,000 / 18,000$
= Rs. 197.00

Room Rent during Season - Rs. 197

Room Rent during Off-season = $\text{Rs. } 197 \times 50\% = \text{Rs. } 98.50$

Q.10. What is Integrated (integral) and non-integrated (non-integral) system of cost accounting? Describe its features and advantages respectively.

Ans. Integrated (integral) and non-integrated (non-integral) cost accounting are two systems for maintaining financial and cost records. The key distinction lies in whether the financial and cost records are kept in a single combined set of books or in two separate sets.

Integrated (Integral) cost accounting : In an integrated system, a single, comprehensive set of books records both financial and cost transactions. This approach avoids duplicating work and eliminates the need to reconcile the profit figure derived from cost accounts with that from financial accounts.

Features :

- (1) **One set of books :** The financial and cost accounts are combined into one master ledger.
- (2) **No reconciliation :** Since there is only one set of accounts and one resulting profit or loss, there is no need to reconcile separate figures.
- (3) **No cost ledger control account :** This system does not use a "General Ledger Adjustment Account" or "Cost Ledger Control Account" to link financial and cost data, as the financial accounts are already integrated.
- (4) **Functional classification :** The transactions are classified based on their function (e.g., office expenses, factory expenses) rather than just their nature, allowing for more detailed cost analysis.
- (5) **Automation-friendly :** This system is well-suited for computerization and mechanized accounting, which reduces manual work and the potential for errors.

Advantages :

- (1) Reduces duplication of entries and clerical effort.
- (2) Saves time and money.
- (3) Provides accurate and reliable data because it avoids the need for reconciliation adjustments.
- (4) Enables quicker reporting and better coordination between the finance and costing departments.

Non-integrated (Non-integral) cost accounting : In a non-integrated system, a company maintains two separate and distinct sets of ledgers: one for cost accounts and one for financial accounts. This system is also known as "Cost Control Accounts" or an "Interlocking" system.

Features :

- (1) **Two sets of books :** Separate books are maintained by the cost and financial accountants.
- (2) **Reconciliation is required :** At the end of an accounting period, a reconciliation statement must be prepared to explain any differences between the profit shown in the cost books and the financial books.
- (3) **Use of control accounts :** The cost ledger uses a "Cost Ledger Control Account" (or "General Ledger Adjustment Account") to complete the double-entry for items that also appear in the financial accounts, such as purchases and sales.

- (4) **Notional expenses :** This system can accommodate notional expenses (e.g., interest on capital or notional rent) for costing purposes, which are not included in the financial accounts.
- (5) **Fewer accounts :** The cost accounts contain fewer accounts than the financial accounts, as they exclude personal accounts (debtors and creditors) and many non-cost items.

Advantages :

- (1) Offers a simpler and more straightforward approach that is easier to manage, particularly for smaller organizations.
- (2) Allows management to focus exclusively on production-related costs without the added complexity of financial accounts.
- (3) Provides a strong internal check by comparing two independent sets of profit and loss calculations.

Q 11. What is the Purpose of reconciling Cost and Financial Accounts? Indicate the possible sources of difference between them.

Or

What is Cost Reconciliation Statement? Give its main Objects.

Or

What do you know by Cost Reconciliation Statement? Mention its main objects and give a proforma for it.

Or

What is a 'Cost Reconciliation Statement.' Discuss the method of preparing Reconciliation Statement.

Or

State the causes of differences in the amount of profit shown by Cost and Financial Accounts.

Or

Indicate the reasons why is it necessary for the cost and financial accounts of an organisation to be reconciled and Explain the main reasons of difference which would you enter in such accounts.

Or

Why is it necessary to Reconcile Cost and Financial Accounts?

Ans. Cost Accounts and Financial Accounts are kept quite separate in a manufacturing concern. The objects and the methods of maintaining both set of books differ. Cost Accounts disclose Cost per unit or job or cost at different stages of completion while Financial Accounts record all the aspects of a business. In case of Financial Accounts, overheads are absorbed in the accounts on the accrual basis (i.e., paid or outstanding). However, in case of Cost Accounts, to derive prompt information (for cost control and cost reduction) the overheads are absorbed at a predetermined

rate. The different assumptions (in few cases) while recording transactions results into different profit disclosed by both the books. The books are maintained by the same concern and therefore, for checking accuracy of the books, the reconciliation of profits is essential.

Objects or Purpose or Need of preparing the reconciliation of Cost and Financial Accounts :

Reconciliation of Cost and Financial Accounts is necessary for the following reasons :

- (1) To find out the reasons for the difference between the profit of cost and financial accounts.
- (2) To ascertain the mathematical accuracy of the cost and financial accounts.
- (3) To ensure that no income or expenditure item has been left over.
- (4) To ensure that no under or over-recovery of overheads is made.
- (5) To test the reliability of cost figures.

(Note : For Reasons of difference please refer to the answer of Q. 16)

Proforma of Cost Reconciliation Statement : The specimen of cost reconciliation statement is given as under :

	Rs.	Rs.
Profit as per cost A/cs		
Add : Over charged expenses in cost accounts		
Less : Under charged items in cost accounts		
Profits as per financial A/cs.		

Q. 12. Mention the main reasons of differences in the profits of Cost Accounts and Financial Accounts.

Or

What are the various reasons of differences in profits of both the Accounts? Explain.

Or

State briefly the Causes of difference between profits shown by Financial Accounts and Cost Accounts.

Ans. Reasons of Differences in two Profits :

(A) Items which are included in Cost Accounts only :

- (i) Interest on self owned capital though not incurred. It is included in cost.
- (ii) Salary for the proprietor, which has not been charged.
- (iii) Rent charges of premises owned by the firm.

(B) Items which are included in Financial Account only :

(I) Purely Financial Charges :

- (i) Losses on the sale of building, investment etc.
- (ii) Discounts on issue of debentures etc.
- (iii) Damages payable at law.
- (iv) Losses due to scrapping of machinery.

- (v) Expenses of company's office.
- (vi) Stamp duty and expenses on issue of shares.
- (vii) Interest on Bank loans, debentures etc.
- (viii) Penalties payable at law.
- (ix) Remuneration paid to the proprietor in excess of fair reward.

(II) Purely Financial Incomes :

- (i) Rents receivables.
- (ii) Dividends received.
- (iii) Interest, dividends received on investments.
- (iv) Brokerage received.
- (v) Interest received on bank deposits, loans etc.
- (vi) Fees received on issue of shares etc.
- (vii) Profits made on the sale of fixed assets.

(III) Appropriation of Profits :

- (i) Charitable donations.
- (ii) Transfer to General Reserves.
- (iii) Additional provision for depreciation of buildings etc.
- (iv) Capital expenditure charged to revenue.
- (v) Taxes on income and profits.
- (vi) Dividend paid.
- (vii) Amounts written off goodwill, debenture discount etc.

(C) Adoption of Different Bases of Valuation of Stock : Due to differences in valuation of stock, the net results of the two sets of books vary and call for reconciliation.

(D) Under and over Absorption of Overhead Charges : In costing, the recovery of overheads is always based on an estimate ratio, while in financial accounting, the actual expenses of overheads are recorded, with the result that there is either under absorption or over absorption of overheads.

(E) Other Reasons :

(i) Different Methods of Charging Depreciation : The methods of charging depreciation may be different in cost books as well as in financial books on account of it, the profits shown by both the books will not tally.

(ii) Abnormal Gains and Losses : Abnormal gains and losses may be taken to costing profit or loss account or it may be excluded from the cost accounts altogether. On account of it there may be difference in profits as shown by cost accounts and financial accounts.

Method of Preparing Reconciliation Statement :

The following steps should be taken to prepare the Reconciliation Statement :

- (1) The points of difference between the two accounts should be determined and noted separately.
- (2) Taking the profit of cost accounts to be the starting point:

(i) Regarding Expenses items :

Add : Items over charged in cost accounts.

Deduct : Items under charged in cost accounts.

If an expenses has been overcharged in cost accounts, it means that the profit of cost accounts is decreased to that extent. So the excess expenses are to be added to reconcile. Same is the analogy applicable to under-charged expenses which should be deducted.

(ii) Regarding Income Items :

Add : Items under-recorded in Cost accounts

Less : Items over-recorded in cost accounts.

If an item of income has been under-recorded in cost accounts, it means the profit stands decreased to that extent in cost accounts. So that should be added to reconcile.

(iii) Regarding Stock Valuation :**(a) Opening stock :**

Add : Over valuation in Cost Accounts.

Less: Under valuation in Cost Accounts

(b) Closing Stock.

Add : Under valuation in Cost Accounts

Less : Over valuation in Cost Accounts.

(iv) Regarding Depreciation :

Add : Over-charge in cost accounts.

Less : Under-charge in cost accounts.

(3) Reverse the treatment prescribed above under point No. 2. if profit of financial accounts is taken to be the starting point.

Q.13. The Profit of a Manufacturing Company as on 31st December, 2025 as per financial accounts are Rs. 1,28,755 and as per cost accounts Rs. 1,72,400. The following differences are extracted from Comprising the both :

	Rs.
Work overheads undercharged in Cost Accounts	3,120
Office overheads overcharged in Cost Accounts	1,700
Depreciation as per Cost Accounts	12,500
Depreciation as per financial Accounts	11,200
Interest on investment not shown in Cost Accounts	8,000
Obsolescence only shown in financial accounts	5,700
Provision for Income Tax in Financial Accounts	40,300
Bank Interest and Dividend Received	750
Depreciation in Stock shown in Financial Accounts	6,750
Stores Adjustment Credited in Cost Accounts	475

Prepare a Profit & Loss reconciliation Statement to reconcile the profits shown by two sets of books.

Solution : Profit and Loss Reconciliation Statement
(as on 31st December, 2025)

Particulars	Rs.	Rs.
Profit as per Cost Statement		1,72,400
Less : Works overhead under recovered	3,120	
Obsolescence not charged	5,700	
Income Tax not provided	40,300	
Depreciation in stock values	6,750	55,870
		1,16,530
Add : Excess depreciation charged	1,300	
Excess office Exp. Charged	1,700	
Interest on investment	8,000	
Bank interest and dividend	750	
Stores Adjustments	475	12,225
Profits as per financial records		1,28,753

Q.14. The costing profit of a concern is Rs. 23,063 and profit of financial records is Rs. 16,624. Prepare a reconciliation statement from the following :

Profit & Loss Account

	Rs.		Rs.
To Opening Stock	2,47,179	By Closing Stocks	75,121
To Purchases	82,154	By Sales	3,46,500
To Direct Wages	23,133		
To Factory Overhead	20,826		
To Gross Profit c/d	48,329		
	4,21,621		4,21,621
To Administration	9,845	By Gross Profits b/d	48,329
To Selling Exp.	22,176	By Sundry Income	316
To Net Profit	16,624		
	48,645		48,645

The costing records show :

1. A stock ledger closing balance Rs. 78,197
2. A direct wages absorption account with a closing credit balance of Rs. 24,867.
3. A factory overhead absorption account with a closing credit balance of Rs. 19,714.
4. Administration expenses calculated at 3% of sales.
5. Selling price includes 5% for selling expenses.
6. No mention of sundry income.

Reconciliation Statement			
Ans.	Particulars	Rs.	Rs.
	Profit as per Cost Books		23,063
Add :			
	1. Direct wages overcharged in Cost Accounts		1,734
	2. Administrative overhead over recovered in Cost A/cs		550
	3. Sundry income not taken into account in Cost A/cs		316
Less :			
	1. Difference in valuation of Closing Stock	3,076	
	2. Factory overheads under recovered in Cost A/c	1,112	
	3. Selling Expenses under recovered in Cost A/c	4,851	
	4. Profit as per Financial Books	16,624	
		25,663	25,663

Q.15. From the following Profit & Loss Account draw up a Memorandum Reconciliation Account :

Profit & Loss Accounts			
	Rs.		Rs.
To Office Salary	11,282	By Gross Profits	54,648
To Office Expenses	6,514	By Dividend Received	400
To Salesman Salaries	4,922	By Interest on Bank Deposits	150
To Sales Expenses	9,304		
To Distribution Expenses	2,990		
To Loss on Sale of Machinery	1,950		
To Fines	200		
To Discount on Debentures	100		
To Net Profit	17,936		
	55,198		55,198
To Income Tax	8,000	By Net Profit	17,936
To Reserve	1,000		
To Dividend	4,000		
To Balance c/d	4,936		
	17,936		17,936

The cost accountant has ascertained a profit of Rs. 19,636.

Memorandum Reconciliation Statement			
Ans.	Rs.		Rs.
To Incomes excluded from Cost Accounts :		By Profit as per P/L A/c	4,936
1. Dividend Received	400	By Items not charged in Cost Accounts :	

2. Interest on Bank Deposits	150	1. Loss on Sale of Machinery	1,950
3. Profit as per Cost A/cs.	19,636	2. Fines	200
		3. Discount on Debentures	100
		4. Income Tax	8,000
		5. Reserve	1,000
		6. Dividend	4,000
	20,186		20,186

Reconciliation Statement		
Particulars	Rs.	Rs.
Profit as per Financial A/cs		17,936
Add : Items not charged in Cost A/cs.		
1. Fines		200
2. Discount on Debentures		100
3. Loss on Sale of Machinery		1,950
Less : Items not credited in Cost A/cs.		
1. Dividend Received	400	
2. Interest on Bank Deposit	150	
Profit as per Cost Accounts	19,636	
	20,186	20,186

Q.16. The Cost Book of Ramesh Ltd. Show a profit of Rs. 84,000. On the basis of the information given below prepare a reconciliation statement and show the profit as per Financial Books.

	Cost Books	Financial Books
	Rs.	Rs.
(a) Opening Stock of Materials	15,800	16,300
(b) Opening Work-in-Progress	9,000	10,000
(c) Closing Stock of Materials	16,000	15,000
(d) Closing Work-in-Progress	9,000	8,000
(e) Interest and Dividend received		850
(f) Loss on Sale of Machinery		850

Keep in view the following additional information :

- Amount of interest Rs. 2,000 is not included in financial books.
- Goodwill Rs. 5,000 has been written off.
- Actual overheads Rs. 56,500 but recovered Rs. 60,000.

Reconciliation Statement

Solution :		Rs.	Rs.
Profits as per Cost Books			84,000
Add: (i) Overheads over-recovered in Cost Books			
	Rs.		
	Cost Books	60,000	
	Financial Books	56,500	
		3,500	
(ii) Interest & Dividend excluded from Cost Books		850	4,350
			88,350
Less: (i) Under valuation of opening Stock in Cost Books			
(a) Raw-materials (Rs. 16,300-15,800) =	500	1,500	
(b) Work in progress (Rs. 10,000-9,000) =	1,000		
(ii) Over valuation of Closing Stock in Cost Books :			
(a) Raw- Materials (Rs. 16,000 : 15,000) =	1,000	2,000	
(b) Work-in-Progress (Rs. 9,000 : 8,000) =	1,000		
(iii) Losses not shown in Cost Books :			
(a) Loss on Sale of Machinery	850		
(b) Goodwill written off	5,000	5,850	9,350
Profit as per financial Books			79,000

Q.17. The net profit shown by financial accounts of a Company amounted to Rs. 28,550 whilst the profit disclosed by company's Cost Accounts for that period was Rs. 38,660. On reconciliation the following differences were discovered :

- The following items were not charged in Cost Accounts :
(a) Director's Fees (Dr.) Rs. 650; (b) Bank Interest (Cr.) Rs. 30; (c) Income Tax (Dr.) Rs. 8,300.
- A provision for Bad and Doubtful debts made in Financial Books did not occur in Cost Accounts Rs. 570.
- Overheads in Cost Accounts absorbed were Rs. 8,500 while the actualise as shown by financial books were Rs. 8,320.
- The expenditure on a work started during the year on a new factory was Rs. 16,000. Depreciation of 5% was provided in Financial Books.

Prepare a statement reconciling the figures of profit as shown by the two sets of Books.

Solution : Reconciliation Statement

	Rs.	Rs.
Profit as shown by Cost Accounts		
Add: Overheads overcharged in Cost accounts:		
Cost Books	8,500	
Financial Books	8,320	
	180	
Bank Interest credited in Financial Books	30	210
		38,870
Less : Expenses not charged in Cost Accounts :		
Income Tax	8,300	
Director's Fees	650	
Bad & Doubts full Debts Reserve	570	
Depreciation @ 5% on Rs. 16,000	800	10,320
Profit as per financial Books		28,550

Objective Type Questions

- Standard costing helps in :
(a) controlling prices (b) measuring efficiency
(c) increasing profits (d) reducing losses
- Ans. (b) measuring efficiency
- It is an advantage of Cost Accounting - Analytical and comparative Study of production Cost :
(a) Advantage to Management and Producer
(b) Advantage to Employees
(c) Advantage to Investors (d) None of the above
- Ans. (a) Advantage to Management and Producer
- In Cost Plus Method of Cost Accounting :
(a) The payment is made after adding a certain percentage of profit in Total Cost
(b) The payment is made after deducting a certain percentage of profit in Total Cost
(c) The value of Total Cost is paid
(d) The total purchase price is paid.
- Ans. (a) The payment is made after adding a certain percentage of profit in Total Cost
- Operating Costing method may be used in :
(a) Sugar Industry
(b) Wherever various managerial decisions are taken
(c) In Electricity, gas and transport Company
(d) In Cloth Industry
- Ans. (c) In Electricity, gas and Transport Company.
- Process Costing system is adopted :
(a) To ascertain the cost of contract

- (b) When an article passes through two or more stages to reach the stage of finished product
 (c) In such firms which render services to the customers
 (d) Where the cost of production in each department is ascertained
- Ans. (b) When an article passes through two or more stages to reach the stage of finished product
- (6) Cost unit in an Electric Supply Company will be :
 (a) Per Ton Kilometer (b) Per Kilo-watt per hour
 (c) Per Cubic meter (d) Per thousand litres
- Ans. (b) Per Kilo-watt per hour
- (7) Comparative study of.....is essential for managerial control :
 (a) Profit & Loss Account (b) Only Expenses
 (c) Cost (d) All the above
- Ans. (c) Cost
- (8) Which method of Costing is used in transport Company? :
 (a) Unit Costing (b) Contract Costing
 (c) Process Costing (d) Operating Costing
- Ans. (d) Operating Costing
- (9) The technique and process of ascertaining costs is called :
 (a) Cost Accounting (b) Cost Accounting
 (c) Costing (d) All the above
- Ans. (c) Costing
- (10) In India Institute of Cost and Works was established in :
 (a) 1956 (b) 1950
 (c) 1932 (d) 1944
- Ans. (d) 1944
- (11) Cost Accounting is used :
 (a) Only in manufacturing industries
 (b) Only in business concerns
 (c) Only in Transport in industry (d) In all the above
- Ans. (d) In all the above
- (12) The Planning department determines the demand for a particular item and prepares a list of materials and then sends it to the purchase department in the form of :
 (a) Purchase order (b) Purchase Requisition
 (c) Quotation (d) Bill
- Ans. (b) Purchase Requisition
- (13) In Bin Card the entry is made :
 (a) For the goods received together with date of receipt
 (b) For the goods issued together with the date of issue
 (c) For the balance of the goods together with the date
 (d) For all the above
- Ans. (d) For all the above

- (14) For determining the issue price, the quantity of material available in the stock and the price both are considered under :
 (a) Simple Average Method (b) Cost Price Method
 (c) Standard Price Method (d) Weighted Average Method

Ans. (d) Weighted Average Method

- (15) The order in which the materials are received in the store are issued at cost price in the same order is called:
 (a) LIFO (b) FIFO
 (c) HIFO (d) None of the above

Ans. (b) FIFO

- (16) The formula given by Wilson for calculating Economic Order Quantity is :

$$(a) EOQ = \frac{2AO}{C} \quad (b) EOQ = \frac{C}{2AO}$$

$$(c) EOQ = \sqrt{\frac{2AO}{C}} \quad (d) EOQ = \sqrt{\frac{C}{2AO}}$$

Ans. (c) $EOQ = \sqrt{\frac{2AO}{C}}$

- (17) In standard costing, standard cost is compared with

- (a) variable cost (b) actual cost
 (c) estimated cost (d) fixed cost

Ans. (b) actual cost

- (18) Under the general conditions of production idle time which is incidental to production is called :

- (a) Abnormal Idle Time (b) Normal Idle Time
 (c) Complex Idle Time (d) None of the above

Ans. (b) Normal Idle Time

- (19) Idle times due to strikes, lock-outs, long spell of electric failure or machine breakdown are to be considered as :

- (a) Normal Idle time (b) Incidental Idletime
 (c) Abnormal Idle Time (d) All the above

Ans. (c) Abnormal Idle Time

- (20) The worker is paid according to the volume of work done by him irrespective of the time taken by him in completing that work, is known :

- (a) Time Rate System (b) Piece Work Rate System
 (c) Time & Piece Work Rate System
 (d) Premium Bonus Method

Ans. (b) Piece Work Rate system

- (21) Under Halsey Plan the total amount of wages will be, when standard time is 15 Hrs, Time Taken is 12 Hrs., Wage rate is Rs. 2 per hour :

- (a) Rs. 24
(c) Rs. 30
Ans. (b) Rs. 27
- (22) Overhead is divided according to elements into :
(a) Indirect material, direct labour, and direct expenses
(b) Direct material and indirect expenses
(c) Indirect material, indirect labour and indirect expenses
(d) Indirect material, indirect labour and direct expenses
Ans. (c) Indirect material, indirect labour and indirect expenses
- (23) The expenses which tend to be unaffected by variation in output, are called :
(a) Semi Variable overheads (b) Fixed overheads
(c) Semi fixed overheads (d) Variable overheads
Ans. (b) Fixed overheads
- (24) All such expenses which are incurred from keeping the goods in the godown to the delivery of goods to the customers are mentioned under the heading :
(a) Factory overhead (b) Selling overhead
(c) Office overhead (d) Distribution overhead
Ans. (d) Distribution overhead
- (25) In a manufacturing concern from the point of view of cost accounts which of the following expense is direct expense :
(a) Expenses on royalties, patents, research, drawings, designs and patents
(b) Rent of factory building, insurance, depreciation, lighting and financial expenses
(c) Salary to factory manager, foreman and technical directors
(d) All the above
Ans. (a) Expenses on royalties, patents, research, drawings, designs and patents
- (26) Charging the full amount of a cost to a cost centre is called :
(a) Absorption of overhead (b) Allocation of overhead
(c) Apportionment of overhead (d) None of the above
Ans. (b) Allocation of overhead
- (27) The overheads of a particular department are to be Allotted to the production units of the department will be :
(a) Absorption of overheads (b) Allocation of overheads
(c) Apportionment of overheads (d) All the above
Ans. (a) Absorption of overheads
- (28) Causes of over or under absorption is/are :
(a) Seasonal fluctuations
(b) Errors in estimating overhead expenses
(c) Error in estimating output or hours to be workers
(d) All the above

- Ans. (d) All the above
- (29) The formula of finding the cost of material consumed is:
(a) Value of opening stock of Raw material + Value of purchase of Raw material + value of closing stock of raw material
(b) Value of opening stock of raw material- value of closing stock of Raw materials- value of Purchase of raw materials
(c) Value of opening stock of raw material + Value of Purchase of raw material- value of closing stock of raw-material
(d) None of the above
Ans. (c) Value of opening stock of raw material + Value of Purchase of raw material- value of closing stock of raw-material
- (30) Prime Cost includes :
(a) Direct Material + Direct Labour - Chargeable Expenses
(b) Direct Material- Indirect - Labour + Chargeable Expenses
(c) Direct Material + Factory Cost + Chargeable Expenses
(d) Direct Material + Direct Labour + Chargeable Expenses
Ans. (d) Direct Material + Direct Labour + Chargeable Expenses
- (31) Cost sheets are prepared under :
(a) Process method of costing (b) Contract method of costing
(c) Unit costing method of costing (d) All the above
Ans. (c) Unit costing method of costing
- (32) The source of receiving material for the Contract is :
(a) By issue from store room
(b) By Direct purchase from the market
(c) By Transfer from other contract (d) By all the above
Ans. (d) By all the above
- (33) Of the following which cost accounting System will be used in a water supply Company?
(a) Contract Costing system (b) Job Costing System
(c) Process Costing system (d) Operating Costing system.
Ans. (d) Operating Costing system
- (34) In unit or output costing the example of Indirect expense of the following is :
(a) Royalty (b) Architect or Surveyor's fees
(c) Excise Duty
(d) Supervision, Inspection and testing fees
Ans. (d) Supervision, Inspection and testing fees
- (35) Where the product of one stage or operation becomes the material of another stage of operation, such a method of costing is known as :
(a) Unit Costing Method (b) Operating Costing Method
(c) Job Costing Method (d) Process Costing Method
Ans. (d) Process Costing Method
- (36) The main object of Process costing method is :
(a) To control the cost
(b) To determine the rate of cost of service

- (c) To control the operating expenses
(d) To ascertain the cost per unit
- Ans. (a) To control the cost
- (37) While producing, by various industries of different commodities alongwith the manufacturing of the products, several other products too are suddenly produced are known as :
- (a) Joint-product (b) By-product
(c) Both the above (d) None of the above
- Ans. (b) By-product
- (38) When the actual production is more than the usual production, the difference is known as :
- (a) Normal loss (b) Abnormal loss
(c) Normal gain (d) Abnormal gain
- Ans. (d) Abnormal gain
- (39) Abnormal gain is written on which side of the process account :
- (a) Debit (b) Credit
(c) Not written (d) Predict
- Ans. (a) Debit
- (40) When actual loss in process is less than normal loss, the difference between the two is known as :
- (a) Abnormal waste (b) Abnormal gain
(c) Normal gain (d) Normal waste
- Ans. (b) Abnormal gain
- (41) When actual loss in process is more than the normal loss, the difference between the two is known as :
- (a) Abnormal wastage (b) Normal wastage
(c) Normal gain (d) Abnormal gain
- Ans. (a) Abnormal wastage
- (42) Normal production cost of 40 units is Rs. 400 then value of abnormal of units will be :
- (a) Rs. 20 (b) Rs. 4
(c) Rs. 8 (d) None of these
- Ans. (a) Rs. 20
- (43) Input in a process is 10,000 units and normal loss is 5%. When finished output in the process is only 9,600 units, there is an :
- (a) Abnormal loss of 100 units (b) Abnormal gain of 100 units
(c) No Abnormal gain or loss (d) None of these
- Ans. (b) Abnormal gain of 100 units
- (44) The difference between actual cost of materials and standard cost of materials is known as :
- (a) Material Price Variance (b) Material Cost Variance
(c) Material Usage Variance (d) Material Yield Variance
- Ans. (b) Material Cost Variance.
- (45) The object of cost audit is :
- (a) Protective (b) constructive

- (c) both of these (d) none of these
- Ans. (c) both of these
- (46) The difference between actual cost and standard cost is known as :
- (a) profit (b) differential cost
(c) variance (d) margin of safety
- Ans. (c) variance
- (47) Which of the following is considered as normal loss of material?
- (a) Pilferage (b) Loss due to accident
(c) Loss due to careless handling of material
(d) None of these.
- Ans. (c) Loss due to careless handling of material.
- (48) The most important element of cost is :
- (a) Material (b) Labour
(c) Overheads (d) All of these.
- Ans. (a) Material.
- (49) Direct material is a :
- (a) Administration Cost (b) Selling and Distribution cost
(c) All of these (d) None of these.
- Ans. (d) None of these.
- (50) Which of the following is considered as accounting record?
- (a) Bin Card (b) Bill of material
(c) Store Ledger (d) None of these.
- Ans. (c) Store Ledger.
- (51) In most of the industries, the most important element of cost is :
- (a) Labour (b) Overheads
(c) Administration Cost (d) All of the above
- Ans. (d) All of the above.
- (52) Which of the following methods smoothes out the effect of fluctuations when material prices fluctuate widely?
- (a) FIFO (b) Simple Average
(c) LIFO (d) None of the above
- Ans. (d) None of the above.
- (53) When overtime is required for meeting urgent orders, overtime premium should be :
- (a) Charged to costing profit and loss A/c
(b) Charged to overhead costs
(c) Charged to respective jobs (d) Ignored.
- Ans. (b) Charged to overhead costs.
- (54) Wages sheet is prepared by :
- (a) Time-keeping department (b) Personnel department
(c) Payroll department (d) Engineering department.
- Ans. (c) Payroll department.

(55) Labour turnover is measured by :

- (a) Number of workers replaced average number of workers
- (b) Number of workers left / number in the beginning plus number at the end
- (c) Number of workers joining / number in the beginning of the period
- (d) All of these.

Ans. (a) Number of workers replaced average number of workers.

(56) Idle time is :

- (a) Time spent by workers in factory
- (b) Time spent by workers in office
- (c) Time spent by workers off their work
- (d) Time spent by workers on their job.

Ans. (c) Time spent by workers off their work.

(57) Over time is :

- (a) Actual hours being more than normal time
- (b) Actual hours being more than standard time
- (c) Standard hours being more than actual hours
- (d) Actual hours being less than standard time.

Ans. (a) Actual hours being more than normal time.

(58) Time keeping refers to :

- (a) Time spent by workers on their job
- (b) Time spent by workers in factory
- (c) Time spent by workers without work
- (d) Time spent by workers on their job.

Ans. (b) Time spent by workers in factory

(59) Labour productivity is measured by comparing :

- (a) Total output with total man-hours
- (b) Added value for the product with total wage cost
- (c) Actual time and standard time
- (d) All of the above.

Ans. (d) All of the above.

(60) Identify, which one of the following, does not account for increasing labour productivity :

- (a) Motivating workers
- (b) Job satisfaction
- (c) Proper supervision and control
- (d) High labour turnover

Ans. (d) High labour turnover.

(61) Which method of absorption of factory overheads do you suggest in a concern which produces only one uniform time of product :

- (a) Percentage of direct wages basis
- (b) Direct labour rate
- (c) Machine hour rate
- (d) A rate per units of output.

Ans. (d) A rate per units of output.

(62) Selling and Distribution overheads are absorbed on the basis of :

- (a) Rate per unit
- (b) Percentage on works cost
- (c) Percentage on selling price of each unit
- (d) Any of these.

Ans. (d) Any of these.

(63) For the purpose of Cost sheet preparation, costs are classified based on:

- (a) Functions
- (b) Relevance
- (c) Variability
- (d) Nature

Ans. (a) Functions.

(64) Salary paid to an office supervisor is a part of:

- (a) Direct expenses
- (b) Administration cost
- (c) Quality control cost
- (d) Factory overheads

Ans. (b) Administration cost.

(65) Audit fees paid to cost auditors is part of:

- (a) Selling & Distribution cost
- (b) Production cost
- (c) Administration cost
- (d) Not recorded in the cost sheet

Ans. (c) Administration cost.

(66) A company pays royalty to State Government on the basis of production, it is treated as:

- (a) Direct Expenses
- (b) Factory overheads
- (c) Direct Material Cost
- (d) Administration cost

Ans. (a) Direct Expenses.

(67) In Reconciliation Statement Expenses shown only in cost accounts are :

- (a) Added to financial profit
- (b) Deducted from financial profit
- (c) Ignored
- (d) Deducted from costing profit.

Ans. (b) Deducted from financial profit.

(68) In Reconciliation Statement, transfers to reserves are :

- (a) Added to financial profit
- (b) Deducted from financial profit
- (c) Ignored
- (d) Added to costing profit.

Ans. (a) Added to financial profit.

(69) In Reconciliation Statement, Incomes shown only in financial accounts are :

- (a) Added to financial profit
- (b) Deducted from financial profit
- (c) Ignored
- (d) Deducted from costing profit.

Ans. (b) Deducted from financial profit.

(70) In Reconciliation Statement, Closing Stock Undervalued in Financial accounts is :

- (a) Added to financial profit
- (b) Deducted from financial profit
- (c) Ignored
- (d) Added to costing profit.

Ans. (a) Added to financial profit.

(71) Under non-integrated accounting system :

- (a) Separate ledgers are maintained for cost and financial accounts
- (b) Same ledger is maintained for cost and financial accounts by accountants
- (c) (a) and (b) both
- (d) None of the above

Ans. (a) Separate ledgers are maintained for cost and financial accounts.

(72) Under non-integrated accounting system, the account made to complete double entry is:

- (a) Finished goods control account
- (b) Work in progress control account
- (c) Stores ledger control account
- (d) General ledger adjustment account

Ans. (d) General ledger adjustment account.

(73) Under non-integrated system of accounting, purchase of raw material is debited to :

- (a) Purchase account
- (b) Material control account/ stores ledger control account
- (c) General ledger adjustment account
- (d) None of the above.

Ans. (b) Material control account/ stores ledger control account.

(74) When costing loss is Rs. 5,600, administrative overhead under-absorbed being Rs. 600, the loss as per financial accounts should be :

- (a) Rs. 5,000
- (b) Rs. 5,600
- (c) Rs. 6,200
- (d) None of the above.

Ans. (c) Rs. 6,200.

(75) Integral accounts eliminate the necessity of operating :

- (a) Cost Ledger control account
- (b) Store Ledger control account
- (c) Overhead adjustment account
- (d) None of the above.

Ans. (a) Cost Ledger control account.

(76) What entry will be passed under integrated system for purchase of stores on credit?

- (a) Dr. Stores
Cr. Creditors
- (b) Dr. Stores ledger control A/c
Cr. Creditors
- (c) Dr. Stores Ledger control A/c

- Cr. General Ledger adjustment A/c
- (d) None of the above.

Ans. (b) Dr. Stores ledger control A/c
Cr. Creditors.

(77) The accounting entry in integrated accounts for recording sales will be:

- (a) Dr. Cost ledger control account
Cr Profit and loss account
- (b) Dr. Sales account
Cr Profit and Loss A/c
- (c) Dr. Cash A/c
Cr. Sales A/c
- (d) None of the above

Ans. (c) Dr. Cash A/c
Cr. Sales A/c.

(78) Job costing is used in :

- (a) Furniture making
- (b) Repair shops
- (c) Printing press
- (d) All of the above.

Ans. (d) All of the above.

(79) In a job cost system, costs are accumulated :

- (a) On a monthly basis
- (b) By specific job
- (c) By department or process
- (d) By kind of material used.

Ans. (b) By specific job.

(80) The most suitable cost system where the products differ in type of material and work performed is :

- (a) Operating Costing
- (b) Job costing
- (c) Process costing
- (d) All of these.

Ans. (b) Job costing.

(81) In case product produced or jobs undertaken are of diverse nature, the system of costing to be used should be:

- (a) Operating Costing
- (b) Process Costing
- (c) Job costing
- (d) None of the above.

Ans. (c) Job costing.

(82) Job Costing is:

- (a) Suitable where similar products are produced on mass-scale
- (b) Method of costing used for non-standard and non-repetitive products
- (c) Technique of costing
- (d) Applicable to all industries regardless of the products or services provided.

Ans. (b) Method of costing used for non-standard and non-repetitive products.

(83) Which of the following documents are used in job costing to record the issue of direct materials to a job:

- (a) Purchase order
- (b) Purchase requisition

(c) Goods received note (d) Material requisition.

Ans. (d) **Material requisition.**

(84) **Spoilage that occurs under inefficient operating conditions and is generally controllable is called :**

- (a) Normal defectives (b) Abnormal spoilage
(c) Normal spoilage (d) None of the above.

Ans. (b) **Abnormal spoilage.**

(85) **In sugar manufacturing industry molasses is also produced along with sugar. Molasses may be of smaller value as compared with the value of sugar and is known as:**

- (a) Joint product (b) Common product
(c) By-product (d) None of them.

Ans. (c) **By-product.**

(86) **Which of the following is an example of by-product:**

- (a) Mustard seeds and mustard oil
(b) Diesel and Petrol in an oil refinery
(c) Edible oils and oil cakes
(d) Curd and butter in a dairy

Ans. (c) **Edible oils and oil cakes.**

(87) **Cost of service under operating costing is ascertained by preparing:**

- (a) Cost sheet (b) Process account
(c) Job cost sheet (d) Production account.

Ans. (a) **Cost sheet.**

(88) **Operating costing is applicable to:**

- (a) Hospitals (b) Cinemas
(c) Transport undertaking (d) All of the above.

Ans. (d) **All of the above.**

(89) **Absolute Tonne-km. is an example of:**

- (a) Composite unit for bus operation
(b) Composite unit of transport sector
(c) Composite unit for oil and natural gas
(d) Composite unit in power sector

Ans. (b) **Composite unit of transport sector.**



Note : In the Examination, some short answer questions will also be asked. So the students are advised to go through all the long answer questions thoroughly as given in this book and write down the answer as much as needed/asked in the question paper.